

# Signals or Smoke? The Determinants and Informativeness of Corporate Artificial Intelligence (AI) Disclosures

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August 2026

## Acknowledgments:

We thank Ahmed Abdalla, Philip Berger, Nathan Bergland, Beth Blankespoor, Chen Chen, Victoria Clout, Anna Costello, Jeff Coulton, Christine Cuny, Robert Czernekowski, Somnath Das, Gus de Franco, Emmanuel de George, Mac Gaulin (discussant), Andrew Jackson, Alibek Korganbekov (discussant), Rick Mergenthaler, Sarah Rice, Mario Schabus, Bryce Schonberger, Nikki Skinner, Stephen Taylor, Jennifer Tucker, Shuyan Wang, and workshop and conference participants at the 2024 BYU Accounting Research Symposium, the 2025 Financial Accounting and Reporting Section (FARS) Midyear Meeting, the 2025 Utah Winter Accounting Conference (UWAC), New England Accounting Research Symposium, Clemson University, University of Cincinnati, University of Illinois Chicago, University of Technology Sydney, Mississippi State University, University of Calgary, University of Minnesota, Monash University and Indiana University. This paper was previously circulated under the title “Artificially Intelligent or Artificially Inflated? Determinants and Informativeness of Corporate AI Disclosures”. We acknowledge financial support from the CBV Institute.

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# **Signals or Smoke? The Determinants and Informativeness of Corporate Artificial Intelligence (AI) Disclosures**

## **ABSTRACT**

Artificial Intelligence (AI) has emerged as a General Purpose Technology (GPT), yet the opacity of AI investments, including the line-item aggregation of direct investments and the intangibility of complementary investments, results in significant uncertainty about firms' AI activities. We examine the characteristics of firms that disclose AI activities, and whether these disclosures provide credible signals of firms' underlying AI investments and their future economic outcomes. We document several key findings. First, firms that disclose AI activities are generally innovative, operate in AI-intensive industries, and face greater investor scrutiny. Second, AI disclosures are positively associated with subsequent operational efficiency and AI patent filings, even after accounting for firms' investment in AI human capital. This indicates that disclosure not only provides a signal about a firm's AI investment levels, but also about managers' confidence in the expected returns of those investments. In cross-sectional tests, we find that AI disclosure appears to provide stronger signals when it concerns the use of AI in a firm's products and services and when firms operate in industries less expected to use AI, and weaker signals when firms appear to be outsourcing their AI capabilities. Overall, our findings suggest that AI disclosures are informative about firms' underlying AI investments and managers' confidence in their payoffs, and that financial statement users could benefit from AI human capital disclosure.

## 1. Introduction

Artificial Intelligence (AI) is rapidly transforming business practices across numerous industries. Broadly defined as machine-based systems capable of processing and analyzing large volumes of data, recognizing patterns, and making predictions or recommendations to meet human-defined objectives (OECD, 2019), AI technologies, such as machine learning, natural language processing, and generative AI, have reshaped industries by automating complex tasks, enhancing productivity, and fostering innovation. Once predominantly the domain of specialized technology companies, AI applications have expanded into mainstream sectors including finance, healthcare, retail, and manufacturing.

Reflecting its broad economic implications, AI is viewed as a General Purpose Technology (GPT), a foundational innovation with broad applicability that drives complementary innovations and fundamentally reshapes the economy (Bresnahan and Trajtenberg, 1995). Consistent with this perspective, Lisa Cook, Member of the Federal Reserve Board of Governors, recently highlighted that “AI, and generative AI in particular, is likely to become a General Purpose Technology” (Cook, 2024). While firms make direct investments in AI (e.g., chips and data centers), realizing AI’s economic potential depends heavily on complementary investments, which are predominantly intangible, such as human capital, organizational capabilities, and proprietary data (Brynjolfsson et al., 2019; Bresnahan et al., 1996). These investments are inherently difficult for investors to measure or verify, creating considerable uncertainty about firms’ true AI capabilities and expected returns. Throughout the paper, we use the term “AI investments” to refer to both components together, “direct investments in AI” for outlays on computing infrastructure and purchased access to third-party models, and “complementary investments in AI” for the human capital, proprietary data, and organizational redesign needed to put that computing to work.

Traditional financial metrics typically do not adequately capture AI investments, both

because of line-item aggregation and because accounting standards fail to recognize most intangible investments (Lev and Zarowin, 1999; Koh and Reeb, 2015). AI hardware is folded into property, plant, and equipment and only reaches the income statement as depreciation, purchased compute is expensed as incurred, and complementary investments are expensed and commingled with wages and overhead. Investors can observe some AI inputs directly, most notably hiring, but no reporting channel separates AI outlays from a firm's other spending, and none conveys managers' own assessment of what those outlays will return. Investors must therefore look to alternative signals of firms' AI adoption. The stakes of doing so are real; Babina et al. (2024) document a positive relationship between external measures of AI labor, firm efficiency, and valuation premiums, highlighting the tangible economic impact underlying these seemingly opaque investments. Consequently, investor demand for greater transparency, combined with the uncertainty surrounding AI investments, has led firms to disclose AI-related activities more frequently and explicitly in their 10-K filings, earnings announcements, and conference calls.

Yet managers themselves may face uncertainty about the expected timing and magnitude of payoffs from their firm-specific AI investments. Historically, AI technology has undergone cycles of rapid enthusiasm, known as "AI booms," followed by periods of stagnation, referred to as "AI winters" (Hendler 2008; Russell and Norvig 2021), driven by computational constraints and subsequent funding cuts. Each boom brought high expectations for the transformative potential of AI, yet these expectations often faded as technical barriers limited practical implementation.<sup>1</sup> Thus, it remains unclear whether the current phase represents a sustainable period of widespread AI adoption and whether firm-level AI initiatives will achieve the significant impacts managers

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<sup>1</sup> See <https://devrev.ai/blog/history-of-ai-part-1> for a brief discussion of the different periods of AI investment.

and investors anticipate.<sup>2</sup>

In this paper, we examine whether AI disclosures inform investors about firms' AI investments and managers' confidence in these investments. Specifically, we address two primary research questions: (1) Which types of firms are more likely to disclose AI-related information? (2) Do these disclosures reliably signal firms' AI adoption, as evidenced by operational efficiency and innovation outcomes?

To answer these questions, we use textual analysis techniques to quantify firms' AI disclosures from 2016 to 2023. Specifically, we use a fine-tuned FinBERT natural language processing model (Huang et al., 2023) to measure the prevalence of AI-related disclosures across corporate communication channels. We also use granular firm-level employment data on AI employees from Revelio, to provide a tangible proxy for firms' otherwise unobservable investments in AI human capital (Babina et al., 2024).

We begin by examining the evolution of AI disclosure and AI employment over our sample period. Both measures exhibit sustained growth, providing initial evidence that AI disclosures, on average, reflect firms' underlying investment activities. We also observe shifts in the content of AI disclosures over time. Initially, firms emphasized capabilities related to “cloud-based” technologies and “data analytics,” but toward the end of the sample, disclosures increasingly referenced technologies such as “machine learning,” “artificial intelligence,” and “generative AI.”

Next, we examine the determinants of AI disclosures. We find that industry affiliation, firm-level innovation intensity, and external monitoring (e.g., analyst coverage) are strongly associated with AI disclosure, consistent with competitive pressures, innovation efforts, and investor expectations within specific industries collectively shaping firms' incentives to

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<sup>2</sup> See <https://www.bostonglobe.com/2023/08/24/opinion/is-ai-really-good-advertised/> for examples of concerns related to AI investments and products.

communicate their AI activities. A Shapley decomposition (Shapley, 1953) indicates that industry-level factors are the primary drivers, explaining approximately 83 to 87% of the explained variation in disclosure decisions, while firm-specific characteristics account for 11 to 16%, and external monitoring mechanisms explain the remaining 2 to 4%. We note, however, that our determinants analysis cannot cleanly separate AI fundamentals from disclosure incentives, and we therefore interpret these results as identifying the firm characteristics associated with AI disclosure, rather than isolating the disclosure incentive effects.

Next, we examine whether AI disclosures are informative about firms' future performance by analyzing subsequent operational efficiency and AI patent filings. If disclosures convey meaningful signals about underlying investments, we expect higher AI disclosure levels to correlate positively with these future outcomes. Furthermore, if disclosures reflect managers' confidence in anticipated investment payoffs, the relationship should remain after controlling for AI employment, a measure of firms' complementary investment in AI-related human capital (as in Babina et al., 2024). Consistent with these expectations, we find that firms with more extensive AI disclosures indeed experience higher future operational efficiency and AI patenting activity. Importantly, these associations persist after controlling for firms' underlying AI employment levels, suggesting that AI disclosures signal not only firms' efforts to integrate AI technologies but also managers' confidence in the expected economic returns to those investments.

We further decompose AI disclosure into product-related and non-product-related components. Product-related AI disclosure includes information about AI-enabled products, services, and business applications, while non-product disclosure pertains to the use of AI in general operations, risk factors, and internal processes. Claims about AI-enabled products and services can be checked by customers, competitors, and analysts against what the firm delivers, so

managers face higher costs of misreporting. In contrast, claims about AI in internal operations and general processes are difficult for outsiders to falsify, leaving more room for cheap talk. If verifiability disciplines disclosure, the product-related component should carry a stronger signal about subsequent outcomes (Ball et al., 2012). We find that product-oriented AI disclosure has a significantly stronger association with both future efficiency and AI patenting than non-product disclosure. These results are consistent with product-oriented AI disclosure, which is more directly verifiable and tied to commercial outcomes, providing a stronger signal of future performance.

It is possible that a firm uses AI without investing in firm-specific labor by outsourcing AI capabilities. To further explore the implications of outsourcing on the informativeness of AI disclosure, we construct a FinBERT-based indicator of AI outsourcing by fine-tuning the model on sentences that co-mention AI and outsourcing language (e.g., third-party vendors, cloud services, named providers such as AWS or OpenAI), and examine whether conditioning on outsourcing moderates the AI disclosure's signal value. Our results suggest that the signal provided by AI disclosures is generally weaker in firms that appear to outsource AI development. Nevertheless, these firms' AI disclosures remain associated with future efficiency, patenting activity, and patent citations.

To strengthen our analysis on how AI disclosures incrementally signal managerial confidence in AI investments, we classify firms into groups based on a composite AI disclosure score (aggregating disclosure across 10-K filings, conference calls, and earnings announcements) and a binary AI employment classification (High, defined as the top tercile among firms with positive AI employment, versus Not-High). Within each employment classification, we find that efficiency and patenting scale monotonically with disclosure. For example, we find that firms with high AI disclosure and high AI employment exhibit the largest efficiency gains while firms with

high AI employment but low disclosure show no significant efficiency gains. This result is consistent with disclosure itself playing an informational role beyond simply serving as a proxy for AI investment. Additionally, we find that firms with high disclosure but not-high employment also show significant gains, consistent with outsourcing or early-stage adoption strategies, but at roughly half the magnitude of high disclosure and high employment firms. Collectively, these results indicate a complementarity between disclosure and employment: disclosure is most informative when backed by substantial AI human capital and provides an incremental signal of managerial confidence in the returns on AI investments.

Our results are robust to a battery of alternative specifications. We consider seniority- and compensation-weighted AI workforce measures to ensure the results are not driven by a simple headcount definition of AI employment. We re-estimate the main specifications in log form and as a negative binomial model for patent counts to confirm that inferences are not sensitive to the choice of scaling or to the zero-inflated nature of patent data. We include lagged dependent variables to rule out that our results merely reflect persistent firm-level differences rather than the information content of disclosure. Finally, we extend the horizon to two and three years ahead to account for the possibility that AI investments may take longer than one year to translate into efficiency and innovation. Across all specifications, AI disclosure remains positively and significantly associated with subsequent efficiency and AI patenting.

We contribute to a rapidly growing literature on AI, which has primarily developed in labor economics and finance. For example, Acemoglu et al. (2022) document substantial growth in AI-related job postings, especially among firms well-positioned to benefit from AI adoption, while Babina et al. (2024) link increased AI-specific hiring to improved firm performance and valuations. In contrast to these studies, which focus on realized labor market outcomes, we examine firms'

*disclosures* about AI investments and evaluate whether these disclosures credibly signal future efficiency gains, innovation, and overall investment quality. Using a comprehensive dataset spanning 2016 to 2023, we show that AI disclosures predict subsequent operational efficiency and patenting activity even after controlling for AI-related employment. This indicates that AI disclosures not only reflect underlying investments but also managerial confidence in anticipated outcomes. Furthermore, we examine three sources of cross-sectional variation in the extent to which AI disclosure provides a signal about future outcomes. We find the results appear stronger for disclosure about the use of AI in a firm's products and services and when firms operate in industries that are less expected to use AI, and weaker when firms appear to be outsourcing their AI capabilities.

Our study is similar to that of two contemporaneous studies, Jia et al. (2026) and Donelson et al. (2025). Jia et al. (2026) study firms' disclosures related specifically to Generative AI (a subset of our broader AI focus) during earnings conference calls in late 2022 and 2023. Their results on the determinants of AI disclosure largely align with ours, showing that AI-related disclosures concentrate in industries positioned to benefit from these technologies. We add to their findings by examining a broader set of AI disclosures (10-K, earnings announcements, and conference calls) over a longer period (2016 to 2023), incorporating detailed firm-level AI *employment* data, and separating what disclosure reveals about how much a firm has invested in AI from what it reveals about managers' confidence that the investment will pay off.

Donelson et al. (2025) study the misalignment between AI 10-K disclosure levels and AI hiring levels, focusing on subsets of firms that appear to be obfuscating or overstating their AI investments. We take a broader perspective and examine whether AI disclosure, on average, is a credible signal of future outcomes, and how that credibility varies with disclosure type,

outsourcing, and industry. That is, we provide evidence that AI disclosure is, on average, informative about future outcomes, and they provide evidence that in some settings the disclosure could represent managerial obfuscation or overstatement.

Collectively, our three papers suggest that AI disclosure is informative. On average, disclosure appears to credibly convey information about firms' AI investments and their future performance. However, the strength of this signal varies, and it is weakest when disclosure diverges sharply from underlying AI labor. Importantly, none of these studies examine the relation between AI investment and realized outcomes directly, nor consider the long-term stock market implications of AI disclosure, investment, or outcomes. Our evidence therefore is limited to whether AI disclosure predicts future outcomes, and not on whether AI investments generate their promised returns or whether the market correctly prices AI disclosure or investment in the short or long run.

Finally, our research directly informs ongoing regulatory and standard-setting debates concerning the improvement of financial reporting on intangible investments and innovation (FASB 2024; Glaeser and Lang 2024). Recent investor surveys indicate a strong desire for enhanced disclosure about firms' innovation activities, though investors often find it challenging to articulate precisely what additional information would be most beneficial (Peters and Winters 2025). Our findings specifically address this gap by demonstrating the economic value of distinguishing between firms' actual AI investments, particularly those reflected in labor market activities, and managerial perceptions of these investments' outcomes. By clearly separating objective measures of investment from subjective managerial signals, enhanced disclosures of AI-related labor investments can assist investors in assessing both the credibility and potential returns of intangible AI initiatives. Our recommendation is economically consistent with Glaeser et al.

(2025) and Campbell et al. (2026), who provide similar evidence that disclosing detailed information on R&D-related labor costs is a credible means for firms to communicate innovation information, balancing transparency with minimal proprietary costs. In addition, our finding that disclosure is less informative when firms appear to outsource suggests that investors might benefit from disclosure around whether and how much a firm pays for AI human capital outside the firm.

## **2. Motivation and Institutional Details**

AI is broadly defined as a machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations, or decisions (OECD, 2019). AI encompasses several core methodologies, including machine learning, natural language processing, and computer vision (Agrawal et al., 2019). These techniques allow AI systems to process large-scale, high-dimensional data (such as text, speech, and images) and perform complex tasks.

The commercialization of AI has accelerated over the past decade, driven by exponential increases in data availability, falling computation costs, and methodological advances in deep learning (Hodson, 2016). In fact, through 2024, by walking around with cellphones, smartwatches, and credit cards, humans have produced and stored 99 percent of all data that had ever existed within the preceding ten years, with 90 percent of that occurring in 2023 and 2024 (Bartley, 2024). It is no surprise that firms want to exploit that data to enhance efficiency and profits. Consequently, while AI adoption was initially limited to the technology sector, it has expanded into finance, healthcare, manufacturing, and consumer services, with corporate executives widely reporting AI integration in decision-making and operations (McKinsey & Company, 2020).

AI's adoption positions it as a GPT capable of stimulating complementary innovations that enhance productivity and transform business models (Bresnahan and Trajtenberg, 1995; Brynjolfsson et al., 2019). However, realizing AI's economic potential depends heavily on

intangible complementary investments, including human capital, proprietary data, and organizational capabilities (Mihet and Philippon, 2019). Additionally, while AI systems can generally be replicated across firms due to their informational nature, competitive advantages primarily accrue to firms possessing superior proprietary data (Fedyk and Hodson, 2023; Jones and Tonetti, 2020).

As AI adoption grows, firms face increasing pressure to signal AI capabilities to investors, given the technology's perceived potential to drive future growth, efficiency gains, and market valuation. However, while many AI investments are tangible, they also require significant intangible investments that are difficult to measure and verify. Further, given the complex nature of AI investments, there is likely significant uncertainty regarding the payoffs of firm-specific AI investments. As suggested by past AI winters (i.e., periods lacking AI investment following an AI boom), investments in AI do not always materialize.<sup>3</sup> Even though optimism is high that AI is here to stay, this sentiment is similar to that around past AI booms. While firms are making significant investments in AI, there is uncertainty from both managers and investors regarding the payoffs related to these investments. Thus, we examine whether firms' AI disclosures signal their confidence in the future payouts from firm-specific AI investments.

AI could be considered a subset of a firm's broader innovation activities. In that sense, our study is related to the stream of research on disclosures about firm innovation. As with AI, innovation activities are generally opaque and proprietary, with significant uncertainty surrounding project viability and the timing of project payouts. To date, the majority of studies in this area examine disclosure from patent filings (e.g., Glaeser et al., 2020; Glaeser and Landsman

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<sup>3</sup> The first AI winter, from roughly 1974 to 1980, followed computational constraints that prevented AI research from delivering on its most ambitious promises, prompting sharp funding cuts by the UK government and DARPA (Lighthill 1973). The second, from roughly 1987 to 1993, followed the collapse of the expert systems market and the failure of the Japanese Fifth Generation Computer Project (Roland and Shiman 2002).

2021; Kim and Valentine 2021; Dyer et al., 2024) or new product announcements (Chu et al., 2024).<sup>4</sup> They generally find that these disclosures are related to peer firm innovation, future revenues, and stock returns. We extend this literature by focusing on AI-specific disclosures in SEC filings and earnings conference calls, channels that directly address investor attention and are more closely linked to financial performance. Additionally, our inclusion of AI-related employment data uniquely allows us to differentiate firms' underlying investment activities from managers' subjective assessments regarding the expected success of these investments.

### **3. Sample Selection and Variable Construction**

Our sample spans 2016 to 2023. We begin our sample in 2016 because Acemoglu et al. (2022) provide evidence that AI adoption across U.S. establishments significantly accelerated after 2015. We capture AI disclosure across multiple sources, including earnings announcement press releases, conference calls, and annual 10-K filings.<sup>5</sup> We employ granular individual-level employment data from Revelio Labs, which sources detailed employment records from online professional profiles (e.g., LinkedIn), as a proxy for firms' complementary investments in AI human capital. After applying our exclusion criteria, our final sample consists of 11,927 firm-year observations.<sup>6</sup> Table 1 outlines our full sample selection procedures.

The underlying Revelio database contains over 300 million global employment records. In a similar spirit to Babina et al. (2024), we identify AI employees by searching for AI-related terms (such as “artificial intelligence,” “machine learning,” “data science,” “deep learning,” “computer vision,” “data mining”, “big data”, and “data analytics”) in both raw job titles and standardized

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<sup>4</sup> A notable exception is Glaeser (2018), who uses disclosure in 10-K filings to develop a measure of trade secrets – an alternative to the patenting process.

<sup>5</sup> Examining disclosure in the 10-K, conference call, and earnings announcements allows us to capture firm communications across different formal channels (Skinner 2024).

<sup>6</sup> Unlike many accounting studies, we include utility and financial firms in our sample, as these industries are also exposed to the impacts and incentives associated with AI.

job classifications in employment records with valid firm identifiers that link with U.S. firms' GVKEY or CIK identifiers. This process mitigates potential measurement errors from job title variations.<sup>7</sup> After identifying the AI employment records, we sum the number of AI employees within the fiscal year and scale AI employment by total employment to capture AI intensity. Thus, we focus on the proportion of human capital devoted to AI, rather than absolute headcount, which would largely reflect firm size. A firm with 10 AI employees out of 20 total makes a fundamentally different organizational commitment to AI than one with 10 AI employees out of 10,000. Absent scaling, our regressions would absorb a confound between AI investment and firm scale. This approach is consistent with the standard in the AI adoption literature (Babina et al., 2024; Acemoglu et al., 2022), facilitating comparability. We define *AI\_Employee* as the number of AI employees at a firm in a given fiscal year, and *AI\_Employee%* as *AI\_Employee* scaled by the firm's total employment. We interpret *AI\_Employee%* as a proxy for the firm's strategic commitment to AI human capital rather than a direct productivity measure.<sup>8</sup>

We identify AI-related disclosures using a fine-tuned FinBERT model (Huang et al., 2023) embedded in a two-stage keyword–FinBERT pipeline. Off-the-shelf FinBERT is pre-trained on financial texts (corporate 10-K/Q filings, analyst reports, and earnings call transcripts) for sentiment classification (positive, negative, neutral) rather than topical classification of technology disclosures. While its financial pre-training enables it to capture domain-specific semantic

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<sup>7</sup> Job titles and classifications in Revelio are short, typically less than 5 words. Thus, there is a general lack of context that make a bag-of-words approach more appropriate than alternative methods. Nevertheless, in untabulated analyses, we fine-tune a FinBERT model, similar to the process we follow to identify AI disclosures. The FinBERT model outputs are highly correlated with the bag-of-words approach (approximately 0.87), yet it classifies significantly more employees as AI employees (e.g., average of approximately 26 AI employees, relative to 7 AI employees using bag-of-words). Specifically, it tends to identify all scientists as AI employees, rather than narrowly defining data scientists as AI employees. Nevertheless, our primary results that AI disclosure is associated with underlying investment in AI, and that the association between disclosure and future performance are incremental to the underlying investments remain qualitatively unchanged.

<sup>8</sup> We recognize that there could be many ways to measure the level of firms' AI investments, including dollar spend on software, token usage, number of AI products, etc. We examine labor because of the availability of the data and because recent research suggests that human input remains an important complement to machine output in producing financial information (Cao et al., 2024); see also Coleman et al. (2022) on the relative performance of robo-analysts and human analysts.

relationships that traditional dictionary methods cannot, the model in its default form is not designed to distinguish substantive AI disclosures. We therefore fine-tune FinBERT on a hand-labeled training set of disclosure sentences using a two-stage pipeline: a broad keyword screen first identifies candidate AI-related sentences, which are manually labeled by human coders to construct the training set, and the fine-tuned model then classifies each candidate as a substantive AI-related disclosure or not.

First, to construct the training set, we extract sentences from Items 1, 1A, and 7 of firms' 10-K filings between 2016 and 2023 and apply the keyword screen to identify candidate sentences containing AI-related terms (e.g., “machine learning,” “artificial intelligence,” “data analytics,” “natural language processing,” “computer vision”). This initial screen ensures that the manually labeled training set includes a sufficient proportion of AI-related sentences, which would otherwise be rare in the full corpus of corporate filings. The keyword-based filter yields 640 candidate sentences containing AI-related terms, which we then add to a random sample of 971 additional sentences, for a total of 1,611 sentences for manual classification.

Two trained research assistants independently categorized each sentence as pertaining to AI or not. To aid in the classification, we provided the research assistants with five AI categories: AI\_Overview (general AI trends), AI\_Use (firm use of AI), AI\_Product/Business/Patentable (AI-related products or technologies), AI\_Acquire/Build (acquiring AI capabilities), and AI\_FLS (forward-looking AI statements). Inter-rater reliability, assessed using Cohen's kappa, exceeded 0.85 across all coder pairs, indicating strong agreement; disagreements were resolved through adjudication discussions with the broader research team. We split the labeled dataset into 80% for training, 10% for validation, and 10% for testing, and fine-tune the FinBERT model over 5 epochs. Model testing on the held-out test set indicates that our fine-tuned model identifies AI-related

disclosures with over 96% accuracy, ensuring high reliability in capturing AI-related corporate communication.

Second, we apply the fine-tuned model to multiple disclosure channels (firms' 10-K annual reports, conference call transcripts, and earnings announcements) to identify AI disclosures at scale.<sup>9</sup> Because the AI-related categories are not mutually exclusive, we construct the main AI disclosure score using a sentence-level AI indicator rather than summing category labels. A sentence is counted once if it is classified as AI-related in any category. We then calculate AI disclosure scores ( $AI\_CC$ ,  $AI\_10K$ ,  $AI\_EA$ ) on a firm-year basis for each disclosure channel by dividing the number of unique AI-related sentences by the total number of sentences in the respective disclosure channel.<sup>10</sup> Scaling AI-related sentences by total sentences controls for overall disclosure verbosity, which varies across firms and filing types, and is the standard approach in the textual analysis literature (Li, 2008; Loughran and McDonald, 2011; Huang et al., 2023). Without scaling, we would confound AI disclosure intensity with a firm's general propensity to write lengthy disclosures. Intuitively, a firm that devotes 10 out of 100 sentences to AI is signaling AI centrality in a way that a firm devoting 10 out of 10,000 sentences is not; we view the sentence-ratio as capturing the prominence of AI in the disclosure. This approach leverages pre-training on financial texts, which overcomes the limitations of traditional dictionary-based methods and enables more accurate identification of AI-related content across various corporate communications (Huang et al., 2023).<sup>11</sup>

We obtain AI patent classifications from the USPTO's official Artificial Intelligence Patent

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<sup>9</sup> For 10-K filings, these include item 1, "Business," item 1A, "Risk Factors," and item 7, "Management Discussion and Analysis of Financial Condition and Results of Operations."

<sup>10</sup> See Appendix OA 2A for examples of each content category by channel (Panel A: 10-K filings; Panel B: earnings conference calls; Panel C: earnings announcements). For the quarterly channels, conference calls and earnings announcements, we average disclosure scores across the four quarters in each fiscal year.

<sup>11</sup> We make the resulting AI disclosure measures available to the research community at <https://github.com/bcjl-research/ai>.

Dataset (Giczy et al., 2022).<sup>12</sup> A patent is classified as an AI patent if the likelihood that it falls within at least one of the following USPTO AI subfield categories exceeds 50%: AI Hardware, evolutionary computation, knowledge processing, machine learning, natural language processing, planning and control, speech, and vision. We scale AI patents by total patents to capture AI innovation intensity while controlling for a firm's overall propensity to patent, which varies widely across industries and firm types (Hall et al., 2005; Giczy et al., 2022).

Before examining detailed descriptive statistics, we present an overview of AI disclosure trends and AI employment patterns in our sample. Figure 1 illustrates the average AI disclosure ratios across firms from 2016 to 2023 for each disclosure channel. Figure 1 reveals a consistent upward trend in AI-related discussions across all three major disclosure channels, with conference calls exhibiting the largest proportion of AI disclosure. Notably, AI mentions in conference calls increased by more than 50% between 2016 and 2023, reflecting a growing emphasis on AI-related strategies in firms' communications with investors. Further, the increase in AI disclosure pre-dates the roll-out of ChatGPT in November 2022, suggesting that our disclosure measures capture broad discussions of AI, rather than solely focusing on Gen-AI.

Figure 2 presents AI employment trends over time. Panel A presents the average number of AI employees per firm, which grew from approximately 3 in 2016 to over 6 by 2023, indicating a rising demand for AI-related talent. Panel B presents AI employees as a percentage of total employees, which increased from 0.06% in 2016 to 0.17% in 2023, suggesting a growing emphasis on AI-related roles within the overall workforce. These employment trends indicate an increase in AI workforce *levels* and a growing importance of AI-related roles relative to overall employment. Moreover, the parallel upward trends in AI disclosure (Figure 1) and AI employment (Figure 2)

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<sup>12</sup> Available at <https://www.uspto.gov/ip-policy/economic-research/research-datasets/artificial-intelligence-patent-dataset>.

are consistent with firms' AI disclosures, on average, reflecting actual AI investments, reinforcing the informativeness of AI-related disclosures.

Figure 3 visualizes the evolution of AI-related language in U.S. public firms' disclosures in our sample using TF-IDF trigrams.<sup>13</sup> The results indicate a clear progression in AI discourse over time. In the earlier years (2016-2017), firms' communications predominantly focused on "cloud-based" technologies and "data analytics." Over time, as technology evolved, firms' discussions shifted toward "machine learning" (2018-2020), followed by explicit references to "artificial intelligence" in recent years (2021-2023).

## 4. Empirical Results

### 4.1 Descriptive Statistics

Table 2, Panel A, presents descriptive statistics for our key variables. Appendix 1 provides a detailed description of all variables used in the study. The mean values for AI disclosures in conference calls (*AI\_CC*), 10-K filings (*AI\_10K*), and earnings announcements (*AI\_EA*) are 1.51%, 1.08%, and 1.07% of sentences, respectively.<sup>14</sup> The standard deviations for these variables (2.61%, 1.97%, and 2.04%) suggest considerable variation in AI disclosure practices among firms, particularly at the upper end of the distributions. The average number of AI employees (*AI\_Employee*) is 7, with a standard deviation of 14, indicating significant variation in firms' AI workforces. The average AI patent intensity in our sample is 11.38%. The average firm in our sample is moderately mature (approximately 10 years old), R&D-intensive (R&D expenditures constituting 6% of total assets), and subject to substantial external scrutiny (7 analysts and 72%

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<sup>13</sup> TF-IDF trigrams is a text analysis method measuring the relative importance of three-word phrases. TF (Term Frequency) counts phrase occurrences in a document, while IDF (Inverse Document Frequency) downweights common phrases across documents. This helps identify distinctive technical terminology in corporate communications.

<sup>14</sup> All AI disclosure and AI employment share variables are expressed in percentage points (i.e., multiplied by 100) throughout the manuscript.

institutional ownership on average).<sup>15</sup>

Table 2, Panel B provides pairwise correlations between AI disclosures across different communication channels (10-K filings, earnings announcements, and conference calls) and AI employment. AI disclosure levels are strongly correlated across communication channels, with the strongest correlation occurring between 10-K filings and conference calls (0.73,  $p < 0.01$ ), followed by earnings announcements and conference calls (0.72,  $p < 0.01$ ). However, correlations between disclosure measures and actual AI employment are relatively lower, ranging from 0.20 to 0.31 (all  $p < 0.01$ ). These weaker correlations suggest potential timing differences or strategic considerations, indicating that while disclosures broadly reflect underlying investments, they do not perfectly align with the current level of AI employment at the time of disclosure.

Table 2, Panel C, presents a breakdown of AI adoption within industries. Specifically, we quantify the percentage of firm-year observations within an industry that disclose any AI information in any disclosure channel. We find that Business Equipment, Telephone and TV, and Consumer Durables have the highest industry concentration of AI disclosers across 10-Ks, Conference Calls, and Earnings Announcements. We find that over 71% of firm-years within these three industries contain discussion of AI in 10-Ks, with 93.5% of observations in the Business Equipment industry discussing AI. We also find that Chemicals, Oil, Gas, and Coal, and Utilities are consistently the industries with the lowest percentage of firms with AI disclosures, with approximately 24% of observations in the Utilities industry disclosing AI information in 10-Ks.

Overall, these results provide evidence consistent with the parallel trends presented in Figures 1 and 2, suggesting a strong link between firms' AI disclosures and their complementary investments in firm-specific AI employment, while also indicating significant variation in AI

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<sup>15</sup> For our baseline sample, we include only firms with available data across all three disclosure channels. Consistent with prior research, these firms typically have higher institutional ownership.

adoption across industries.

## 4.2 The Determinants of AI Disclosure

To identify the drivers of AI disclosure, we estimate an OLS regression incorporating key firm-level variables, industry-wide characteristics, and external monitoring factors:

$$AIDisclosure_{i,t} = \alpha + \beta_1 R\&D_{i,t} + \beta_2 Investment_{i,t} + \beta_3 Age_{i,t} + \beta_4 Size_{i,t} + \beta_5 Profitability_{i,t} + \beta_6 Tangibility_{i,t} + \beta_7 HHI\_Seg_{i,t} + \beta_8 MTB_{i,t} + \beta_9 Leverage_{i,t} + \beta_{10} N\_Analyst_{i,t} + \beta_{11} Insti\_Own_{i,t} + \beta_{12} AI\_Dis\_Ind_{j,t} + Industry\ FE + Year\ FE + \mu_{i,t} \quad (1)$$

where  $i$  indexes firm,  $j$  indexes firm  $i$ 's industry, and  $t$  the year. *AIDisclosure* is a set of variables capturing AI disclosures in earnings announcements, conference calls, and annual 10-K reports (*AI\_10K*, *AI\_CC*, *AI\_EA*). We include variables related to firm characteristics including R&D expenses (*R&D*), investment ratio (*Investment*), firm age (*Age*), firm size (*Size*), performance (*Profitability*), asset tangibility (*Tangibility*), the concentration of operations (*HHI\_Seg*), market to book ratio (*MTB*), debt ratio (*Leverage*), and information environments such as analyst following (*N\_Analyst*) and institutional ownership (*Insti\_Own*). We also control for industry-wide AI disclosure and the percentage of employees in AI-related roles within the firm.<sup>16</sup> Lastly, we include industry and year fixed effects to remove unobservable time- or industry-invariant factors that may lead to spurious associations between our variables of interest.<sup>17</sup> We cluster standard errors at the firm level and winsorize continuous variables at the 1<sup>st</sup> and 99<sup>th</sup> percentiles.

Drawing on the voluntary disclosure literature (Verrecchia, 1983; Healy and Palepu, 2001; Dye, 1985) and the innovation disclosure literature (Glaeser et al., 2020; Kim and Valentine, 2021), we predict the following directional effects. Innovation intensity should positively predict AI disclosure, as firms with greater intangible investment face higher investor uncertainty and thus

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<sup>16</sup> We measure industry-wide AI disclosure (*AI\_Dis\_Ind*) as the count of firms making AI-related claims across all communication channels, aggregated by three-digit SIC code annually.

<sup>17</sup> We use the Fama-French 12 industry classification as our industry fixed effects following prior research (e.g., Jia et al., 2026).

have stronger incentives to communicate their AI activities to reduce information asymmetry. External monitoring (analyst following, institutional ownership) should positively predict disclosure, as analysts and institutional investors demand transparency and impose reputational costs on firms that fail to communicate coherently about growth strategies (Lang and Lundholm, 1993). Asset tangibility should negatively predict AI disclosure, as firms with fewer tangible assets rely more heavily on intangible value drivers that investors cannot observe from balance sheets, strengthening disclosure incentives. Industry-level AI disclosure intensity should positively predict firm-level disclosure, reflecting both competitive mimicry and the co-evolution of industry-wide AI adoption (DiMaggio and Powell, 1983; Seo, 2021).

Table 3 presents the results of estimating Equation 1, which analyzes the determinants of AI disclosure across three key communication channels: 10-K filings, earnings conference calls, and earnings announcements. To facilitate comparisons, we standardize all independent variables. We begin by examining the determinants of AI disclosure without explicitly considering firms' AI employment (columns 1–3). Results consistently highlight that industry characteristics, asset tangibility, and R&D intensity are the strongest determinants. Specifically, firms in innovation-intensive industries and those with significant intangible investments (as reflected in higher R&D expenditures and lower asset tangibility) are more likely to disclose AI-related information. These findings underscore that firms' disclosure behavior is primarily driven by industry conditions and their innovation orientation.

We then extend our analysis by incorporating firms' AI employment intensity as an additional determinant (columns 4 through 6), using *AI\_Employee%*, the share of a firm's workforce in AI-related roles, as the key measure. Across all three disclosure channels, *AI\_Employee%* loads significantly positively (0.33 for 10-Ks, 0.49 for conference calls, and 0.39

for earnings announcements; all  $p < 0.01$ ), indicating that firms with greater AI workforce intensity disclose more about AI across all venues. Importantly, the inclusion of these variables does not materially alter the coefficients of other key determinants, indicating that AI employment captures variation in AI adoption distinct from the firm characteristics already in the model. The positive association between AI disclosure and AI workforce intensity is consistent with AI disclosure serving as a signal of genuine AI adoption, but we cannot rule out that both respond to common unobserved fundamentals.

Specifically, we note that several of our determinant variables, particularly R&D intensity, industry AI concentration, and AI employment, likely capture firms' underlying AI capabilities (i.e., AI fundamentals) as well as disclosure incentives (Böke et al., 2025). Cleanly separating the two channels would require an instrument that shifts disclosure incentives without affecting AI fundamentals, a demanding identification task that, to our knowledge, no existing study has accomplished. We therefore interpret our determinants analysis as identifying the firm characteristics associated with AI disclosure, recognizing that these characteristics shape disclosure through both channels. We view this as informative: understanding which firms discuss AI, whether because they have more to disclose or because they face stronger incentives to do so, is a necessary first step toward the narrower identification that future work may achieve.

To further clarify the relative importance of each determinant, we perform a Shapley decomposition of our model's explanatory power ( $R^2$ ) for this set of determinants. The primary goal of this decomposition is to characterize whether AI disclosure is primarily driven by a firm's industry context (consistent with competitive herding and industry-level AI adoption pressures) or by firm-specific investment choices (consistent with a more deliberate, firm-level signaling motivation). As shown in Panel A of Figure 4, industry-level factors dominate across all three

disclosure channels: the combination of industry fixed effects and industry-level AI adoption measures accounts for 83-87% of the explained variation. Within this industry component, pure industry fixed effects account for 42 to 45%, while industry-level AI adoption measures add another 35 to 42%, indicating that AI disclosure patterns are primarily driven by sector-wide technological conditions rather than firm-specific strategy. Firm-specific characteristics account for 11 to 16% of explained variation, with the contribution rising modestly from 10-K filings (11.1%) to earnings announcements (15.9%), consistent with managers exercising greater discretion in more interactive disclosure settings. External monitoring mechanisms, which most cleanly proxy for disclosure incentives, contribute incrementally but modestly, explaining just 2.2 to 4.0% of variation across channels. This pattern suggests that while governance and information environment characteristics shape disclosure behavior at the margin, the dominant determinant of whether and how firms discuss AI is the broader industry-level adoption trend.

Figure 4, Panel B provides a more complete picture of AI employment's explanatory power by reporting three measures side by side for each disclosure channel: (1) the standalone  $R^2$ , which regresses AI disclosure on only AI employment and fixed effects, attributing all shared variance to employment (the least conservative measure); (2) the Shapley value, which decomposes the full model's  $R^2$  by computing AI employment's average marginal contribution across all possible orderings of variable groups, distributing shared variance proportionally; and (3) the incremental  $R^2$ , which subtracts the full model's  $R^2$  without AI employment from the full model's  $R^2$  with it, capturing only the variance unique to AI employment after all other determinants are included (the most conservative measure). For the 10-K specification, these values are approximately 37.23% (standalone), 12.21% (Shapley), and 1.21% (incremental). For conference calls, the values are 36.22% (standalone), 12.65% (Shapley), and 1.04% (incremental). For earnings announcements,

the values are 33.39% (standalone), 15.34% (Shapley), and 1.43% (incremental). The progression from standalone to incremental makes clear that AI employment shares substantial explanatory power with other firm characteristics, yet retains a meaningful unique contribution: the Shapley value, which fairly allocates this overlap, attributes roughly 12 to 15% of total explained variance to AI employment across all three channels. Even under the most conservative incremental approach, AI employment adds approximately 1 to 1.4 percentage points of  $R^2$  beyond what all other determinants can explain. These results collectively suggest that while strategic considerations, industry dynamics, and internal innovation investments predominantly shape AI disclosure decisions, firm-level AI employment is a robust, independently informative determinant.<sup>18</sup>

### 4.3 AI Disclosures and Firm Performance

#### 4.3.1 Firm Efficiency

To test whether AI disclosures convey economically meaningful information about future firm performance, we first examine their association with subsequent operational efficiency. We estimate the following regression model:

$$Efficiency_{i,t+1} = \alpha + \beta_1 AI\_Variables_{i,t} + \beta_2 R\&D_{i,t} + \beta_3 Investment_{i,t} + \beta_4 AGE_{i,t} + \beta_5 Size_{i,t} + \beta_6 Profitability_{i,t} + \beta_7 Tangibility_{i,t} + \beta_8 HHI\_Seg_{i,t} + \beta_9 MTB_{i,t} + \beta_{10} Leverage_{i,t} + \beta_{11} N\_Analyst_{i,t} + \beta_{12} Insti\_Own_{i,t} + \beta_{13} AI\_Dis\_Ind_{j,t} + Industry\ FE + Year\ FE + \mu_{i,t+1} \quad (2)$$

where  $i$  indexes firm and  $t$  the year. *Efficiency* is measured using stochastic frontier analysis (SFA) (Aigner et al., 1977; Meeusen and Van den Broeck, 1977). SFA infers efficiency by assessing firms' resource use to generate revenue within each industry. As a parametric method, it allows

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<sup>18</sup> Our primary determinants model (i.e., Equation 1) considers the effects of industry, firm, and monitoring characteristics. However, it is possible that manager characteristics and litigation play a role in determining firms' AI disclosure practices. Thus, we augment Equation 1 to include other measures of external monitoring, such as board independence and mutual fund ownership; manager characteristics, such as managerial ability and CEO stock compensation; and litigation risk. In the results tabulated in Online Appendix Table OA 1, we find that many of these effects are significantly associated with AI disclosure when included on their own, but their effects are largely subsumed when included alongside the original variables in Equation 1.

direct statistical assessment of model parameters, accommodating random errors (Dopuch et al., 2003; Callen et al., 2005; Baik et al., 2013). Following Demerjian et al. (2012), inputs include cost of goods sold, net PP&E, net R&D, and employee count, with revenue as the output. Firms can enhance efficiency through process optimization, automation, and enhanced decision-making, all of which can be facilitated by AI adoption. Given that our SFA model controls for industry-wide relations between inputs and outputs, our efficiency measure captures firm-specific productivity advantages, making it well-suited to test whether AI disclosures signal real efficiency gains. All other variables are as previously defined.

Table 4 presents the results of estimating Equation 2. Similar to Table 3, initially, we examine whether AI disclosures alone (excluding explicit measures of underlying AI investment) predict future firm efficiency. Results in columns 1–3 consistently demonstrate a positive and highly significant association ( $p < 0.01$ ) between AI disclosures across all communication channels (10-K filings, earnings announcements, and conference calls) and future efficiency. This finding supports the interpretation that firms disclosing more extensively about their AI initiatives tend to realize greater operational efficiency in subsequent periods relative to their industry peers.

Next, we consider whether the positive relationship between AI disclosure and efficiency remains incremental after explicitly controlling for firms' investments in AI human capital. Columns 4–6 show that the association between AI disclosure and future efficiency remains strongly significant, even when controlling for underlying AI employment. The persistence of these results indicates that AI disclosures do not merely reflect observable AI investments. Instead, disclosures appear to provide an additional signal of managers' confidence regarding the potential success and payoffs of their AI investments. Firms with higher disclosure likely possess greater certainty about payoffs from their AI-related strategies, beyond what investors could infer from

observable investment measures alone.

### 4.3.2 Innovation Output

We next examine the association between AI disclosures and AI innovation output, specifically AI patents. We estimate the following equation:

$$AI\_Patent_{i,t+1} = \alpha + \beta_1 AI\_Variables_{i,t} + \beta_2 R\&D_{i,t} + \beta_3 Investment_{i,t} + \beta_4 AGE_{i,t} + \beta_5 Size_{i,t} + \beta_6 Profitability_{i,t} + \beta_7 Tangibility_{i,t} + \beta_8 HHI\_Seg_{i,t} + \beta_9 MTB_{i,t} + \beta_{10} Leverage_{i,t} + \beta_{11} N\_Analyst_{i,t} + \beta_{12} Insti\_Own_{i,t} + \beta_{13} AI\_Dis\_Ind_{j,t} + Industry\ FE + Year\ FE + \mu_{i,t+1} \quad (3)$$

where *AI\_Patent* is the proportion of all firms' patent filings that pertain to AI-related innovation.

We identify AI patents as patent filings with a greater than 50% likelihood of being classified in at least one of the following categories by the USPTO: AI Hardware, evolutionary computation, knowledge processing, machine learning, natural language processing, planning/control, speech, and vision (following Giczy et al., 2022).<sup>19</sup>

We present the results of estimating Equation 3 in columns 1-3 of Table 5, Panel A. We find that AI disclosures across all channels are positively and significantly associated with AI patent ratios ( $p < 0.01$ ). This robust relationship suggests that firms disclosing more about their AI initiatives are also more likely to incorporate AI into their innovations, indicating a strong link between AI disclosure and AI innovation outputs. We augment Equation 3 by adding *AI\_Employee%* to evaluate whether the association between AI disclosure and AI innovation is incremental to firms' underlying investments in AI employment. We present these results in columns 4 through 6. We continue to find that AI disclosure is positively associated with AI innovation, suggesting that disclosure is an important mechanism by which managers convey information about the viability of firm-specific AI investments.<sup>20</sup>

<sup>19</sup> Appendix OA 2B provides detailed examples of AI patents for each category.

<sup>20</sup> In untabulated analysis, we remove firm-year observations that do not have any patent filings during the period to minimize the possibility that outsourcing decisions impact our inferences. Our results are robust to this research design choice.

### 4.3.3 Product vs. Non-Product AI Disclosure

We next examine whether AI disclosures are more informative when they specifically pertain to the firm's products. We anticipate that the verifiability of product-related disclosures will discipline managers against misreporting their AI activities and will provide a stronger signal of underlying AI adoption than non-product AI disclosures (Ball et al., 2012). Claims about AI-enabled products and services can be checked by customers, competitors, and analysts against what the firm delivers, whereas claims about AI in internal operations, risk factors, and general processes are far more difficult for outsiders to falsify. If verifiability disciplines disclosure, product-related AI disclosure should carry a stronger signal about subsequent outcomes than non-product disclosure. To test this assertion, we decompose AI disclosure into product-related disclosure (*AI\_Product\_10K*, capturing sentences about AI-enabled products, services, and business applications) and non-product disclosure (*AI\_Non\_Product\_10K*, capturing sentences about AI in general operations, risk factors, and internal processes).<sup>21</sup> Table 6, Panel A shows that AI employment is a significant determinant of product-related AI disclosure (coefficient = 0.18,  $p < 0.01$ ) and non-product disclosure (coefficient = 0.09,  $p < 0.01$ ). We also find that the coefficient on product-related disclosure is approximately twice that on non-product-related disclosure. This result suggests that firms with a larger share of AI employees are more likely to disclose information about the business applications of AI capabilities than about AI in general.

In Table 6, Panel B (*Efficiency*), product-related AI disclosure has a coefficient of 0.02 ( $p < 0.01$ ), while non-product disclosure has a coefficient of 0.01 ( $p < 0.01$ ) on its own but attenuates to 0.00 ( $p < 0.05$ ) when both are included alongside AI employment. A formal cross-equation

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<sup>21</sup> While we fine-tuned FinBERT for AI disclosure detection, the product vs. non-product classification is a more conceptually nuanced distinction better served by encoding the decision rule explicitly in an LLM prompt. Information about the prompt used for the model can be found in the online appendix.

coefficient equality test rejects the null that the product and non-product coefficients are equal ( $\chi^2 = 26.79$ ,  $p < 0.001$  without AI employment controls;  $\chi^2 = 23.37$ ,  $p < 0.001$  with controls). In Table 6, Panel C (*AI Patent*), a similar pattern emerges: product disclosure (8.54,  $p < 0.01$ ) loads more strongly than non-product disclosure (4.05,  $p < 0.01$ ), with the difference confirmed by formal testing ( $\chi^2 = 20.96$ ,  $p < 0.001$ ;  $\chi^2 = 15.90$ ,  $p < 0.001$  with *AI\_Employee%* as an additional control). These results are consistent with product-oriented AI disclosure, which is more directly verifiable and tied to commercial outcomes, provides a stronger signal of future performance.

#### 4.3.4 AI Outsourcing

Firms that obtain AI capabilities from external providers (e.g., consultants, cloud platforms, APIs) may disclose information about AI without maintaining large internal AI workforces. Additionally, firms with internal AI investments likely have different information asymmetries regarding AI capabilities than firms that outsource their AI activities. To test whether outsourcing impacts the relative information conveyed by managers' disclosures, we construct a textual indicator of AI outsourcing using a two-stage approach. First, we develop a rule-based classifier that identifies sentences co-mentioning AI-related terms with outsourcing language, including generic outsourcing phrases (e.g., third-party, vendor, partner, license, cloud service, API) and named vendors (e.g., AWS, Azure, OpenAI, Palantir, Accenture), using proximity-window matching and negation filtering. A graduate-level research assistant manually validated a stratified sample of flagged sentences, yielding approximately 77% precision for the rule-based approach. Second, using these labels, we construct a training sample of 1,528 sentences and fine-tune a FinBERT model for binary outsourcing classification, achieving 89% accuracy and an F1 score of 0.81 on a held-out test set. We apply this model to all AI sentences across 10-K filings, conference

calls, and earnings announcements to generate firm-year outsourcing indicators.<sup>22</sup>

We first examine the determinants of AI disclosure for firms identified as potential outsourcers. We present the results of re-estimating Equation 1 in the subsample of firms identified as potential outsourcers in Table 7, Panel A. Overall, the determinants of AI disclosure in the subsample of firms identified as potential AI outsourcers largely align with the results on the full sample. However, there are some meaningful differences. While the results using the full sample, presented in Table 3, suggest that *R&D* and *Investment* are significant predictors of AI disclosure, this is not true for the subsample of firms identified as potential AI outsourcers. This result suggests that the R&D and capital expenditures of firms identified as potential outsourcers are largely unassociated with AI-related disclosure. This result provides credibility to our classification procedure, as traditional R&D and capital investment measures are unlikely to capture the amount companies spend to acquire AI capabilities from third parties, while they would likely capture spending on internal AI development.<sup>23</sup>

We then augment Equations 2 and 3 by including the indicator for suspected AI outsourcing and interacting it with AI disclosure in the 10-K. Table 7, Panel B, presents the results with three main findings. First, AI disclosures made by firms that appear to outsource AI are significantly less associated with future efficiency at  $t+1$  and  $t+2$ , but are insignificantly different at  $t+3$ . These results potentially suggest that the near-term costs of external AI services offset some of the short-term operational efficiency gains. However, F-tests of the combination of AI disclosure and the interaction between disclosure and outsourcing suggest that there remains a

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<sup>22</sup> A detailed list of the keywords used and sentences identified by the Fine-Tuned FinBert model to identify AI outsourcing can be found in the online appendix.

<sup>23</sup> In Figure OA 2 of the Online Appendix, we also re-estimate the Shapley Decomposition presented in Figure 4, Panel A, and find that the relative explanatory power of industry characteristics in the subsample of firms identified as AI outsourcers largely reflects that of the full sample. However, the information environment/monitoring factors appear to have much higher explanatory power in this subsample than in the full sample.

significant positive association between disclosure and efficiency for outsourcing firms across all three periods.

Second, there is marginal evidence that the association between AI disclosure and future patent activities varies between suspected internal developers and suspected outsourcers, but this result is only concentrated in  $t+1$ , with no differences found in  $t+2$ , and  $t+3$ . This result suggests that outsourcing firms still generate patentable AI innovations, potentially by combining external AI tools with internal domain expertise. Third, when we examine AI patent citations, we find that the association between AI disclosure and patent citations is significantly mitigated for suspected outsourcing firms. For example, in 10-K filings, the disclosure-citation slope falls by roughly 60% (from 0.08 to 0.03) for suspected outsourcing firms relative to suspected internal developers (Column (7) in Panel B of Table 7). However, consistent with the results on efficiency and patent filings, F-tests indicate that the combined effect remains positive up to  $t+3$ , when it becomes insignificant. This result suggests that AI disclosures continue to provide an informative signal of future patent citations among suspected outsourcing firms.<sup>24</sup>

This mechanism plays out differently across the three disclosure channels. Table OA 2 in the online appendix reports the results when examining the conference call and earnings announcement disclosure channels. The moderation is most evident in the 10-K, with weaker results in the conference call, and the weakest results in the earnings announcement, which is the order predicted by the proprietary cost framework. The 10-K is the formal, legally scrutinized, audited disclosure in which misleading proprietary claims are most costly, and firms are most likely to officially discuss AI outsourcing activities. However, conference calls and earnings

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<sup>24</sup> In untabulated analyses, the moderating effect of outsourcing does not vary for firms with high AI employment relative to firms with lower AI employment, suggesting that high AI employment firms are not able to “rescue” the effects of outsourcing, likely due to the commoditized, rather than proprietary, nature of the AI capabilities.

announcements are softer venues where these distinctions do not apply as strongly. Nevertheless, the overall inferences remain the same. Whether firms internally develop or outsource AI, their AI disclosures provide meaningful signals about future efficiency, patenting activity, and patent citations.

Overall, AI disclosure remains a credible signal of future efficiency, AI patenting, and AI patent quality regardless of whether the firm builds AI in-house or sources it externally. However, the marginal informational content of disclosure is largely attenuated for suspected outsourcing firms. This pattern is consistent with disclosure being more informative when it conveys proprietary information about firm-specific AI capability than when it describes the use of externally provided AI inputs.

#### **4.4 Interactive Effect of AI Disclosure and AI Employment**

Our results suggest that AI disclosure provides an incremental signal of firm performance relative to a measure of the firms' underlying investments in AI, *AI\_Employee%*. We argue this is due to the disclosure signaling managers' confidence in the payoffs from underlying AI investments. If disclosure signals investment payoff, we argue that higher disclosure for a given level of investment would be more strongly associated with firm performance. To test this, we construct a composite AI disclosure score that aggregates disclosure across all three communication channels (10-K filings, conference calls, and earnings announcements). For each channel (*AI\_10K*, *AI\_CC*, *AI\_EA*), we assign firms with non-zero disclosure into terciles (Low, Mid, High), creating nine binary indicators (3 channels  $\times$  3 tercile levels). We then sum these indicators to produce a composite score ranging from 0 (no AI disclosure in any channel) to 9 (high tercile in all three channels). This composite measure captures both the breadth (how many channels) and depth (how prominently) a firm discusses AI. Firms are classified into four

disclosure groups: No Disclosure (zero composite score) and Low, Mid, and High terciles computed among firms with non-zero composite scores. For AI employment, we simplify to two groups: High (top tercile among firms with positive AI employment) versus Not-High (bottom two terciles and zero AI employment). This simplification reflects the highly right-skewed distribution of AI employment documented in Table 2. The reference group in all models is firms with no AI disclosure and not-high AI employment.<sup>25</sup> We present the results of this analysis in Table 8 and plot the coefficients of interest in Figure 5.

In Table 8, Panel A, we first find that, within the high employment group, efficiency gains scale monotonically with disclosure: *HighDis\_HighEmp* (0.07) is more than three times *MidDis\_HighEmp* (0.02), while *LowDis\_HighEmp* is indistinguishable from zero. Second, the same pattern holds within the not-high employment group, where *HighDis\_NotHighEmp* (0.04) and *MidDis\_NotHighEmp* (0.02) are both significantly positive relative to the reference group, while *LowDis\_NotHighEmp* is not. Third, comparing across employment groups at the same disclosure level, firms with high AI employment realize larger efficiency gains than firms with not-high employment, consistent with disclosure and employment operating as complements.

Panel B shows a consistent pattern for AI patents. Within each employment group, the disclosure gradient is monotone, with *HighDis\_HighEmp* (22.69) and *HighDis\_NotHighEmp* (13.84) substantially exceeding their mid- and low-disclosure counterparts, all relative to firms with no disclosure and not-high employment. Collectively, these results indicate that disclosure is the primary differentiator within employment groups. Conditional on AI employment, firms that communicate more about their AI strategy are associated with meaningfully better future outcomes, consistent with managerial disclosures conveying the firm's confidence in its returns.

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<sup>25</sup> Table OA 3 of the Online Appendix presents the sample composition of the different groups.

## 4.5 Industry Variation

Given the variation in industry-level adoption presented in Table 2, Panel C, we separately examine our results for firm-years in the top-3 adopting industries, which we label high-disclosure industries, and for all other firm-years, which we label low-disclosure industries. We present these results in Table 9.

We find that, while disclosure is positively related to efficiency and patent filings in both samples, the association between disclosure and efficiency is much stronger in *low*-disclosure industries, and there is no difference between groups in patent activity. Interestingly, we find that the associations between AI employment and efficiency, and between AI employment and patent filings, are much stronger for firms in the *high*-disclosure group. Thus, there is some evidence that in the industries where disclosure is less common, AI disclosure is a stronger signal of future performance. Whereas in industries where disclosure is common, firms' underlying investments in AI employment serve as a stronger signal of future performance.

## 4.6 Robustness Checks

We subject our main findings to a battery of robustness checks reported in Online Appendix Tables OA 4 through OA 8. The tests examine alternative AI workforce measures, alternative functional forms, persistence of outcomes, and longer horizons.

### 4.6.1 *Alternative AI Workforce Measures*

To capture potential heterogeneity in the impact of AI employees across different levels, we construct three additional AI employment measures in addition to the baseline headcount share. First, we disaggregate AI employees using Revelio's seniority classification. Revelio assigns each position a seniority score on a five-level ordinal scale, constructed using an ensemble model that incorporates job title, company, industry, and career history. We construct two measures based on

this ranking: the share of AI employees at seniority levels 3–5 (mid-to-senior), who are more likely to command higher compensation and drive substantive AI implementation, and the share at seniority levels 1–2 (junior), who hold more routine AI-related roles. Figure OA 1 plots the average number of AI employees per firm by seniority tier over the sample period. The average number of junior AI employees (levels 1–2) rises from 1.1 in 2016 to 2.5 in 2023, while the average number of mid-to-senior AI employees (levels 3–5) rises from 1.5 to 3.7 over the same period. Mid-to-senior AI hiring exhibits a steeper absolute increase, consistent with growing organizational demand for experienced AI professionals alongside the broader expansion of AI workforces.

Second, we construct two compensation-weighted AI employment measures. First, we use the estimated salaries provided by Revelio. Second, we map Revelio AI job titles to U.S. Bureau of Labor Statistics (BLS) Standard Occupational Classification (SOC) codes and merge in the corresponding median annual wages from the BLS Occupational Employment and Wage Statistics (OEWS) survey.

We create a weighted salary measure for both estimates by summing annual salaries across a firm’s AI workforce and scaling by total firm employment:

$$CompWeighted\_AI\_Share = \Sigma (AI\ employee\ wage) / total\ firm\ employees \quad (5)$$

These measures capture AI wage intensity rather than headcount, placing more weight on AI employees in higher-skill, higher-compensation occupations. Firm-years with AI employees but no Revelio estimated salary or BLS match are set to missing.

Panels A and B of Online Appendix Table OA 4 report descriptive statistics and pairwise correlations. The junior share averages 0.06 percent, the mid-to-senior share 0.08 percent, and the compensation-weighted ratio of 0.19, compared to a baseline AI share of 0.15 percent. Correlations with the baseline share are high (0.87, 0.91, and 0.84, respectively), indicating that the alternative

measures capture the same underlying variation in the AI workforce while adding granularity in seniority and wage intensity.

Panels C-E of Table OA 4 report regression results. In Panel C, all three alternative workforce variables are positively and significantly associated with AI disclosure across all three channels ( $p < 0.01$ ), with the compensation-weighted measure exhibiting the largest standardized coefficients (0.31 to 0.60), followed by the mid-to-senior share (0.32 to 0.50) and the junior share (0.28 to 0.41).

More importantly, regardless of the employment measure used, our inferences regarding the informativeness of managers' AI disclosures for future efficiency and patenting remain qualitatively unchanged. These results suggest that disclosure remains an important signal of managers' confidence in their underlying AI investments, and that these associations are not due to mismeasurement of AI employees' importance.

#### ***4.6.2 Alternative Functional Forms***

Online Appendix Table OA 5 reports estimates using alternative functional forms. Panels A and B replace the scaled measures with log-transformed variables for AI disclosure and AI employment. Log AI disclosure remains positively and significantly associated with both one-year-ahead efficiency (Panel A; coefficients of 0.02 to 0.03;  $p < 0.01$ ) and log AI patents (Panel B; coefficients of 0.25 to 0.33 with AI employment included;  $p < 0.01$ ). Panel C estimates a negative binomial model for raw AI patent counts and confirms that AI disclosure remains positively and significantly associated with AI patenting (coefficients of 0.41 to 0.50;  $p < 0.01$ ). Thus, our results are likely not driven by our choice in scaling or functional forms.

#### ***4.6.3 Control for Lagged Dependent Variables***

One concern is that our findings reflect persistent firm-level differences rather than the informational content of AI disclosure. Table OA 6 augments our main specifications with the lagged dependent variable. Efficiency and AI patenting are highly persistent (coefficients of 0.79 to 0.80 on lagged efficiency and 0.62 to 0.64 on lagged AI patent share;  $p < 0.01$ ). Even with these stringent controls, AI disclosure across all three channels remains positively and significantly associated with subsequent efficiency and AI patenting, indicating that disclosure conveys information about future outcomes incremental to past realizations.

#### ***4.6.4 Longer Horizons***

In our main tests, we measure outcomes one year ahead. A concern is that AI investments may take longer than one year to affect efficiency and innovation. Tables OA 7 and OA 8 extend the horizon to two and three years. AI disclosure remains positively and significantly associated with future efficiency (coefficients of 0.01;  $p < 0.01$ ) and future AI patenting (coefficients of 2.08 to 3.53;  $p < 0.01$ ) at both horizons, incremental to AI employment. The AI patent magnitudes at  $t+2$  and  $t+3$  are comparable to those at  $t+1$ , consistent with AI investments producing persistent innovation benefits that unfold over multiple years. Figures OA 3 and OA 4 in the online appendix present the corresponding coefficient plots from re-estimating Table 8 at these horizons; the patterns track those in Figure 5, with slightly larger cross-group differences at longer horizons. Across all specifications, AI disclosure continues to provide an incremental signal of future efficiency and innovation beyond AI employment, reinforcing the informativeness of corporate AI disclosures.

## **5. Conclusion**

AI has emerged as a transformative GPT, reshaping business practices across a broad array of industries. However, significant uncertainty remains regarding the extent and quality of firms'

AI investments, largely due to their opaque nature and the intangibility of complementary investments in AI. In this paper, we examine which firms choose to disclose AI activities, and whether these disclosures credibly inform investors about firms' underlying AI investments and future economic performance.

Using advanced textual analysis combined with detailed firm-level data on AI employment spanning 2016 to 2023, we document three key findings. First, AI disclosures are concentrated among firms operating in innovation-intensive industries, firms that invest heavily in intangible assets, and firms subject to greater external monitoring pressures from investors and analysts. This highlights that disclosures reflect strategic considerations, competitive pressures, and the broader informational environment surrounding firms.

Second, we demonstrate that higher levels of AI disclosure are positively and robustly associated with subsequent operational efficiency and AI-related innovation, as measured by AI patents. Crucially, these associations hold even after controlling for firms' actual investments in AI-related human capital. This incremental relationship suggests that disclosures are not merely reflections of observable investments; rather, they provide valuable incremental signals about managerial confidence in the eventual success and payoffs from AI initiatives.

Third, the signal provided by managers' AI disclosures varies cross-sectionally by industry-level AI intensity, by the type of AI disclosure (i.e., product-related and non-product-related), and by whether the firm appears to outsource AI development to third parties.

Collectively, our findings emphasize the vital informational role of AI disclosure in the marketplace. As AI continues to shape corporate strategies and regulatory frameworks, improving transparency around intangible AI investments will become increasingly important. Our results contribute directly to ongoing policy discussions around enhancing financial reporting standards

for innovation and intangible assets, ultimately helping investors better evaluate firms' AI capabilities. An important caveat is that our determinants analysis cannot cleanly separate AI fundamentals from disclosure incentives, and our findings are associative rather than causal. Future research could exploit exogenous shocks to disclosure incentives or employ experimental designs to identify causal mechanisms more precisely.

## References

- Acemoglu, D., D. Autor, J. Hazell, and P. Restrepo. 2022. Artificial intelligence and jobs: Evidence from online vacancies. *Journal of Labor Economics* 40 (S1): 293-340.
- Agrawal, A., J. S. Gans, and A. Goldfarb. 2019. Artificial intelligence: The ambiguous labor market impact of automating prediction. *Journal of Economic Perspectives* 33 (2): 31-50.
- Aigner, D., C. K. Lovell, and P. Schmidt. 1977. Formulation and estimation of stochastic frontier production function models. *Journal of Econometrics* 6 (1): 21-37.
- Babina, T., A. Fedyk, A. He, and J. Hodson. 2024. Artificial intelligence, firm growth, and product innovation. *Journal of Financial Economics* 151.
- Baik, B., J. Chae, S. Choi, and D. B. Farber. 2013. Changes in operational efficiency and firm performance: A frontier analysis approach. *The Accounting Review* 30 (3): 996-1026.
- Ball, R., Jayaraman, S. and Shivakumar, L., 2012. Audited financial reporting and voluntary disclosure as complements: A test of the confirmation hypothesis. *Journal of Accounting and Economics*, 53(1-2), pp.136-166.
- Bartley, K., 2024. Big data statistics: How much data is there in the world? Rivery blog, accessed May 16, 2025.
- Böke, J., D. De la Parra, J. Gallemore, and S. Glaeser. 2025. Artificial intelligence and white-collar work: Evidence from the accounting profession.
- Bresnahan, T. F., and M. Trajtenberg. 1995. General purpose technologies “Engines of growth”? *Journal of Econometrics* 65 (1): 83-108.
- Bresnahan, T., S. Greenstein, D. Brownstone, and K. Flamm. 1996. Technical progress and co-invention in computing and in the uses of computers. *Brookings Papers on Economic Activity: Microeconomics*: 1-83.
- Brynjolfsson, E., D. Rock, and C. Syverson. 2019. Artificial intelligence and the modern productivity paradox: A clash of expectations and statistics. *The Economics of Artificial Intelligence*. University of Chicago Press: 23-60.
- Callen, J. L., M. Morel, and C. Fader. 2005. Productivity measurement and the relationship between plant performance and JIT intensity. *Contemporary Accounting Research* 22 (2): 271-309.
- Campbell, J., H. Chen, J. Guan, and K. Ye. 2026. The amount and quality of R&D capitalization under International Financial Reporting Standards (IFRS). *Working paper*.
- Cao, S., W. Jiang, J. Wang, and B. Yang. 2024. From Man vs. Machine to Man + Machine: The art and AI of stock analyses. *Journal of Financial Economics* 160.
- Cao, S., W. Jiang, B. Yang, and A. L. Zhang. 2023. How to talk when a machine is listening: Corporate disclosure in the age of AI. *The Review of Financial Studies* 36 (9): 3603-3642.
- Chu, J., Y. He, K. Hui, and R. Leavy. 2024. New product announcements, innovation disclosure, and future firm performance. *Review of Accounting Studies* 30: 352-383.
- Coleman, B., K. Merkley, and J. Pacelli. 2022. Human versus machine: A comparison of robo-analyst and traditional research analyst investment recommendations. *The Accounting Review* 97 (5): 221-244.
- Cook, L. D. 2024. Artificial intelligence, big data, and the path ahead for productivity. *Federal Reserve Speech*.

- Demerjian, P., B. Lev, and S. McVay. 2012. Quantifying managerial ability: A new measure and validity tests. *Management Science* 58 (7): 1229-1248.
- DiMaggio, P. J., and W. W. Powell. 1983. The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review* 48 (2): 147–160.
- Donelson, D.C., MJ Kim, S.S. Kim, and M.A. Yip. 2025. Strategic AI Disclosures: Determinants and Consequences of Obfuscated and Overstated AI Investments. *Working Paper*.
- Dopuch, N., M. Gupta, D. A. Simunic, and M. T. Stein. 2003. Production efficiency and the pricing of audit services. *Contemporary Accounting Research* 20 (1): 47-77.
- Dye, R. A. 1985. Disclosure of nonproprietary information. *Journal of Accounting Research* 23 (1): 123–145.
- Dyer, T. A., S. Glaeser, M. H. Lang, and C. Sprecher. 2024. The effect of patent disclosure quality on innovation. *Journal of Accounting and Economics*. 77(2-3).
- FASB, 2024. Invitation to comment: Recognition of Intangibles. December 19, 2024.
- Fedyk, A., and J. Hodson. 2023. Trading on talent: Human capital and firm performance. *Review of Finance* 27 (5): 1659-1698.
- Giczy, A. V., N. A. Pairolero, and A. A. Toole. 2022. Identifying artificial intelligence (AI) invention: A novel AI patent dataset. *The Journal of Technology Transfer* 47: 476-505.
- Glaeser, S. 2018. The effects of proprietary information on corporate disclosure and transparency: Evidence from trade secrets. *Journal of Accounting and Economics* 66 (1): 163-193.
- Glaeser, S., and W. Landsman. 2021. Deterrent disclosure. *The Accounting Review* 96 (5): 291-315.
- Glaeser, S., and M. Lang. 2024. A review of the accounting literature on innovation. *Journal of Accounting and Economics* 78 (2-3): 101720.
- Glaeser, S., Lang, M. H., Omartian, J. D., & Pae, A. (2025). Opening the Black Box of Corporate R&D. *Working paper*.
- Glaeser, S., J. Michels, and R. Verrecchia. 2020. Discretionary disclosure and manager horizon: Evidence from patenting. *Review of Accounting Studies* 25 (1): 597-635.
- Hall, B. H., A. Jaffe, and M. Trajtenberg. 2005. Market value and patent citations. *RAND Journal of Economics* 36 (1): 16–38.
- Healy, P. M., and K. G. Palepu. 2001. Information asymmetry, corporate disclosure, and the capital markets: A review of the empirical disclosure literature. *Journal of Accounting and Economics* 31 (1–3): 405–440.
- Hendler, J. 2008. Avoiding another AI winter. *IEEE Intelligent Systems*, 23(2), 2- 4.
- Hodson, J. 2016. How to make your company machine learning ready. *Harvard Business Review*: 1-4.
- Huang, A. H., H. Wang, and Y. Yang. 2023. FinBERT: A large language model for extracting information from financial text. *Contemporary Accounting Research* 40 (2): 806-841.
- Jia, N., N. Li, G. Ma, and D. Xu. 2026. Corporate Responses to Generative AI: Early Evidence from Conference Calls. *Review of Accounting Studies* 31: 649-703.
- Jones, C. I., and C. Tonetti. 2020. Nonrivalry and the economics of data. *American Economic Review* 110 (9): 2819-2858.

- Kim, J., and K. Valentine. 2021. The innovation consequences of mandatory patent disclosures. *Journal of Accounting and Economics* 71(2-3).
- Kogan, L., D. Papanikolaou, A. Seru, and N. Stoffman. 2017. Technological innovation, resource allocation, and growth. *Quarterly Journal of Economics* 132 (2): 665–712.
- Koh, P.-S., and D. M. Reeb. 2015. Missing R&D. *Journal of Accounting and Economics* 60 (1): 73-94.
- Lang, M., and R. Lundholm. 1993. Cross-sectional determinants of analyst ratings of corporate disclosures. *Journal of Accounting Research* 31 (2): 246–271.
- Lev, B., and P. Zarowin. 1999. The boundaries of financial reporting and how to extend them. *Journal of Accounting Research* 37 (2): 353-385.
- Lighthill, J. 1973. *Artificial intelligence: A general survey*. British Science Research Council.
- Li, F. 2008. Annual report readability, current earnings, and earnings persistence. *Journal of Accounting and Economics* 45: 221–247.
- Loughran, T., and B. McDonald. 2011. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance* 66 (1): 35–65.
- McKinsey & Company. 2020. The state of AI in 2020: Survey. *McKinsey Digital/Quantum Black* (November).
- Meeusen, W., and J. van Den Broeck. 1977. Efficiency estimation from Cobb-Douglas production functions with composed error. *International Economic Review* 18 (2): 435-444.
- Mihet, R., and T. Philippon. 2019. The economics of big data and artificial intelligence. *Disruptive Innovation in Business and Finance in the Digital World*: 29-43.
- OECD. 2019. Artificial intelligence in society. *OECD Publishing*.
- Peters, S., and M. Winters., 2025. Investor Perspectives: Intangible Assets, Before Recognition, Improved Disclosure and Disaggregation are Needed. Chartered Financial Analyst (CFA) Institute Research & Policy Center. February 2025.
- Roland, A., and P. Shiman. 2002. Strategic Computing: DARPA and the quest for machine intelligence, 1983 to 1993. MIT Press.
- Russell, S., and Norvig, P. 2021. *Artificial intelligence: A modern approach* (4th ed.). Pearson.
- Seo, H. 2021. Peer effects in corporate disclosure decisions. *Journal of Accounting and Economics* 71 (1): 101364.
- Shapley, L. S. 1953. A value for n-person games. In *Contributions to the Theory of Games, Vol. II, Annals of Mathematics Studies*, Vol. 28: 307-317. Princeton University Press.
- Skinner, A. N. 2024. Subject matter complexity and disclosure channel richness. *The Accounting Review* 99 (1): 393-425.
- Verrecchia, R. E. 1983. Discretionary disclosure. *Journal of Accounting and Economics* 5: 179–194.

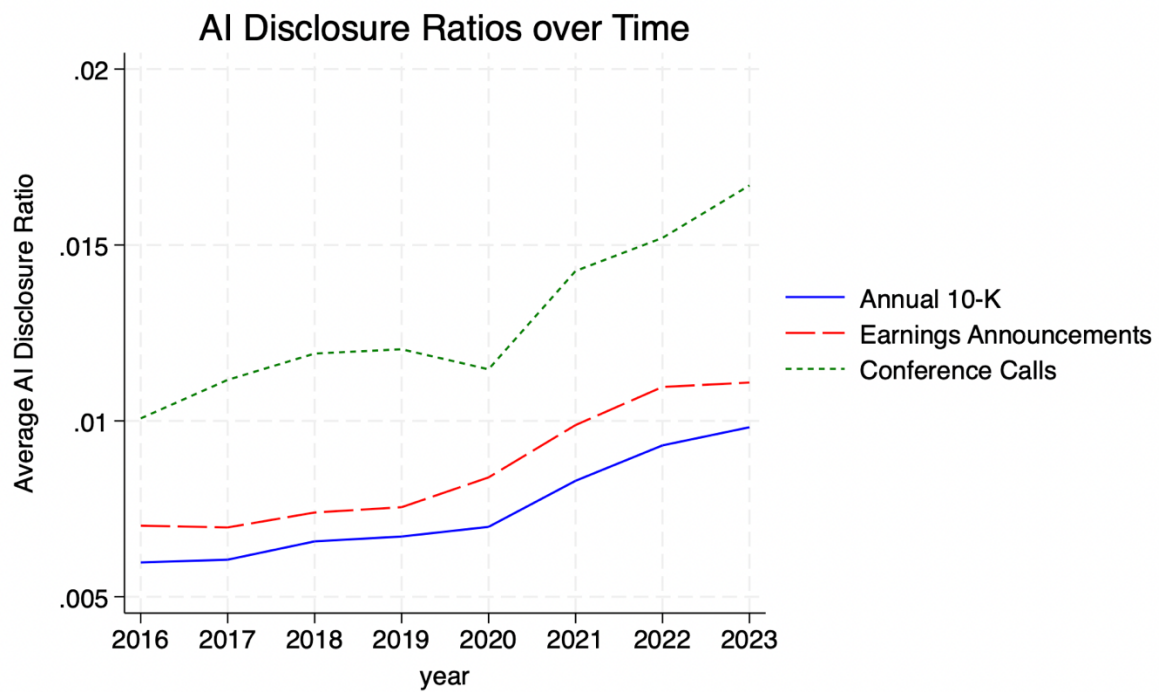
## Appendix 1. Variable Definitions

| Variable           | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Note/Source                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Age                | Firm age                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | COMPUSTAT                          |
| AI_10K             | The number of sentences classified as AI-related by the fine-tuned FinBERT model in Items 1, 1A, and 7 of the firm's annual 10-K filing, divided by the total number of sentences across these three items, expressed as a percentage.                                                                                                                                                                                                                                                                                                                   | Loughran and McDonald 10-K filings |
| AI_CC              | The number of sentences classified as AI-related by the fine-tuned FinBERT model across the firm's quarterly earnings conference call transcripts, divided by the total number of sentences across those transcripts, expressed as a percentage. The annual measure is the mean of the four quarterly ratios within the firm's fiscal year.                                                                                                                                                                                                              | Capital IQ-Transcripts             |
| AI_Dis_Ind         | The total AI disclosure score for each 3-digit SIC industry-year, calculated as the sum of firms' AI disclosures across all three communication channels (earnings announcements, conference call transcripts, and 10-K annual reports) within each industry-year group. This measure captures the overall intensity of AI-related discussions at the industry level.                                                                                                                                                                                    | Author Construction                |
| AI_EA              | for each firm-quarter, the number of sentences classified as AI-related by the fine-tuned FinBERT model (see OA 1B for details) in the firm's quarterly earnings announcement, divided by the total number of sentences in that announcement, expressed as a percentage. The annual measure is the mean of the four quarterly ratios within the firm's fiscal year.                                                                                                                                                                                      | SEC 8-K Filings                    |
| AI_Employee(raw)   | Total number of employees in AI-related roles at the firm-fiscal year level. This includes titles such as "artificial intelligence", "machine learning", "data science", "deep learning", "computer vision", "data mining", "big data", and "data analytics".                                                                                                                                                                                                                                                                                            | Revelio                            |
| AI_Employee(log)   | The natural logarithm of one plus the total number of employees in AI-related roles at the firm-fiscal year level (i.e., $\ln(1 + \text{AI\_Employee})$ ).                                                                                                                                                                                                                                                                                                                                                                                               | Revelio                            |
| AI_Employee%       | The percent share of AI employees to total employees, where total employees is measured as the total number of employees reported in Compustat.                                                                                                                                                                                                                                                                                                                                                                                                          | Revelio and Compustat              |
| AI_Patent(%)       | The percent share of firms' AI-related patent filings divided by total patent filings (in Percent). A patent is classified as an AI patent if the likelihood that the patent falls into one of the following categories exceeds 50% per the USPTO: AI Hardware, evolutionary computation, knowledge processing, machine learning, natural language processing, planning/control, speech, vision.                                                                                                                                                         | USPTO Office                       |
| Total_Patent(log)  | The natural logarithm of one plus the total number of patent filings by the firm in a given fiscal year (i.e., $\ln(1 + \text{Total\_Patent})$ ).                                                                                                                                                                                                                                                                                                                                                                                                        | USPTO Office                       |
| AI_Product_10K     | The percent share of product-related AI sentences (sentences about AI-enabled products, services, and business applications, as classified by Llama3.3-70B model) to total sentences in items 1, 1A, and 7 of the 10-K filing. Analogous measures are constructed for conference calls (AI_Product_CC) and earnings announcements (AI_Product_EA). The prompt can be found in the Online Appendix 1C.                                                                                                                                                    | Author Construction                |
| AI_Non_Product_10K | The percent share of non-product-related AI sentences (sentences about AI in general operations, risk factors, and internal processes, as classified by Llama3.3-70B model) to total sentences in items 1, 1A, and 7 of the 10-K filing. Analogous measures are constructed for conference calls (AI_Non_Product_CC) and earnings announcements (AI_Non_Product_EA). The prompt can be found in the Online Appendix 1C.                                                                                                                                  | Author Construction                |
| AI_Outsource_10K   | Binary indicator equal to one if the firm's 10-K filing contains sentences classified as AI outsourcing by the FinBERT-based outsourcing classifier, and zero otherwise. AI outsourcing sentences co-mention AI-related terms with outsourcing language (e.g., third-party, vendor, partner, cloud service, API) or named vendors (e.g., AWS, Azure, OpenAI, Palantir). Analogous measures are constructed for conference calls (AI_Outsource_CC) and earnings announcements (AI_Outsource_EA). The full dictionary can be found in the Online Appendix. | Author Construction                |

|                       |                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                  |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI_Patent_Cite(log)   | The natural logarithm of one plus the mean forward citation count of a firm's AI-related patents granted within the 365 days prior to the fiscal year-end date. Forward citation counts are obtained from the Kogan, Papanikolaou, Seru, and Stoffman (KPSS, 2017) patent database, which provides patent-level citation for U.S. patents matched to publicly traded firms. | Noah Stoffman's website<br><a href="https://github.com/KPSS2017/Technological-Innovation-Resource-Allocation-and-Growth">https://github.com/KPSS2017/Technological-Innovation-Resource-Allocation-and-Growth</a> |
| Efficiency            | Operational Efficiency using Stochastic Frontier Analysis Method (Aigner et al., 1977; Meeusen and van den Broeck, 1977)<br>Input: Cost of Goods Sold; Net Property, Plant, and Equipment (PP&E); Net R&D; Number of Employees<br>Output: Revenue                                                                                                                           | COMPUSTAT                                                                                                                                                                                                        |
| HHI_Seg               | The Herfindahl-Hirschman Index (HHI) for firm diversification. The HHI is calculated by first dividing each segment's share of the firm's total sales. These shares are then squared and summed across all segments of the firm. The resulting HHI ranges from 0 to 1, with higher values indicating greater concentration and less diversification.                        | COMPUSTAT                                                                                                                                                                                                        |
| Insti_Own             | The number of shares owned by institutions divided by the total shares outstanding.                                                                                                                                                                                                                                                                                         | Thomson Reuters Institutional Holdings (13F)                                                                                                                                                                     |
| Investment            | Capital expenditure divided by lagged tangible assets.                                                                                                                                                                                                                                                                                                                      | COMPUSTAT                                                                                                                                                                                                        |
| Leverage              | Ratio of total debt divided by total assets.                                                                                                                                                                                                                                                                                                                                | COMPUSTAT                                                                                                                                                                                                        |
| MTB                   | Market value of equity divided by book value of equity.                                                                                                                                                                                                                                                                                                                     | COMPUSTAT                                                                                                                                                                                                        |
| N_Analyst             | Number of Analysts providing EPS estimates for the firm.                                                                                                                                                                                                                                                                                                                    | I/B/E/S                                                                                                                                                                                                          |
| Profitability         | Operating income before depreciation and amortization divided by total assets at the current period.                                                                                                                                                                                                                                                                        | COMPUSTAT                                                                                                                                                                                                        |
| R&D                   | R&D expenditure divided by total assets.                                                                                                                                                                                                                                                                                                                                    | COMPUSTAT                                                                                                                                                                                                        |
| Size                  | Natural logarithm of one plus total assets at the end of the fiscal year.                                                                                                                                                                                                                                                                                                   | COMPUSTAT                                                                                                                                                                                                        |
| Tangibility           | Ratio of net property, plant, and equipment divided by total assets at the current period.                                                                                                                                                                                                                                                                                  | COMPUSTAT                                                                                                                                                                                                        |
| HighDis_HighEmp       | Indicator equal to 1 if firm has high composite AI disclosure (top tercile) and high AI employment (top tercile among firms with positive AI employment); 0 if firm has no AI disclosure and not-high AI employment.                                                                                                                                                        | Author Construction                                                                                                                                                                                              |
| HighDis_NotHighEmp    | Indicator equal to 1 if firm has high composite AI disclosure (top tercile) and not-high AI employment (bottom two terciles among firms with positive AI employment, or zero AI employment); 0 if firm has no AI disclosure and not-high AI employment.                                                                                                                     | Author Construction                                                                                                                                                                                              |
| MidDis_HighEmp        | Indicator equal to 1 if firm has middle tercile composite AI disclosure and high AI employment; 0 if firm has no AI disclosure and not-high AI employment.                                                                                                                                                                                                                  | Author Construction                                                                                                                                                                                              |
| MidDis_NotHighEmp     | Indicator equal to 1 if firm has middle tercile composite AI disclosure and not-high AI employment; 0 if firm has no AI disclosure and not-high AI employment.                                                                                                                                                                                                              | Author Construction                                                                                                                                                                                              |
| LowDis_HighEmp        | Indicator equal to 1 if firm has low tercile composite AI disclosure and high AI employment; 0 if firm has no AI disclosure and not-high AI employment.                                                                                                                                                                                                                     | Author Construction                                                                                                                                                                                              |
| LowDis_NotHighEmp     | Indicator equal to 1 if firm has low tercile composite AI disclosure and not-high AI employment; 0 if firm has no AI disclosure and not-high AI employment.                                                                                                                                                                                                                 | Author Construction                                                                                                                                                                                              |
| HighDisInd_10K(CC/EA) | Binary indicator equal to one for firms in Fama-French 12 industries with above-P75 AI disclosure incidence (share of firm-years with any AI disclosure) in the respective channel (10-K, conference call, or earnings announcement); zero otherwise. Used in Table 9.                                                                                                      | Author Construction                                                                                                                                                                                              |

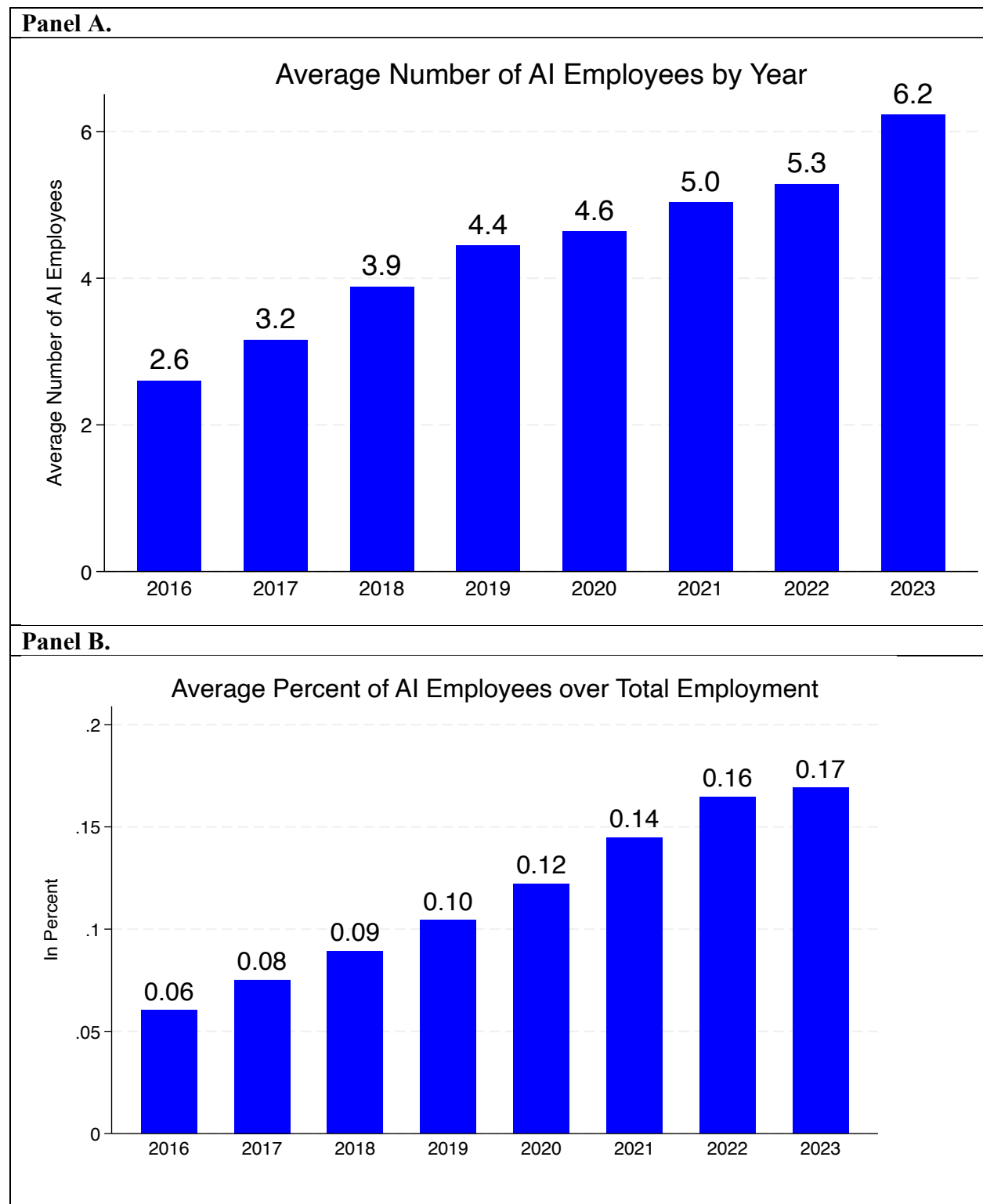
## Figure 1. AI Disclosure Over Time

Figure 1 presents the ratio of AI-related sentences to total sentences across different communication channels (10-K filings, earnings announcements, and conference call transcripts) from 2016 to 2023. All variable definitions are in Appendix 1.



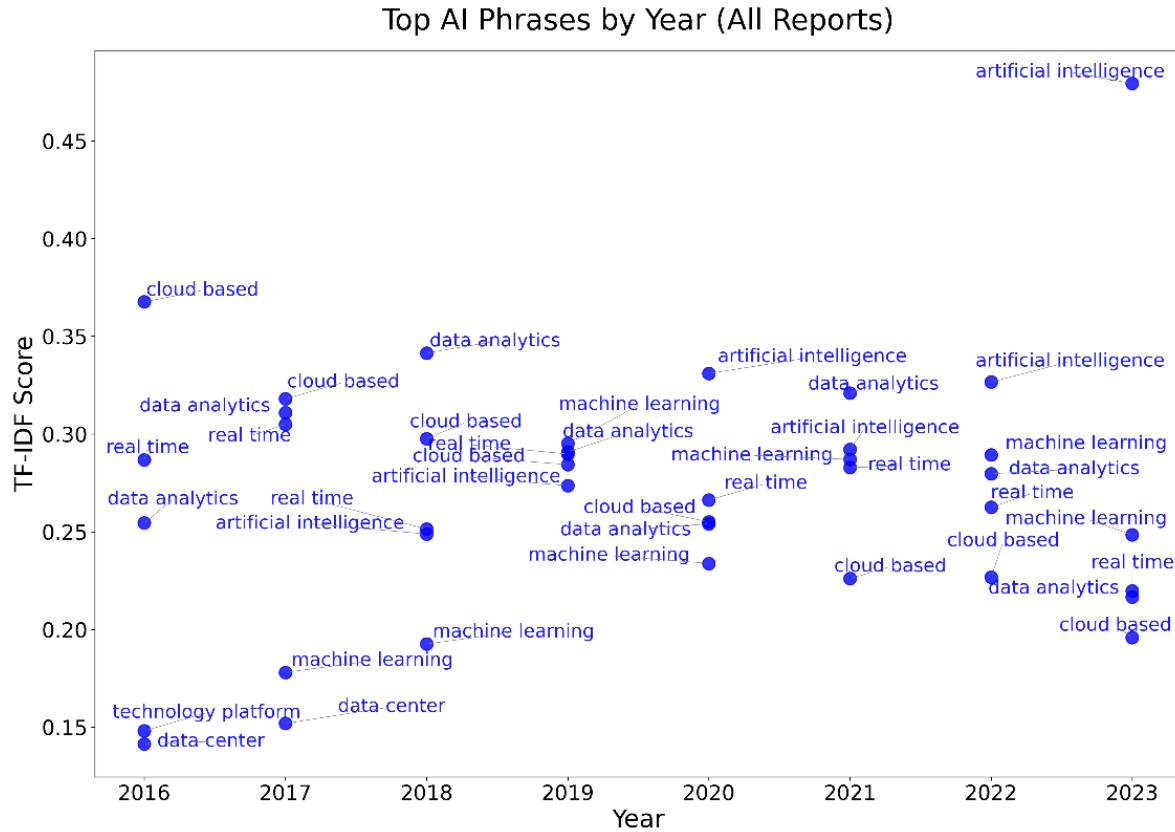
## Figure 2. AI Employees

Panel A presents the average number of AI employees over the sample period from 2016 to 2023. Panel B presents the average percentage of AI employees relative to total employment over the sample period. All variable definitions are provided in Appendix 1.



### Figure 3. Top AI Phrases by Year (All Reports)

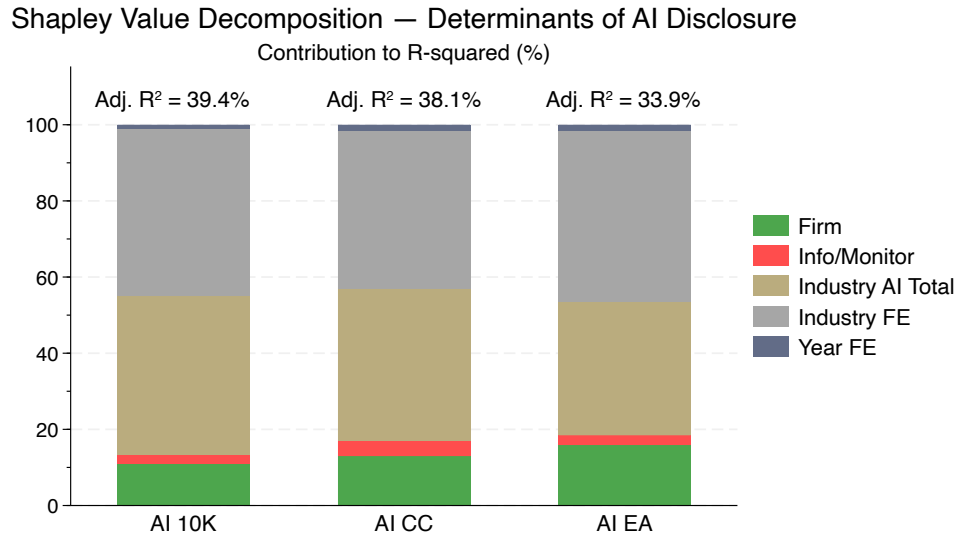
This figure presents the top AI-related phrases mentioned in all reports (10-K, conference call, and earnings announcements) from 2016 to 2023, comparing their term frequency-inverse document frequency (TF-IDF). The score is measured on a pooled basis.



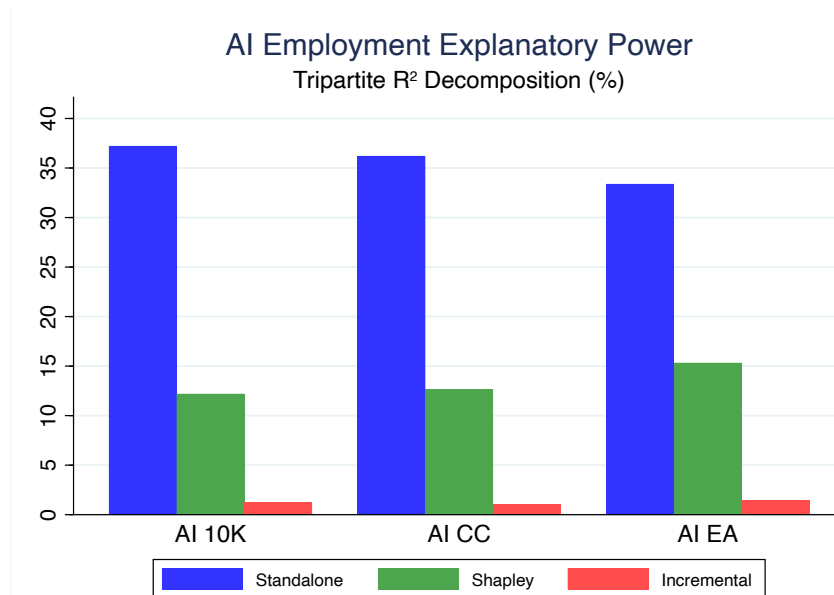
## Figure 4. Contribution of Determinants to AI Disclosure

Panel A presents Shapley value decomposition of each factor group's contribution to the adjusted R-squared across three AI disclosure measures: 10-K filings (AI\_10K), conference call transcripts (AI\_CC), and earnings announcements (AI\_EA). Panel B presents an R<sup>2</sup> decomposition reporting standalone, Shapley, and incremental contributions of AI employment across the same three measures. All variable definitions are provided in Appendix 1.

### Panel A. Contribution of Determinants to AI Disclosure R<sup>2</sup>



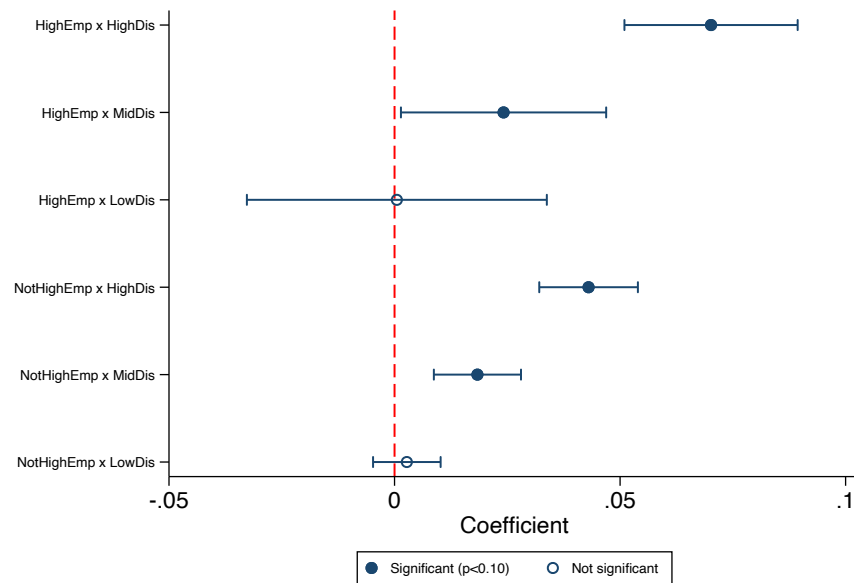
### Panel B. Contribution of AI Employment to AI Disclosure R<sup>2</sup>



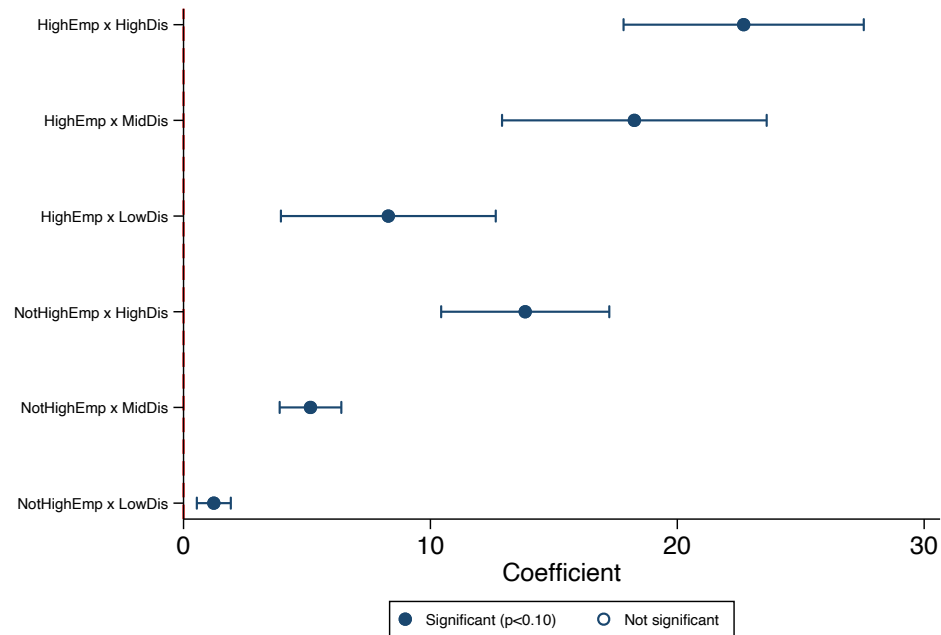
## Figure 5. Coefficient Confidence Interval Plot

This figure presents 90% confidence intervals for the coefficients of interest from Table 8. Panel A reports estimates for one-year-ahead efficiency and Panel B for one-year-ahead AI patent ratio. Each row represents a combination of AI employment level (HighEmp versus NotHighEmp) and AI disclosure intensity (High, Mid, or Low). The omitted reference group is firms with not-high AI employment and no AI disclosure. Filled markers indicate significance at the 10% level. All variable definitions are provided in Appendix 1.

### Panel A. Efficiency



### Panel B. AI Patents



### Table 1. Sample Selection

This table presents our sample selection criteria and its impact on our final sample. The variables required for our primary model include *R&D*, *Investment*, *Age*, *Size*, *Profitability*, *Tangibility*, *HHI\_Seg*, *Leverage*, *N\_Analyst*, and *Insti\_Own*. Variable definitions are in Appendix 1.

| Selection Criteria                                               | Observations | Unique Firms |
|------------------------------------------------------------------|--------------|--------------|
| Compustat Data (non-missing assets/sales or CIK; FY 2016 – 2023) | 57,186       | 9,961        |
| Available 10-K                                                   | 31,037       | 6,614        |
| Available Earnings Announcements                                 | 20,648       | 4,749        |
| Available Conference Call Transcripts                            | 16,276       | 3,667        |
| Available Employment Data from Revelio                           | 14,955       | 3,572        |
| Non-missing variables for primary models                         | 11,927       | 2,899        |

**Table 2. Summary Statistics and Correlations**

This table reports descriptive statistics, pairwise correlations, and industry-level disclosure incidence for the variables used in the main analyses. The sample spans 2016 to 2023. Panel A reports N, mean, SD, and 25th/50th/75th percentiles. Panel B reports pairwise correlations among AI\_10K, AI\_CC, AI\_EA, and AI\_Employee. Panel C reports the share of firm-years with any AI disclosure in each channel by Fama-French 12 industry. All variables are defined in Appendix 1. Continuous variables are winsorized at the 1% and 99% levels. \*\*\*, \*\*, \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

**Panel A. Summary Statistics**

| Variable            | N     | Mean  | SD    | P25  | Median | P75   |
|---------------------|-------|-------|-------|------|--------|-------|
| AI_CC               | 11927 | 1.51  | 2.61  | 0.00 | 0.50   | 1.75  |
| AI_10K              | 11927 | 1.08  | 1.97  | 0.00 | 0.29   | 1.25  |
| AI_EA               | 11927 | 1.07  | 2.04  | 0.00 | 0.00   | 1.25  |
| AI_Employee(raw)    | 11927 | 7.03  | 14.02 | 0.00 | 0.00   | 5.00  |
| AI_Employee%        | 11927 | 0.15  | 0.35  | 0.00 | 0.00   | 0.10  |
| AI_Employee(log)    | 11927 | 1.02  | 1.32  | 0.00 | 0.00   | 1.79  |
| AI_Patent(%)        | 11927 | 11.38 | 26.52 | 0.00 | 0.00   | 0.00  |
| Total_Patent(log)   | 11927 | 0.97  | 1.56  | 0.00 | 0.00   | 1.61  |
| AI_Patent_Cite(log) | 11927 | 0.26  | 0.65  | 0.00 | 0.00   | 0.00  |
| Efficiency          | 11925 | 0.69  | 0.13  | 0.67 | 0.70   | 0.76  |
| AI_Product_10K      | 11927 | 0.39  | 0.85  | 0.00 | 0.00   | 0.38  |
| AI_Product_CC       | 11927 | 0.59  | 1.12  | 0.00 | 0.00   | 0.78  |
| AI_Product_EA       | 11927 | 0.30  | 0.83  | 0.00 | 0.00   | 0.00  |
| AI_Non_Product_10K  | 11927 | 0.54  | 0.98  | 0.00 | 0.14   | 0.61  |
| AI_Non_Product_CC   | 11927 | 0.83  | 1.43  | 0.00 | 0.25   | 1.00  |
| AI_Non_Product_EA   | 11927 | 0.70  | 1.27  | 0.00 | 0.00   | 1.00  |
| AI_Outsource_10K    | 10567 | 0.18  | 0.39  | 0.00 | 0.00   | 0.00  |
| R&D                 | 11927 | 0.06  | 0.13  | 0.00 | 0.01   | 0.07  |
| Size                | 11927 | 7.25  | 1.96  | 5.95 | 7.27   | 8.58  |
| HHI_Seg             | 11927 | 0.54  | 0.29  | 0.33 | 0.45   | 0.80  |
| Tangibility         | 11927 | 0.46  | 0.44  | 0.13 | 0.30   | 0.67  |
| Leverage            | 11927 | 0.31  | 0.29  | 0.11 | 0.28   | 0.44  |
| Age                 | 11927 | 9.74  | 3.05  | 8.00 | 10.00  | 12.00 |
| Investment          | 11927 | 0.28  | 0.42  | 0.10 | 0.18   | 0.32  |
| Profitability       | 11927 | 0.03  | 0.27  | 0.02 | 0.09   | 0.14  |
| MTB                 | 11927 | 2.49  | 2.28  | 1.23 | 1.72   | 2.84  |
| N_Analyst           | 11927 | 7.32  | 7.50  | 1.00 | 5.00   | 11.00 |
| Insti_Own           | 11927 | 0.72  | 0.27  | 0.57 | 0.80   | 0.92  |
| AI_Dis_Ind          | 11927 | 4.94  | 10.61 | 0.07 | 0.40   | 2.83  |

**Panel B. AI-related Measures Pairwise Correlation Table**

| Variables        | (1)     | (2)     | (3)     | (4)  |
|------------------|---------|---------|---------|------|
| (1) AI_10K       | 1.00    |         |         |      |
| (2) AI_CC        | 0.73*** | 1.00    |         |      |
| (3) AI_EA        | 0.67*** | 0.72*** | 1.00    |      |
| (4) AI_Employee% | 0.30*** | 0.31*** | 0.20*** | 1.00 |

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

**Panel C. Industry AI Disclosure Intensity**

AI Disclosure Incidence by FF12 Industry — Share of Firm-Years with Any AI Disclosure

| Rank | Industry<br><i>10-K Annual Reports</i>   | %<br>Disclosing | Industry<br><i>Conference Calls</i>       | %<br>Disclosing | Industry<br><i>Earnings Announcements</i> | %<br>Disclosing | No.<br>Obs. |
|------|------------------------------------------|-----------------|-------------------------------------------|-----------------|-------------------------------------------|-----------------|-------------|
| 1    | Business Equipment                       | 93.5%           | Business Equipment                        | 93.2%           | Business Equipment                        | 85.4%           | 2,666       |
| 2    | Telephone and TV                         | 78.5%           | Telephone and TV                          | 85.4%           | Telephone and TV                          | 65.9%           | 205         |
| 3    | Consumer Durables                        | 71.1%           | Consumer Durables                         | 71.7%           | Consumer Durables                         | 49.6%           | 339         |
| 4    | Finance                                  | 64.1%           | Wholesale, Retail, and Services           | 70.2%           | Manufacturing                             | 43.3%           | 1,367       |
| 5    | Manufacturing                            | 61.2%           | Finance                                   | 67.6%           | Finance                                   | 40.8%           | 743         |
| 6    | Wholesale, Retail, and Services          | 59.9%           | Consumer NonDurables                      | 61.8%           | Wholesale, Retail, and Services           | 37.3%           | 1,353       |
| 7    | Healthcare, Medical Equipment, and Drugs | 56.9%           | Healthcare, Medical, Equipment, and Drugs | 59.1%           | Healthcare, Medical, Equipment, and Drugs | 37.2%           | 1,955       |
| 8    | Other                                    | 56.8%           | Manufacturing                             | 58.9%           | Other                                     | 37.0%           | 1,492       |
| 9    | Consumer NonDurables                     | 50.9%           | Other                                     | 56.9%           | Consumer NonDurables                      | 31.5%           | 568         |
| 10   | Chemicals                                | 41.5%           | Chemicals                                 | 43.6%           | Chemicals                                 | 24.2%           | 388         |
| 11   | Oil, Gas, and Coal                       | 27.8%           | Oil, Gas, and Coal                        | 33.8%           | Utilities                                 | 18.0%           | 300         |
| 12   | Utilities                                | 24.0%           | Utilities                                 | 29.3%           | Oil, Gas, and Coal                        | 16.7%           | 551         |

**Table 3. Determinants of Disclosure**

This table reports the results of regressions of determinants of AI disclosure. We report the industry fixed effect coefficients and use Industry 1: Consumer Non-Durables as the reference group captured by the intercept. Industry classification information can be found on Ken French's website. Variables employed in this table are standardized. All variables are defined in Appendix 1. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

| VARIABLES          | Pred. | (1)<br>AI 10K       | (2)<br>AI CC        | (3)<br>AI EA        | (4)<br>AI 10K       | (5)<br>AI CC        | (6)<br>AI EA        |
|--------------------|-------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| AI Employee%       | +     |                     |                     |                     | 0.33***<br>(6.87)   | 0.49***<br>(6.95)   | 0.39***<br>(7.29)   |
| R&D                | +     | 0.13***<br>(3.44)   | 0.28***<br>(5.05)   | 0.20***<br>(4.00)   | 0.09**<br>(2.55)    | 0.22***<br>(4.47)   | 0.16***<br>(3.35)   |
| Investment         | ?     | 0.03<br>(1.36)      | 0.07**<br>(2.33)    | 0.07***<br>(2.93)   | 0.02<br>(1.01)      | 0.06**<br>(2.04)    | 0.06***<br>(2.62)   |
| Age                | ?     | 0.04<br>(1.29)      | -0.11**<br>(-2.45)  | -0.08**<br>(-2.40)  | 0.07**<br>(2.19)    | -0.06<br>(-1.52)    | -0.05<br>(-1.49)    |
| Size               | ?     | -0.03<br>(-0.66)    | -0.07<br>(-1.24)    | -0.20***<br>(-3.94) | -0.02<br>(-0.47)    | -0.06<br>(-1.07)    | -0.19***<br>(-3.93) |
| Profitability      | ?     | 0.00<br>(0.08)      | 0.06<br>(1.36)      | -0.01<br>(-0.12)    | 0.02<br>(0.48)      | 0.08*<br>(1.84)     | 0.01<br>(0.24)      |
| Tangibility        | -     | -0.24***<br>(-8.13) | -0.21***<br>(-5.61) | -0.19***<br>(-6.51) | -0.22***<br>(-7.93) | -0.19***<br>(-5.31) | -0.17***<br>(-6.21) |
| HHI_Seg            | ?     | -0.06**<br>(-2.10)  | -0.12***<br>(-3.41) | -0.04<br>(-1.49)    | -0.06**<br>(-2.14)  | -0.12***<br>(-3.54) | -0.05<br>(-1.56)    |
| MTB                | ?     | 0.04<br>(1.17)      | 0.07*<br>(1.69)     | 0.07*<br>(1.87)     | 0.02<br>(0.61)      | 0.05<br>(1.07)      | 0.05<br>(1.29)      |
| Leverage           | ?     | -0.05**<br>(-2.11)  | -0.09***<br>(-3.31) | -0.06**<br>(-2.30)  | -0.03<br>(-1.50)    | -0.07***<br>(-2.64) | -0.05*<br>(-1.76)   |
| N_Analyst          | +     | 0.16***<br>(3.74)   | 0.34***<br>(5.76)   | 0.21***<br>(4.65)   | 0.15***<br>(3.50)   | 0.32***<br>(5.54)   | 0.19***<br>(4.40)   |
| Insti_Own          | +     | -0.00<br>(-0.16)    | 0.04<br>(0.93)      | -0.07*<br>(-1.86)   | -0.02<br>(-0.74)    | 0.01<br>(0.26)      | -0.09**<br>(-2.56)  |
| AI_Dis_Ind         | +     | 0.62***<br>(8.83)   | 0.78***<br>(8.25)   | 0.44***<br>(6.20)   | 0.52***<br>(7.45)   | 0.63***<br>(6.72)   | 0.32***<br>(4.56)   |
| Consumer Durables  |       | 0.50***<br>(3.91)   | 0.23<br>(1.06)      | 0.42***<br>(2.88)   | 0.49***<br>(4.19)   | 0.22<br>(1.04)      | 0.41***<br>(3.01)   |
| Manufacturing      |       | 0.35***<br>(4.46)   | -0.00<br>(-0.02)    | 0.33***<br>(3.64)   | 0.36***<br>(4.77)   | 0.00<br>(0.03)      | 0.34***<br>(3.86)   |
| Oil, Gas, and Coal |       | 0.37***<br>(4.31)   | -0.01<br>(-0.06)    | 0.34***<br>(3.39)   | 0.30***<br>(3.64)   | -0.11<br>(-0.89)    | 0.26***<br>(2.61)   |
| Chemicals          |       | -0.03<br>(-0.51)    | -0.35***<br>(-3.00) | 0.04<br>(0.40)      | -0.03<br>(-0.55)    | -0.35***<br>(-3.14) | 0.04<br>(0.43)      |
| Business Equipment |       | 1.45***<br>(10.48)  | 1.57***<br>(7.47)   | 1.52***<br>(9.12)   | 1.44***<br>(10.92)  | 1.57***<br>(7.73)   | 1.52***<br>(9.37)   |

|                                                |                     |                     |                     |                     |                     |                     |
|------------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Telephone and TV                               | 0.79***<br>(5.66)   | 0.82***<br>(4.28)   | 1.04***<br>(5.27)   | 0.76***<br>(5.44)   | 0.77***<br>(4.18)   | 1.01***<br>(4.97)   |
| Utilities                                      | 0.32***<br>(3.17)   | 0.28*<br>(1.75)     | 0.63***<br>(4.94)   | 0.24**<br>(2.51)    | 0.16<br>(1.09)      | 0.54***<br>(4.41)   |
| Wholesale, Retail,<br>and Services             | 0.26***<br>(3.98)   | 0.32***<br>(2.72)   | 0.26***<br>(2.71)   | 0.26***<br>(4.12)   | 0.32***<br>(2.78)   | 0.25***<br>(2.71)   |
| Healthcare, Medical<br>Equipment, and<br>Drugs | -0.34***<br>(-4.70) | -0.64***<br>(-5.06) | -0.46***<br>(-4.67) | -0.29***<br>(-4.29) | -0.57***<br>(-4.77) | -0.40***<br>(-4.26) |
| Finance                                        | 0.44***<br>(3.89)   | 0.60***<br>(3.24)   | 0.59***<br>(3.53)   | 0.32***<br>(3.06)   | 0.42**<br>(2.43)    | 0.44***<br>(2.83)   |
| Other                                          | 0.57***<br>(5.89)   | 0.45***<br>(3.01)   | 0.45***<br>(4.42)   | 0.51***<br>(5.64)   | 0.37***<br>(2.61)   | 0.38***<br>(3.96)   |
| Constant                                       | 0.59***<br>(10.60)  | 1.12***<br>(10.68)  | 0.58***<br>(8.07)   | 0.61***<br>(11.21)  | 1.14***<br>(11.20)  | 0.60***<br>(8.31)   |
| Observations                                   | 11,927              | 11,927              | 11,927              | 11,927              | 11,927              | 11,927              |
| Adj. R-squared                                 | 0.39                | 0.38                | 0.34                | 0.42                | 0.41                | 0.37                |
| Year FE                                        | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Industry FE                                    | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Cluster                                        | Firm                | Firm                | Firm                | Firm                | Firm                | Firm                |

**Table 4. Efficiency**

This table reports regression results examining the relationship between future firm efficiency and AI disclosures. Specifically, we examine AI disclosure in the 10-K (AI\_10K), earnings conference calls (AI\_CC), and earnings announcements (AI\_EA). All variables are defined in Appendix 1. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

| VARIABLES      | (1)                         | (2)                | (3)                | (4)                | (5)                | (6)                |
|----------------|-----------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                | Efficiency <sub>(t+1)</sub> |                    |                    |                    |                    |                    |
| AI_10K         | 0.01***<br>(4.42)           |                    |                    | 0.01***<br>(4.12)  |                    |                    |
| AI_CC          |                             | 0.01***<br>(8.25)  |                    |                    | 0.01***<br>(7.40)  |                    |
| AI_EA          |                             |                    | 0.01***<br>(7.22)  |                    |                    | 0.01***<br>(6.42)  |
| AI_Employee%   |                             |                    |                    | 0.02***<br>(3.03)  | 0.02***<br>(2.66)  | 0.02***<br>(2.81)  |
| Constant       | 0.79***<br>(46.59)          | 0.78***<br>(46.92) | 0.78***<br>(46.81) | 0.79***<br>(46.81) | 0.78***<br>(47.01) | 0.78***<br>(46.93) |
| Observations   | 9,516                       | 9,516              | 9,516              | 9,516              | 9,516              | 9,516              |
| Adj. R-squared | 0.26                        | 0.26               | 0.26               | 0.26               | 0.26               | 0.26               |
| Controls       | Yes                         | Yes                | Yes                | Yes                | Yes                | Yes                |
| Year FE        | Yes                         | Yes                | Yes                | Yes                | Yes                | Yes                |
| Industry FE    | Yes                         | Yes                | Yes                | Yes                | Yes                | Yes                |
| Cluster        | Firm                        | Firm               | Firm               | Firm               | Firm               | Firm               |

**Table 5. AI Patent**

This table reports regression results examining the relationship between AI disclosure and future patenting activity. AI disclosure is measured across three channels: 10-K filings (AI\_10K), conference call transcripts (AI\_CC), and earnings announcements (AI\_EA). All variable definitions are provided in Appendix 1. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses.\*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

| VARIABLES      | (1)                           | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|----------------|-------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                | AI Patent(%) <sub>(t+1)</sub> |                      |                      |                      |                      |                      |
| AI_10K         | 3.35***<br>(6.17)             |                      |                      | 2.76***<br>(5.89)    |                      |                      |
| AI_CC          |                               | 2.78***<br>(10.57)   |                      |                      | 2.27***<br>(8.55)    |                      |
| AI_EA          |                               |                      | 2.72***<br>(8.56)    |                      |                      | 2.05***<br>(6.26)    |
| AI_Employee%   |                               |                      |                      | 15.07***<br>(7.78)   | 14.38***<br>(7.47)   | 15.30***<br>(7.71)   |
| Constant       | -13.66***<br>(-4.07)          | -15.60***<br>(-4.83) | -15.97***<br>(-4.78) | -15.38***<br>(-4.75) | -16.85***<br>(-5.33) | -17.04***<br>(-5.24) |
| Observations   | 9,646                         | 9,646                | 9,646                | 9,646                | 9,646                | 9,646                |
| Adj. R-squared | 0.26                          | 0.27                 | 0.25                 | 0.29                 | 0.30                 | 0.28                 |
| Controls       | Yes                           | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Year FE        | Yes                           | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Industry FE    | Yes                           | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Cluster        | Firm                          | Firm                 | Firm                 | Firm                 | Firm                 | Firm                 |

**Table 6. AI Product vs Non-AI Product Disclosure**

This table decomposes AI disclosure into product-related and non-product-related components using the Llama3.3-70B model. Product-related AI disclosure captures sentences about AI-enabled products, services, and business applications; non-product disclosure captures sentences about AI in general operations, risk factors, and internal processes. Panel A reports determinants tests for product-related AI disclosure. Panels B and C report efficiency and AI patent ratio regressions, respectively, comparing the informativeness of product versus non-product AI disclosure, with formal tests of cross-equation coefficient equality (suest). All variables are defined in Appendix 1. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

**Panel A. Determinants Test**

| VARIABLES      | (1)<br>AI Product 10K | (2)<br>AI nonProduct 10K | (3)<br>AI Product 10K | (4)<br>AI nonProduct 10K |
|----------------|-----------------------|--------------------------|-----------------------|--------------------------|
| AI Employee%   |                       |                          | 0.18***<br>(7.36)     | 0.09***<br>(3.96)        |
| R&D            | 0.08***<br>(3.92)     | 0.04**<br>(2.17)         | 0.06***<br>(3.17)     | 0.03*<br>(1.66)          |
| Investment     | 0.01<br>(1.16)        | 0.01<br>(1.16)           | 0.01<br>(0.76)        | 0.01<br>(0.89)           |
| Age            | 0.01<br>(0.51)        | 0.05***<br>(2.79)        | 0.02<br>(1.58)        | 0.06***<br>(3.18)        |
| Size           | -0.06***<br>(-3.10)   | 0.02<br>(0.62)           | -0.06***<br>(-3.02)   | 0.02<br>(0.73)           |
| Profitability  | 0.01<br>(0.54)        | 0.02<br>(0.95)           | 0.02<br>(1.01)        | 0.02<br>(1.13)           |
| Tangibility    | -0.08***<br>(-7.57)   | -0.13***<br>(-7.07)      | -0.08***<br>(-7.38)   | -0.12***<br>(-6.96)      |
| HHI_Seg        | -0.03***<br>(-2.87)   | -0.02<br>(-1.33)         | -0.03***<br>(-2.94)   | -0.02<br>(-1.34)         |
| MTB            | 0.03*<br>(1.70)       | 0.01<br>(0.55)           | 0.02<br>(1.09)        | 0.00<br>(0.25)           |
| Leverage       | -0.02**<br>(-2.21)    | -0.01<br>(-1.04)         | -0.01<br>(-1.50)      | -0.01<br>(-0.73)         |
| N_Analyst      | 0.08***<br>(4.34)     | 0.06***<br>(2.80)        | 0.07***<br>(4.10)     | 0.06***<br>(2.66)        |
| Insti_Own      | -0.00<br>(-0.26)      | 0.00<br>(0.08)           | -0.01<br>(-1.00)      | -0.00<br>(-0.20)         |
| AI_Dis_Ind     | 0.21***<br>(6.34)     | 0.29***<br>(7.83)        | 0.16***<br>(4.70)     | 0.27***<br>(7.03)        |
| Constant       | 0.17***<br>(7.78)     | 0.29***<br>(9.07)        | 0.18***<br>(8.47)     | 0.30***<br>(9.29)        |
| Observations   | 11,927                | 11,927                   | 11,927                | 11,927                   |
| Adj. R-squared | 0.35                  | 0.35                     | 0.39                  | 0.36                     |
| Year FE        | Yes                   | Yes                      | Yes                   | Yes                      |
| Industry FE    | Yes                   | Yes                      | Yes                   | Yes                      |
| Cluster        | Firm                  | Firm                     | Firm                  | Firm                     |

### Panel B. Efficiency

| VARIABLES            | (1)                | (2)                | (3)                | (4)                |
|----------------------|--------------------|--------------------|--------------------|--------------------|
|                      | Efficiency(t+1)    |                    |                    |                    |
| AI_Product_10K       | 0.02***<br>(6.33)  |                    | 0.02***<br>(5.80)  |                    |
| AI_Non_Product_10K   |                    | 0.01***<br>(2.93)  |                    | 0.00**<br>(2.44)   |
| AI_Employee%         |                    |                    | 0.02**<br>(2.54)   | 0.03***<br>(3.67)  |
| Constant             | 0.78***<br>(46.65) | 0.79***<br>(46.51) | 0.78***<br>(46.83) | 0.79***<br>(46.74) |
| Observations         | 9,516              | 9,516              | 9,516              | 9,516              |
| Adj. R-squared       | 0.26               | 0.25               | 0.26               | 0.26               |
| Controls             | Yes                | Yes                | Yes                | Yes                |
| Year FE              | Yes                | Yes                | Yes                | Yes                |
| Industry FE          | Yes                | Yes                | Yes                | Yes                |
| Cluster              | Firm               | Firm               | Firm               | Firm               |
| Prod=NonProd Chi2    |                    | 26.79              |                    | 23.37              |
| Prod=NonProd p-value |                    | 0.000              |                    | 0.000              |

### Panel C. AI Patent

| VARIABLES            | (1)                  | (2)                  | (3)                  | (4)                  |
|----------------------|----------------------|----------------------|----------------------|----------------------|
|                      | AI_Patent(%) (t+1)   |                      |                      |                      |
| AI_Product_10K       | 8.54***<br>(8.75)    |                      | 7.01***<br>(7.77)    |                      |
| AI_Non_Product_10K   |                      | 4.05***<br>(4.99)    |                      | 3.38***<br>(4.41)    |
| AI_Employee%         |                      |                      | 13.96***<br>(7.10)   | 16.85***<br>(8.85)   |
| Constant             | -14.92***<br>(-4.54) | -12.16***<br>(-3.55) | -16.27***<br>(-5.09) | -14.38***<br>(-4.39) |
| Observations         | 9,646                | 9,646                | 9,646                | 9,646                |
| Adj. R-squared       | 0.28                 | 0.24                 | 0.30                 | 0.28                 |
| Controls             | Yes                  | Yes                  | Yes                  | Yes                  |
| Year FE              | Yes                  | Yes                  | Yes                  | Yes                  |
| Industry FE          | Yes                  | Yes                  | Yes                  | Yes                  |
| Cluster              | Firm                 | Firm                 | Firm                 | Firm                 |
| Prod=NonProd Chi2    |                      | 20.96                |                      | 15.90                |
| Prod=NonProd p-value |                      | 0.000                |                      | 0.000                |

**Table 7. AI Outsourcing and the Informativeness of AI Disclosure**

This table examines the role of AI outsourcing in AI disclosures, Panel A, and the association between AI disclosure and future performance, Panel B. We measure AI outsourcing using a textual indicator constructed via a two-stage rule-based and FinBERT classification approach applied to 10-K filings. Specifically, we re-estimate Tables 4 and 5 to see if outsourcing moderates the associations between AI disclosure and future efficiency, AI patent ratio, and log AI patent citations over multiple time horizons. All variable definitions are provided in Appendix 1. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

**Panel A. Determinants**

| VARIABLES      | (1)<br>AI 10K       | (2)<br>AI CC       | (3)<br>AI EA        |
|----------------|---------------------|--------------------|---------------------|
| AI Employee%   | 0.24***<br>(3.24)   | 0.63***<br>(5.08)  | 0.38***<br>(4.04)   |
| R&D            | 0.10<br>(0.78)      | 0.34<br>(1.33)     | 0.37*<br>(1.96)     |
| Investment     | 0.02<br>(0.33)      | -0.02<br>(-0.23)   | 0.09<br>(0.98)      |
| Age            | 0.30***<br>(3.34)   | -0.13<br>(-0.80)   | 0.12<br>(1.00)      |
| Size           | -0.05<br>(-0.28)    | 0.13<br>(0.49)     | -0.24<br>(-0.95)    |
| Profitability  | -0.04<br>(-0.40)    | -0.11<br>(-0.38)   | 0.08<br>(0.44)      |
| Tangibility    | -0.67***<br>(-3.70) | -0.08<br>(-0.34)   | -0.60***<br>(-3.26) |
| HHI_Seg        | 0.03<br>(0.21)      | -0.19<br>(-1.01)   | 0.05<br>(0.39)      |
| MTB            | -0.03<br>(-0.40)    | 0.06<br>(0.58)     | 0.03<br>(0.29)      |
| Leverage       | -0.01<br>(-0.10)    | -0.35**<br>(-2.15) | 0.00<br>(0.00)      |
| N_Analyst      | 0.18<br>(1.33)      | 0.62***<br>(3.26)  | 0.49**<br>(2.47)    |
| Insti_Own      | -0.13<br>(-0.96)    | -0.04<br>(-0.23)   | -0.43**<br>(-2.36)  |
| AI_Dis_Ind     | 0.46***<br>(4.46)   | 0.41**<br>(2.36)   | 0.11<br>(0.77)      |
| Constant       | 1.08***<br>(3.54)   | 2.00***<br>(3.37)  | 2.34**<br>(2.35)    |
| Observations   | 1,935               | 1,218              | 1,217               |
| Adj. R-squared | 0.25                | 0.29               | 0.23                |
| Year FE        | Yes                 | Yes                | Yes                 |
| Industry FE    | Yes                 | Yes                | Yes                 |
| Cluster        | Firm                | Firm               | Firm                |

**Panel B: Outcomes**

| VARIABLES                      | (1)                 | (2)                | (3)                | (4)                  | (5)                  | (6)                  | (7)                        | (8)                 | (9)                 |
|--------------------------------|---------------------|--------------------|--------------------|----------------------|----------------------|----------------------|----------------------------|---------------------|---------------------|
|                                | <b>Efficiency</b>   |                    |                    | <b>AI Patent (%)</b> |                      |                      | <b>AI Patent Cite(Log)</b> |                     |                     |
|                                | t+1                 | t+2                | t+3                | t+1                  | t+2                  | t+3                  | t+1                        | t+2                 | t+3                 |
| AI_10K                         | 0.01***<br>(3.46)   | 0.01***<br>(2.89)  | 0.01**<br>(2.43)   | 3.21***<br>(4.86)    | 3.10***<br>(4.31)    | 2.72***<br>(3.54)    | 0.08***<br>(4.62)          | 0.07***<br>(5.19)   | 0.05***<br>(3.87)   |
| AI_Outsource_10K               | 0.02***<br>(2.58)   | 0.01**<br>(2.06)   | 0.02**<br>(2.33)   | 6.27***<br>(3.21)    | 7.54***<br>(3.10)    | 6.57**<br>(2.36)     | 0.12***<br>(2.82)          | 0.13***<br>(2.63)   | 0.11**<br>(2.30)    |
| AI_10K x<br>AI_Outsource_10K   | -0.01***<br>(-2.65) | -0.00*<br>(-1.66)  | -0.00<br>(-1.47)   | -1.10*<br>(-1.77)    | -1.14<br>(-1.52)     | -0.72<br>(-0.92)     | -0.05***<br>(-2.98)        | -0.05***<br>(-3.18) | -0.03**<br>(-2.30)  |
| AI_Employee(%)                 | 0.02***<br>(2.60)   | 0.03***<br>(3.04)  | 0.03***<br>(3.13)  | 13.93***<br>(6.84)   | 16.77***<br>(7.17)   | 18.53***<br>(6.65)   | 0.16***<br>(4.16)          | 0.18***<br>(4.32)   | 0.16***<br>(4.04)   |
| Constant                       | 0.78***<br>(42.81)  | 0.75***<br>(35.32) | 0.74***<br>(29.45) | -13.88***<br>(-3.98) | -14.35***<br>(-3.74) | -15.01***<br>(-3.59) | -0.39***<br>(-5.00)        | -0.37***<br>(-4.46) | -0.33***<br>(-3.94) |
| Observations                   | 8,478               | 6,609              | 5,123              | 8,598                | 6,704                | 5,206                | 8,598                      | 6,704               | 5,206               |
| Adj. R-squared                 | 0.26                | 0.26               | 0.24               | 0.31                 | 0.33                 | 0.33                 | 0.22                       | 0.21                | 0.20                |
| Controls                       | Yes                 | Yes                | Yes                | Yes                  | Yes                  | Yes                  | Yes                        | Yes                 | Yes                 |
| Year FE                        | Yes                 | Yes                | Yes                | Yes                  | Yes                  | Yes                  | Yes                        | Yes                 | Yes                 |
| Industry FE                    | Yes                 | Yes                | Yes                | Yes                  | Yes                  | Yes                  | Yes                        | Yes                 | Yes                 |
| Cluster                        | Firm                | Firm               | Firm               | Firm                 | Firm                 | Firm                 | Firm                       | Firm                | Firm                |
| F-Test ( $\beta_1 + \beta_3$ ) | 5.46**              | 5.81**             | 4.30**             | 15.81***             | 11.22***             | 11.41***             | 7.73**                     | 2.99*               | 1.84                |

**Table 8. Interaction between AI Disclosure and AI Employment**

This table examines the performance implications from Tables 4 and 5 for different combinations of disclosure and employment levels. AI disclosure is measured using a composite score that aggregates disclosure across 10-K filings, conference calls, and earnings announcements, with firms classified into four categories: None, Low (bottom tercile based on non-zero composite scores), Mid (middle tercile), and High (top tercile). AI employment is classified into two groups: High (top tercile among firms with positive AI employment) and Not-High (bottom two terciles and zero AI employment). Panel A reports results for efficiency and Panel B for AI patent ratio. The reference group in all models is firms with no AI disclosure and not-high AI employment. Control variable output suppressed for brevity. All models use firms with no disclosure and no AI employment as the comparison group to facilitate comparisons across models. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, and \* denote statistical significance at the 1%, 5%, and 10% levels.

**Panel A. Efficiency (t+1)**

|                        | (1)                | (2)                | (3)                | (4)                | (5)                | (6)                |
|------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| VARIABLES              | Efficiency (t+1)   |                    |                    |                    |                    |                    |
| HighDis_HighE<br>mp    | 0.07***<br>(6.01)  |                    |                    |                    |                    |                    |
| MidDis_HighE<br>mp     |                    | 0.02*<br>(1.75)    |                    |                    |                    |                    |
| LowDis_HighE<br>mp     |                    |                    | 0.00<br>(0.03)     |                    |                    |                    |
| HighDis_NotHi<br>ghEmp |                    |                    |                    | 0.04***<br>(6.47)  |                    |                    |
| MidDis_NotHig<br>hEmp  |                    |                    |                    |                    | 0.02***<br>(3.13)  |                    |
| LowDis_NotHi<br>ghEmp  |                    |                    |                    |                    |                    | 0.00<br>(0.60)     |
| Constant               | 0.73***<br>(24.97) | 0.73***<br>(21.67) | 0.73***<br>(20.50) | 0.78***<br>(29.49) | 0.75***<br>(32.38) | 0.74***<br>(30.16) |
| Observations           | 2,805              | 2,123              | 2,029              | 3,206              | 3,795              | 4,688              |
| Adj. R-squared         | 0.34               | 0.28               | 0.28               | 0.29               | 0.27               | 0.27               |
| Controls               | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Year FE                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Industry FE            | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Cluster                | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               |

**Panel B. AI Patent (t+1)**

|                 | (1)                | (2) | (3) | (4) | (5) | (6) |
|-----------------|--------------------|-----|-----|-----|-----|-----|
|                 | AI_Patent(%) (t+1) |     |     |     |     |     |
| HighDis_HighEmp | 22.69***<br>(7.67) |     |     |     |     |     |

|                    |          |          |         |          |           |          |
|--------------------|----------|----------|---------|----------|-----------|----------|
| MidDis_HighEmp     |          | 18.26*** |         |          |           |          |
|                    |          | (5.60)   |         |          |           |          |
| LowDis_HighEmp     |          |          | 8.30*** |          |           |          |
|                    |          |          | (3.14)  |          |           |          |
| HighDis_NotHighEmp |          |          |         | 13.84*** |           |          |
|                    |          |          |         | (6.69)   |           |          |
| MidDis_NotHighEmp  |          |          |         |          | 5.14***   |          |
|                    |          |          |         |          | (6.77)    |          |
| LowDis_NotHighEmp  |          |          |         |          |           | 1.23***  |
|                    |          |          |         |          |           | (2.93)   |
| Constant           | -12.30** | -10.45** | -8.57** | -12.30** | -15.27*** | -9.99*** |
|                    | (-2.19)  | (-2.30)  | (-2.29) | (-2.32)  | (-4.66)   | (-3.88)  |
| Observations       | 2,822    | 2,134    | 2,038   | 3,239    | 3,835     | 4,738    |
| Adj. R-squared     | 0.45     | 0.23     | 0.10    | 0.21     | 0.11      | 0.06     |
| Controls           | Yes      | Yes      | Yes     | Yes      | Yes       | Yes      |
| Year FE            | Yes      | Yes      | Yes     | Yes      | Yes       | Yes      |
| Industry FE        | Yes      | Yes      | Yes     | Yes      | Yes       | Yes      |
| Cluster            | Firm     | Firm     | Firm    | Firm     | Firm      | Firm     |

**Table 9. Industry Heterogeneity Effects**

This table examines whether the relationship between AI disclosure and firm outcomes varies across industries with different disclosure intensity. Panels A and B focus on real effect efficiency and AI patent share regressions, respectively, comparing the informativeness of AI disclosure in high- versus low-disclosure industries, with formal tests of cross-equation coefficient equality (suest). All variables are defined in Appendix 1. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

**Panel A. Efficiency (t+1)**

| VARIABLES          | (1)                                  | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
|--------------------|--------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    | Dependent Variable: Efficiency (t+1) |                     |                     |                     |                     |                     |
|                    | HighDisInd<br>10K                    | LowDisInd<br>10K    | HighDisInd<br>CC    | LowDisInd<br>CC     | HighDisInd<br>EA    | LowDisInd<br>EA     |
| AI_10K             | 0.003***<br>(2.60)                   | 0.011***<br>(5.02)  |                     |                     |                     |                     |
| AI_CC              |                                      |                     | 0.003***<br>(3.79)  | 0.008***<br>(5.01)  |                     |                     |
| AI_EA              |                                      |                     |                     |                     | 0.002***<br>(2.65)  | 0.008***<br>(4.11)  |
| AI_Employee<br>%   | 0.018***<br>(2.66)                   | 0.010<br>(0.83)     | 0.017**<br>(2.53)   | 0.009<br>(0.72)     | 0.019***<br>(2.71)  | 0.010<br>(0.84)     |
| Constant           | 0.728***<br>(29.68)                  | 0.765***<br>(38.34) | 0.725***<br>(29.89) | 0.763***<br>(38.39) | 0.723***<br>(28.89) | 0.764***<br>(38.50) |
| Observations       | 2,506                                | 7,010               | 2,506               | 7,010               | 2,506               | 7,010               |
| Adj. R-squared     | 0.42                                 | 0.24                | 0.42                | 0.24                | 0.42                | 0.24                |
| Controls           | Yes                                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Year FE            | Yes                                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Industry FE        | Yes                                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Cluster            | Firm                                 | Firm                | Firm                | Firm                | Firm                | Firm                |
| Chi2 AI_10K<br>H=L | 12.60                                |                     |                     |                     |                     |                     |
| Chi2 AI_CC<br>H=L  | 10.02                                |                     |                     |                     |                     |                     |
| Chi2 AI_EA<br>H=L  | 7.47                                 |                     |                     |                     |                     |                     |

**Panel B. AI Patent (t+1)**

|                    | (1)                                     | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|--------------------|-----------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                    | Dependent Variable: AI_Patent (t+1) (%) |                      |                      |                      |                      |                      |
| VARIABLES          | HighDisInd<br>10K                       | LowDisInd_<br>10K    | HighDisInd<br>CC     | LowDisInd<br>CC      | HighDisInd<br>EA     | LowDisInd<br>EA      |
| AI_10K             | 2.40***<br>(4.52)                       | 2.67***<br>(3.64)    |                      |                      |                      |                      |
| AI_CC              |                                         |                      | 2.08***<br>(6.37)    | 1.92***<br>(4.05)    |                      |                      |
| AI_EA              |                                         |                      |                      |                      | 1.95***<br>(4.67)    | 1.70***<br>(3.20)    |
| AI_Employee<br>%   | 19.49***<br>(6.83)                      | 10.19***<br>(4.29)   | 18.95***<br>(6.73)   | 9.95***<br>(4.07)    | 19.95***<br>(6.85)   | 10.65***<br>(4.25)   |
| Constant           | -28.65***<br>(-3.37)                    | -10.63***<br>(-3.29) | -31.01***<br>(-3.79) | -11.04***<br>(-3.43) | -32.84***<br>(-3.90) | -10.69***<br>(-3.27) |
| Observations       | 2,558                                   | 7,088                | 2,558                | 7,088                | 2,558                | 7,088                |
| Adj. R-squared     | 0.32                                    | 0.12                 | 0.32                 | 0.12                 | 0.31                 | 0.11                 |
| Controls           | Yes                                     | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Year FE            | Yes                                     | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Industry FE        | Yes                                     | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Cluster            | Firm                                    | Firm                 | Firm                 | Firm                 | Firm                 | Firm                 |
| Chi2 AI_10K<br>H=L |                                         | 0.09                 |                      |                      |                      |                      |
| Chi2 AI_CC<br>H=L  |                                         |                      |                      | 0.08                 |                      |                      |
| Chi2 AI_EA<br>H=L  |                                         |                      |                      |                      |                      | 0.14                 |

# Online Appendix for “Signals or smoke? The Determinants and Informativeness of Corporate Artificial Intelligence (AI) Disclosures”

## Table of Contents

- **Appendix OA 1 – Variable Construction**
  - **1A** – AI Employment
  - **1B** – AI Disclosure
  - **1C** - Product vs. Non-Product Disclosure
  - **1D** – Outsourcing
- **Appendix OA 2 – Classification Examples**
  - **2A** – Examples of AI Disclosure by Disclosure Format
  - **2B** – AI Patents
  - **2C** – Outsourcing
- **Figure OA 1** - Employment by Seniority and Salary
- **Figure OA 2** – Shapley Decomposition for Outsourcing Firms
- **Figure OA 3** - Confidence plots t+2
- **Figure OA 4** - Confidence plots t+3
- **Table OA 1** - Alternative Determinants Framework
- **Table OA 2** – Outsourcing using Conference Calls and Earnings Announcements
- **Table OA 3** - Sample Decomposition of Disclosure/Employment Groups
- **Table OA 4** - Robustness Checks: Alternative AI Worker Measures
- **Table OA 5** - Efficiency and AI Patent in Alternative Functional Forms
- **Table OA 6** - Control for Lagged Dependent Variables
- **Table OA 7** - Efficiency over Longer Horizons
- **Table OA 8** - AI Patent over Longer Horizons

## **Appendix OA 1 – Variable Construction**

### **Appendix OA 1A. AI Employment**

We measure AI employees using the comprehensive employment data from Revelio Labs, which sources detailed employment records from professional networking sites (e.g., LinkedIn), job postings, and government records.<sup>26</sup> This data source offers several distinct advantages for identifying AI-related positions. First, unlike traditional employment databases that typically provide aggregate workforce statistics or limited samples, Revelio’s data contains granular individual-level employment records with detailed job titles, positions, and standardized job classifications. This granularity enables precise identification of AI-related roles across organizational hierarchies and functional areas. Second, traditional databases often suffer from sparse coverage or selection bias toward specific industries or firm sizes, while Revelio provides extensive coverage across a broad cross-section of firms. This comprehensive coverage allows us to track employment patterns and AI talent deployment across firms of different sizes and industries over fiscal years. Third, the database’s standardized job classification system helps address the challenge of inconsistent job titles across firms and over time, providing a more structured and reliable framework for identifying AI-related positions. Fourth, by sourcing from LinkedIn profiles, the data captures real-time changes in employment and job responsibilities, offering a more dynamic and current view of firms’ AI talent composition compared to traditional annual reporting or survey-based measures.

In a similar spirit to Babina et al. (2024), we search for AI-related terms (such as “artificial intelligence,” “machine learning,” “data science,” “deep learning,” “computer vision,” “data mining,” “big data”, and “data analytics”) in both raw job titles and standardized job classifications. This dual approach helps mitigate potential measurement errors from job title variations or misclassifications. The raw title search captures the diverse ways firms designate AI roles, while the standardized classification provides a more structured identification framework.

We implement several filtering steps starting with the comprehensive Revelio individual position database (approximately 4 terabytes). After retaining only records with valid firm identifiers globally, the sample is 306,143,646 observations. After removing duplicates and merging with U.S firms with GVKEY or CIK, we obtain 152,623,519 observations. Matching to Compustat firm-year shows 55,194 firm-year observations with non-missing total assets and sales from 2016 to 2023. Of these Compustat observations, 45,477 firm-years (82.3%) have at least one employee record in Revelio. This comprehensive coverage and detailed classification system allows us to construct a reliable measure of firms’ AI human capital intensity, capturing technical and analytical AI roles.

### **Appendix OA 1B. AI Disclosure**

We employ the FinBERT model to identify AI-related corporate disclosures, which offers several distinct advantages in text vectorization and semantic understanding (Yang et al., 2020; Huang et al., 2023). First, while traditional bag-of-words or dictionary-based approaches treat words as independent units, FinBERT employs sophisticated contextual embeddings where each word is represented as a high-dimensional vector that captures its semantic relationship with the surrounding text. This vector representation allows FinBERT to understand that the same word can carry different meanings in different contexts. Second, FinBERT’s pre-training on financial

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<sup>26</sup> Babina et al (2024) use the resume data from Cognisum.Inc, which is no longer accessible. Barrios (2022), Ahn et al. (2025), Lin et al (2026), and Baker et al. (2024), among others, have used Revelio data to study questions related to diversity, accountants, and auditors.

texts (corporate 10-K/Q filings, analyst reports, and earnings call transcripts) enables it to learn domain-specific semantic relationships and generate more accurate vector representations of financial terminology. For instance, in vectorizing phrases like “artificial intelligence investment” or “machine learning deployment,” FinBERT captures not just the individual word meanings but also their collective implications in a financial context.

Third, FinBERT’s transformer architecture enables it to generate contextual embeddings at multiple levels - words, phrases, and entire sentences. This hierarchical representation allows the model to capture both local semantic relationships (within phrases) and broader contextual information (across sentences). Fourth, through its self-attention mechanism, FinBERT can dynamically weigh the importance of different words and phrases in a sentence, creating context-aware vector representations that are particularly effective for identifying AI-related discussions in varied contexts. For example, when analyzing phrases like “investing in intelligence systems,” FinBERT can distinguish between general business intelligence and artificial intelligence based on the broader document context.

To fine-tune the FinBERT model for classifying AI-related sentences, we start by collecting corporate disclosures from 10-K filings between 2016 and 2023. Specifically, we extracted sentences from Items 1, 1A, and 7 from firms’ Form 10-K filings, as these sections often contain information about a firm’s operations, risks, and strategies. Using the NLTK package in Python, we split the documents into individual sentences. Because not all sentences are useful for fine-tuning, we focus on sentences relevant to AI. We use keyword-based filtering to identify sentences containing terms like “machine learning,” “data analytics,” and “artificial intelligence.” Using this process, we extract 640 sentences that contain these terms. To ensure diversity and balance in the dataset, we randomly select an additional 971 sentences, resulting in a total of 1,611 sentences for manual classification.

Each of the 1,611 sentences is manually reviewed and categorized into six categories. AI\_Overview discusses general trends or developments in AI. AI\_Use describes firms’ use of AI without creating AI-related products. AI\_Product/Business/Patentable refers to AI-related products, businesses, or patentable technologies developed by the firm. AI\_Acquire/Build describes firms acquiring or building AI-related businesses instead of creating them in-house. AI\_FLS refers to forward-looking statements about AI-related actions or plans. NON\_AI refers to sentences that are unrelated to AI. These categories were not mutually exclusive; a sentence could belong to multiple categories, except for NON\_AI, which was exclusive.<sup>27</sup> After classification, the sentences are distributed across the categories as follows: **7%** are categorized as AI\_Overview, **31%** as AI\_Use, **25%** as AI\_Product/Business/Patentable, **3%** as AI\_Acquire/Build, and **15%** as AI\_FLS. Overall, **40%** of the sentences were AI-related, while **60%** were classified as NON\_AI.

We split the human classified dataset into 80% for training, 10% for validation, and 10% for testing. Using the Transformers library in Python, we fine-tune the FinBERT model over 5 epochs, meaning the model passes through the entire training dataset 5 times. During the training process, raw model predictions are converted to probabilities using the sigmoid function, mapping the outputs to values between 0 and 1. A probability threshold of 0.5 is applied to classify sentences. Probabilities above 0.5 are assigned a label of 1, and 0 otherwise. The fine-tuned model achieves a training accuracy of 96%, indicating it can effectively classify AI-related sentences within the

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<sup>27</sup> See Appendix 1C for examples of sentences in each category.

training set. This high accuracy demonstrates that the model successfully learned the patterns in the data.<sup>28</sup>

After fine-tuning, the model can now classify AI-related sentences in corporate disclosures. In our context, we use this fine-tuned FinBERT model to classify sentences that are AI-related by applying the model to earnings announcements, conference call transcripts, and Form 10-K annual filings.<sup>29</sup> We calculate AI disclosure scores (AI\_CC, AI\_10K, AI\_EA) on a firm-year basis for each disclosure channel by counting sentences identified as AI-related and dividing by the total number of sentences in the respective disclosure channel.<sup>30,31</sup>

This domain-specific fine-tuning enhances the model's ability to generate precise vector representations of AI-related disclosures across different corporate communication channels. The resulting model can effectively identify and analyze AI-related content in 10-K reports, earnings announcements, and conference call transcripts, providing a more sophisticated and advanced measurement of firms' AI-related communications compared to traditional text analysis methods.

### **Appendix OA 1C. Product vs. Non-Product Disclosure**

We used the following information within a prompt using Llama 3.3-70b to determine whether firms' AI disclosures were related to product innovation. After identifying product-related AI sentences, all other sentences were classified as not-product related.

#### **PRODUCT\_INNOVATION**

Definition: Use of AI to create, improve, or differentiate customer-facing products, services, or revenue-generating features that change the value proposition for end users.

Includes (non-exhaustive):

- New AI-native products or modules offered to customers (e.g., AI copilot in our app)
- New or enhanced features (personalization, recommendations, search, content generation for users, vision/speech features)
- Customer-facing risk tools sold as a product (e.g., fraud detection SaaS)
- Improved accuracy/quality/safety of an existing customer feature (if experienced by the end user)
- Platform/SDK/API that exposes AI capabilities to customers/partners

Typical signals: “customers/users”, “feature”, “product”, “launch”, “offer”, “pricing plans”, “API”, “SDK”, “pilot with clients”, “revenue”, “monetize”.

### **Appendix OA 1D – Outsourcing**

To identify firms that may outsource AI capabilities, we use the following bag-of-words dictionaries to identify generic outsourcing terms, specific AI vendors, consultants, or external hardware mentioned in AI disclosures, and exclude sentences that contain negation.

Generic Outsourcing Terms:

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<sup>28</sup> We make the resulting AI disclosure measures available to the research community at <https://github.com/bcjl-research/ai>.

<sup>29</sup> For 10-K annual filings, we only extract sentences from Item 1, “Business,” Item 1A, “Risk Factors,” and Item 7, “Management Discussion and Analysis of Financial Condition and Results of Operations.”

<sup>30</sup> Because our AI-related sentences are not mutually exclusive, the AI score for each disclosure channel is calculated as one minus the percentage of sentences that are not AI-related, to avoid double counting.

<sup>31</sup> For AI\_CC and AI\_EA variables, we calculate firm-level AI scores by averaging the quarterly reports within each fiscal year.

outsource / outsourced / outsourcing, third-party, external (provider/vendor/partner/service/supplier/platform), outside (vendor/provider/partner/firm), license / licensed / licensing, subscribe / subscription, contract / contracted / contractor, procure / procurement, purchase (service/capability/solution/platform), white-label, co-develop / jointly develop, API / application programming interface, webhook, SDK, software-as-a-service (SaaS), platform-as-a-service (PaaS), infrastructure-as-a-service (IaaS), managed (service/solution/platform), hosted (service/solution/platform/model), cloud (service/platform/solution/provider/hosted/based/native/infrastructure/compute/deployment), cloud computing / cloud storage, as-a-service, partner / partnership, alliance, collaboration (with/agreement), agreement (with/from), rely on (external/third/outside), provided by a (third/external/outside), powered by, built on, run on, integrated with, embedded in / from, accessed through / via / using, pay-per-use / pay-per-call / pay-per-query / pay-per-token, usage-based (pricing/billing/fee), token (cost/pricing/fee)

#### Named Vendors:

Cloud infrastructure: Amazon Web Services (AWS), Microsoft Azure, Google Cloud (GCP), IBM Cloud, Oracle Cloud, Alibaba Cloud, Huawei Cloud, Tencent Cloud

AI foundation model providers: OpenAI, Anthropic, Cohere, Mistral AI, Hugging Face, Replit, Stability AI, Midjourney, Perplexity AI, AI21 Labs, Aleph Alpha, Inflection AI, Databricks, Scale AI, Anyscale, Together AI, vLLM

AI platforms and MLOps: Palantir, C3.ai, DataRobot, H2O.ai, Vertex AI, SageMaker, Azure ML, Snowflake, dbt Labs, Dataiku, Domino Data, Weights & Biases (wandb), MLflow, ClearML, Veritone, SoundHound, Cresta, Gong, Clari, 6sense, Cognitive Scale

SaaS AI products: Salesforce Einstein, SAP AI / HANA, Workday AI, ServiceNow AI, Zendesk AI, HubSpot AI, Adobe Sensei / Firefly, Oracle AI, Microsoft Copilot / 365 Copilot / Power Platform / Power Automate / Power BI, Tableau

Robotic process automation: UiPath, Automation Anywhere, Blue Prism, Pegasystems, Appian, Workato, Zapier

Consulting firms: Deloitte, Accenture, McKinsey, Boston Consulting Group (BCG), Bain, PwC, KPMG, EY / Ernst & Young, Cognizant, Infosys, Wipro, TCS / Tata Consultancy, Capgemini

Semiconductor / hardware: NVIDIA, AMD / Advanced Micro Devices, Intel, Qualcomm AI, TPU / Tensor Processing Unit

#### Negation Filters (excluded):

not, no, never, neither, nor, n't, without, avoid, eliminate, replace (third-party/vendor/external), no longer, cease, discontinue, internalize, in-house, insource, proprietary, built in-house, developed internally / in-house / by us / by our team

## Appendix OA 2 – Classification Examples

### Appendix OA 2A – Examples of AI Disclosure by Disclosure Format

*This appendix presents representative sentences that our fine-tuned FinBERT model classified as AI-related in each of the three disclosure channels we examine. The model is available at <https://github.com/bcjl-research/ai>. Within each panel, examples are organized by the five AI content categories used in our classification scheme, and each example is identified by firm, Fama-French 12 industry, and the date of the document it comes from.*

#### Panel A. Annual Report (10-K)

##### ***AI\_Overview***

- In addition, the emergence of big data and new tools such as machine learning and artificial intelligence that capitalize on the availability of large data sets is leading semiconductor manufacturers and equipment manufacturers to pursue new products and approaches that exploit those tools to advance technology development.

*Lam Research Corp (Manufacturing) — 10-K, period ending 28 June 2020*

- We believe the market for solutions that can power commerce using AI and machine learning is a large, growing market that spans most major industries.

*PROS Holdings Inc (Business Equipment) — 10-K, period ending 31 December 2018*

##### ***AI\_Use***

- We also accelerate information modernization with intelligent tools and services for moving off paper, automating classification and building clean data lakes for Artificial Intelligence (AI), analytics and automation.

*Open Text Corp (Business Equipment) — 10-K, period ending 30 June 2021*

- We are investing to further integrate big data and automated intelligence technologies within our core business lines and to create new product and service offerings.

*NantHealth Inc (Business Equipment) — 10-K, period ending 31 December 2019*

##### ***AI\_Product/Business/Patentable***

- Our web-based software employs predictive technology enhanced by artificial intelligence to find volatility and unusual market activity that may result in the rapid change in the trading price of a stock or option.

*Blackboxstocks Inc (Business Equipment) — 10-K, period ending 31 December 2021*

- We currently incorporate artificial intelligence and machine learning (AI) solutions into our intelligence cloud products including our deal intelligence application, and these applications may become important in our operations over time.

*Model N Inc (Business Equipment) — 10-K, period ending 30 September 2023*

##### ***AI\_Acquire/Build***

- We acquired MindMeld, Inc. to bring artificial intelligence capabilities into our collaboration portfolio focused on the future of conversational user interfaces.

*Cisco Systems Inc (Business Equipment) — 10-K, period ending 29 July 2017*

- We are working with Descartes Labs to infuse the power of artificial intelligence, machine learning, predictive intelligence and insight monitoring across Comtech's product offerings.

*Comtech Telecommunications Corp. (Business Equipment) — 10-K, period ending 31 July 2023*

### ***AI\_FLS***

- To best take advantage of these continued advances in technology, we are investing in AWS, which offers a broad set of on-demand technology services, including compute, storage, database, analytics, and machine learning, and other services, to developers and enterprises of all sizes.

*Amazon.com Inc (Wholesale, Retail, and Services) — 10-K, period ending 31 December 2020*

- It is possible that new laws and regulations will be adopted in the jurisdictions in which we operate, or existing laws and regulations may be interpreted in new ways, that would affect the way in which AI and other machine learning technology is used in our products, systems or solutions.

*Open Text Corp (Business Equipment) — 10-K, period ending 30 June 2023*

## **Panel B. Earnings Conference Call**

### ***AI\_Overview***

- For example, machine learning enables Siri to understand words as well as the intent behind them.

*Apple Inc (Business Equipment) — earnings call, 26 July 2016*

- The common message from participants was that the ecosystem needs to work together in new ways to bring AI technologies to market faster and at lower cost.

*Applied Materials Inc (Manufacturing) — earnings call, 16 August 2018*

### ***AI\_Use***

- Recall that CACI has over 100 projects incorporating AI capabilities across our business.

*CACI International Inc (Business Equipment) — earnings call, 22 April 2021*

- For example, in search, we leveraged new deep learning models to better understand purchase intent, leading to increased conversion at the top of search.

*eBay Inc. (Wholesale, Retail, and Services) — earnings call, 2 November 2022*

### ***AI\_Product/Business/Patentable***

- As part of these efforts, we launched our new AI-based pricing engine, which is already providing improved, actionable data, enabling us to optimize pricing structures with customers and facilitate our goals toward margin expansion.

*Farmer Bros Co (Consumer NonDurables) — earnings call, 10 May 2023*

- At the same time, we launched a parallel effort to create an AI platform to provide predictive pricing on materials for our supply chain.

*Fluor Corp (Other) — earnings call, 3 November 2023*

### ***AI\_Acquire/Build***

- Last week, we announced the acquisition of Digital Surgery, a pioneer in artificial intelligence and analytics for surgery.

*Medtronic PLC (Healthcare, Medical Equipment, and Drugs) — earnings call, 18 February 2020*

- During the quarter, we also added key technology and talent with the acquisition of ID R&D, an award winning provider of AI-based voice and face biometrics and liveness detection.

*Mitek Systems Inc (Business Equipment) — earnings call, 29 July 2021*

### ***AI\_FLS***

- Technology will continue to play a pivotal role in the company's future success with analytics, big data and machine learning being significant areas of opportunity for the industry.

*Helmerich & Payne Inc. (Oil, Gas, and Coal) — earnings call, 27 July 2017*

- We believe we are just scratching the surface when it comes to machine learning and machine vision.

*Shutterstock Inc (Business Equipment) — earnings call, 31 October 2017*

## **Panel C. Earnings Announcement**

### ***AI\_Overview***

- The Fire Safe 3.0 program involves partnering with academic, government and public safety professionals to implement artificial intelligence, satellite wildfire alerts and a new Vegetation Risk index, among other strategies.

*Sempra (Utilities), then named Sempra Energy — earnings announcement, 27 February 2020*

- The contract includes Unisys IntelliServe, which uses artificial intelligence and robotic process automation to provide a first-rate user experience for 39,000 employees globally.

*Unisys Corp (Business Equipment) — earnings announcement, 22 February 2021*

### ***AI\_Use***

- We use artificial intelligence and machine learning to help buyers more easily browse, filter and find the item they desire.

*Etsy Inc (Business Equipment) — earnings announcement, 27 February 2018*

- We apply cloud computing, artificial intelligence, advanced sensors, advisory services and managed services to achieve safe, efficient and sustainable mobility.

*Iteris Inc (Business Equipment) — earnings announcement, 9 June 2020*

### ***AI\_Product/Business/Patentable***

- AMD made significant progress expanding its AI software ecosystem, including unveiling the latest version of its open-source ROCm 6, software stack optimized for generative AI.

*Advanced Micro Devices Inc (Business Equipment) — earnings announcement, 30 January 2024*

- Announced the general availability of RingCX, a native, AI-first contact center with new capabilities powered by its RingSense AI platform.

*RingCentral Inc (Business Equipment) — earnings announcement, 20 February 2024*

### ***AI\_Acquire/Build***

– In April, we acquired a small machine learning company focused on novel applications of deep neural networks to signal processing that we anticipate will augment our product offerings in the voice and speech domains.

*Cirrus Logic Inc. (Business Equipment) — earnings announcement, 3 May 2017*

– With the acquisition of Orma Health last month, we have made great strides already in integrating their AI-driven analytics tools into our technology platform.

*Astrana Health Inc (Healthcare, Medical Equipment, and Drugs), then named Apollo Medical Holdings — earnings announcement, 24 February 2022*

## ***AI\_FLS***

– The feedback from those customers with initial ProFound AI installs has been overwhelmingly positive, and we believe we are well on our way to achieving our goals with this product.

*iCAD Inc (Healthcare, Medical Equipment, and Drugs) — earnings announcement, 18 March 2019*

– I am proud of our global team, and excited about 2021 as we will continue delivering industry-leading technology and expertise aligned with our vision for the convergence of simulation, high-performance computing, and artificial intelligence.

*Altair Engineering Inc (Business Equipment) — earnings announcement, 25 February 2021*

## Appendix OA 2B – AI Patents

### Machine Learning:

Qualcomm - <https://patents.google.com/patent/US8433665B2/en?q=8433665>

Toyota - <https://patents.google.com/patent/US8688356B2/en?q=8688356>

### Evolutionary computation:

IBM - <https://patents.google.com/patent/US9058564B2/en?q=9058564>

Adobe - <https://patents.google.com/patent/US8014615B2/en?q=8014615>

### Natural language processing:

Facebook - <https://patents.google.com/patent/US9830386B2/en?q=9830386>

Nokia - <https://patents.google.com/patent/US9274646B2/en?q=9274646>

### Speech:

Apple Inc. - <https://patents.google.com/patent/US9582608B2/en?q=9582608>

Amazon - <https://patents.google.com/patent/US8996372B1/en?q=8996372>

### Vision:

Microsoft - <https://patents.google.com/patent/US8483436B2/en?q=8483436>

Honeywell - <https://patents.google.com/patent/US7925117B2/en?q=7925117>

### Knowledge Processing:

AT&T - <https://patents.google.com/patent/US8892443B2/en?q=8892443>

Oracle - <https://patents.google.com/patent/US9672080B2/en?q=9672080>

### Planning:

Google - <https://patents.google.com/patent/US8655901B1/en?q=8655901>

Yahoo - <https://patents.google.com/patent/US8754848B2/en?q=8754848>

### Hardware:

Salesforce - <https://patents.google.com/patent/US9471666B2/en?q=9471666>

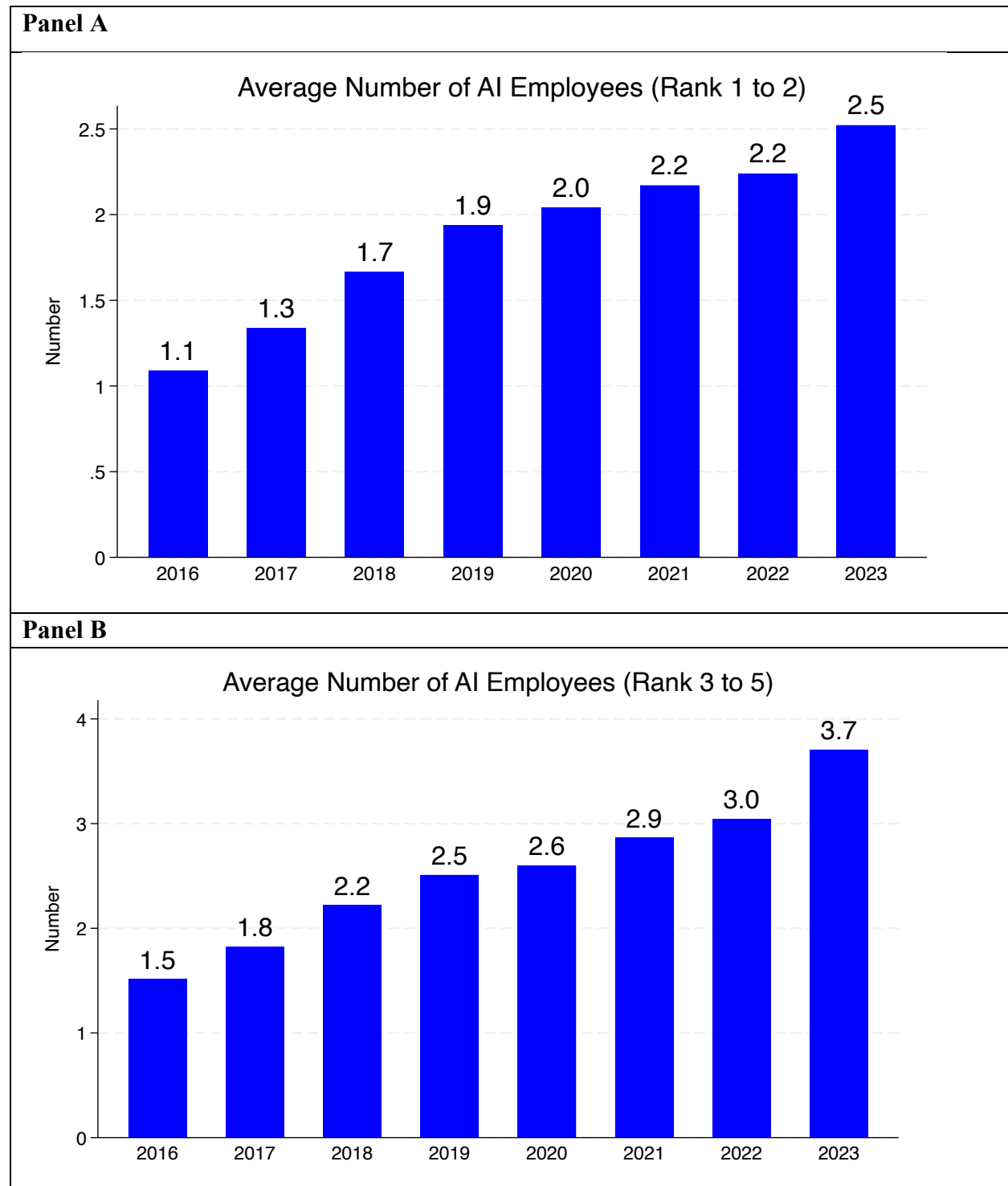
SanDisk - <https://patents.google.com/patent/US8848430B2/en?q=8848430>

## **Appendix OA 2C – Outsourcing**

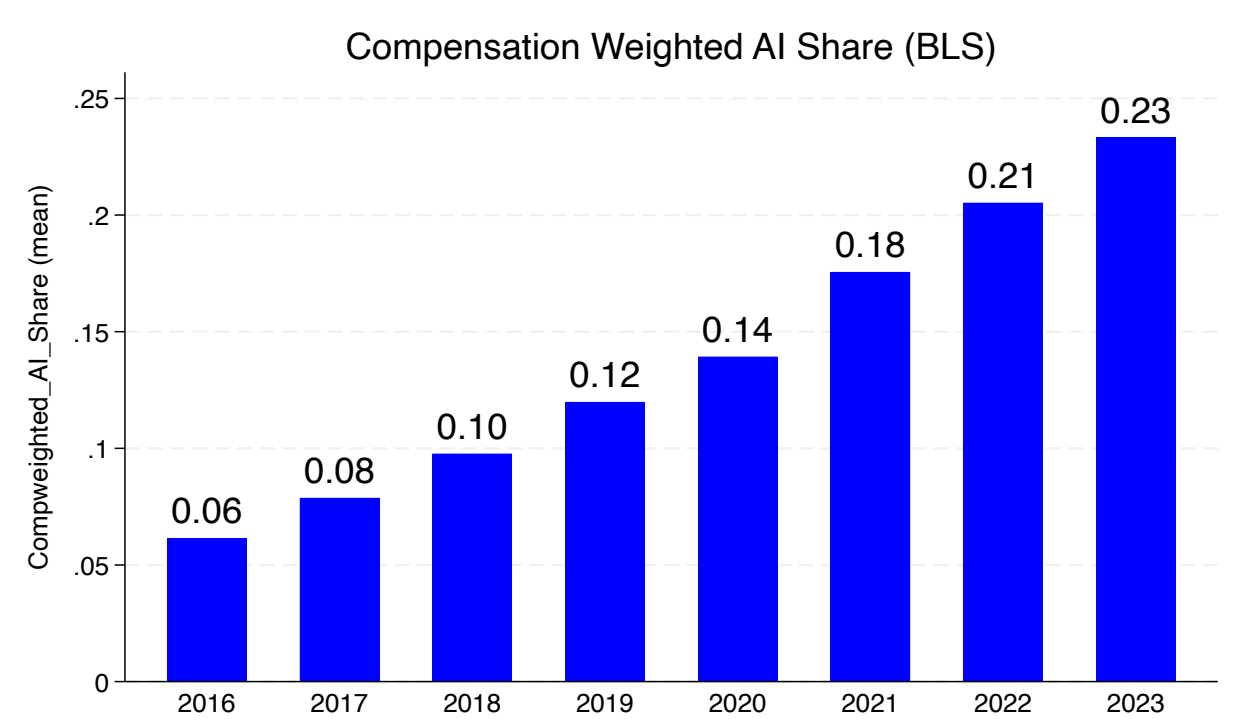
- Silicon Valley-based provider accelerated compute platforms application-optimized high performance high-efficiency server storage systems variety markets, including enterprise data centers, cloud computing, artificial intelligence, 5G edge computing.
- Verisure will also purchase associated cloud services from Arlo, including artificial intelligence, computer vision and other features.
- We achieved a major milestone on the cloud innovation road map with Alteryx Machine Learning now being integrated on the Alteryx Analytics Cloud platform.
- In fact, earlier today at the AI Summit, we announced the preview of our next-generation, cloud-native machine learning platform powered by Kubernetes.

**Figure OA 1**

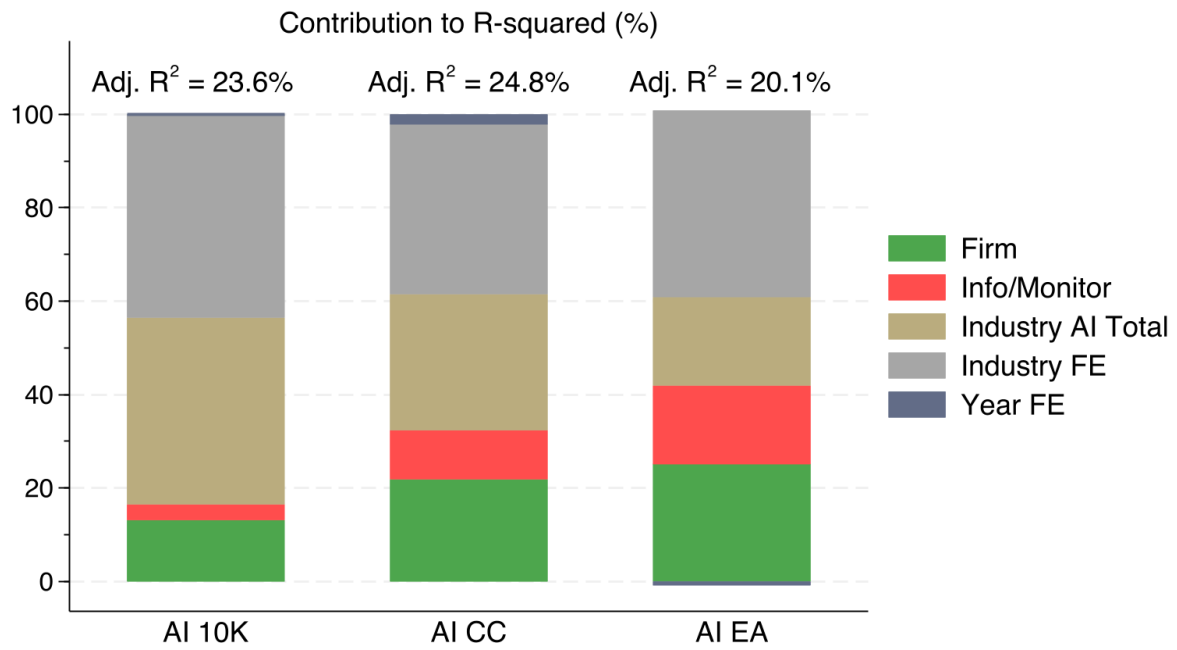
This figure plots the average number of AI employees per firm across Revelio's seniority tiers from 2016 to 2023. Junior AI employees correspond to Revelio seniority levels 1-2, and mid-to-senior AI employees correspond to levels 3-5.



Panel C

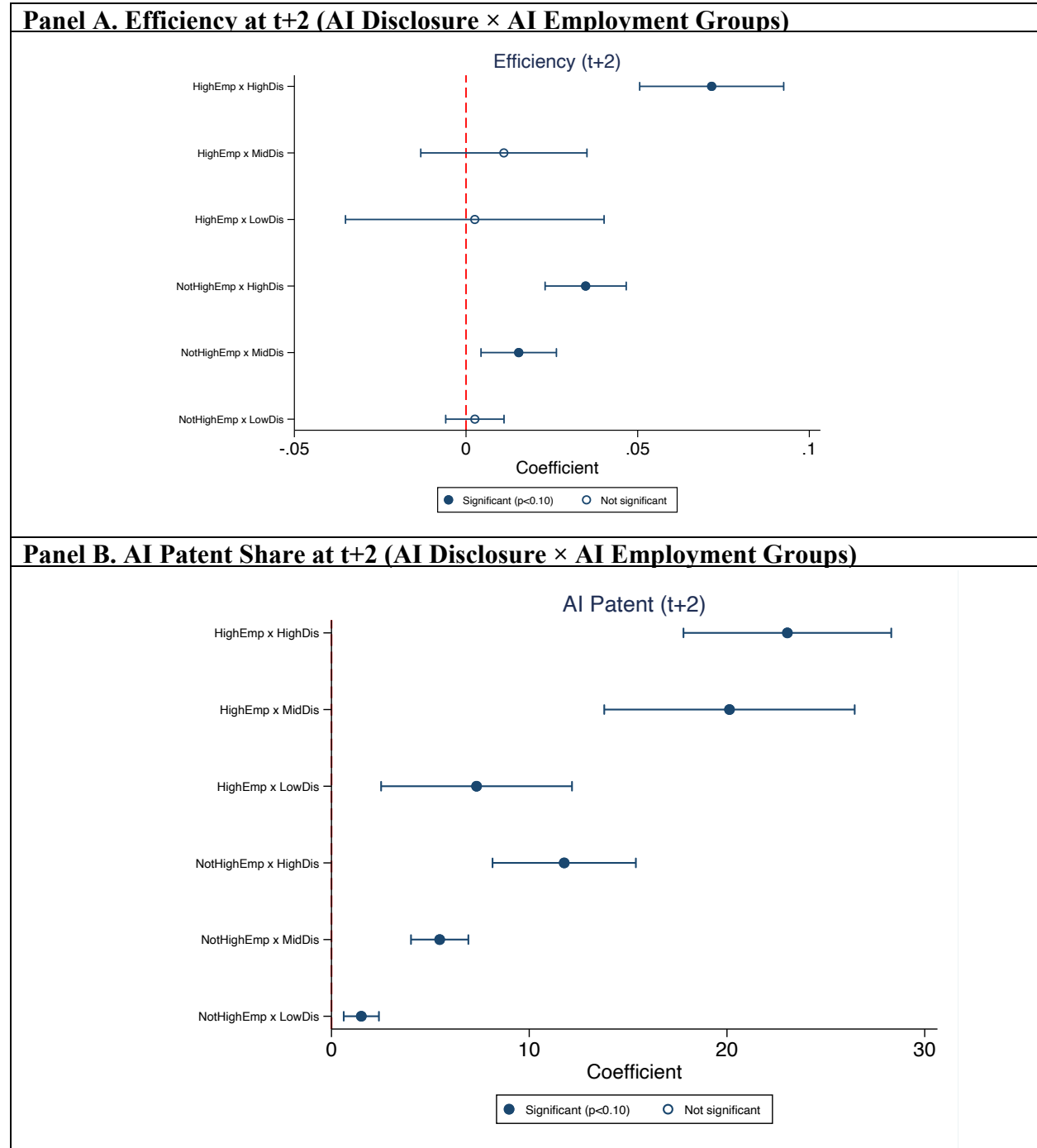


**Figure OA 2 – Shapley Decomposition for Outsourcing Firms**



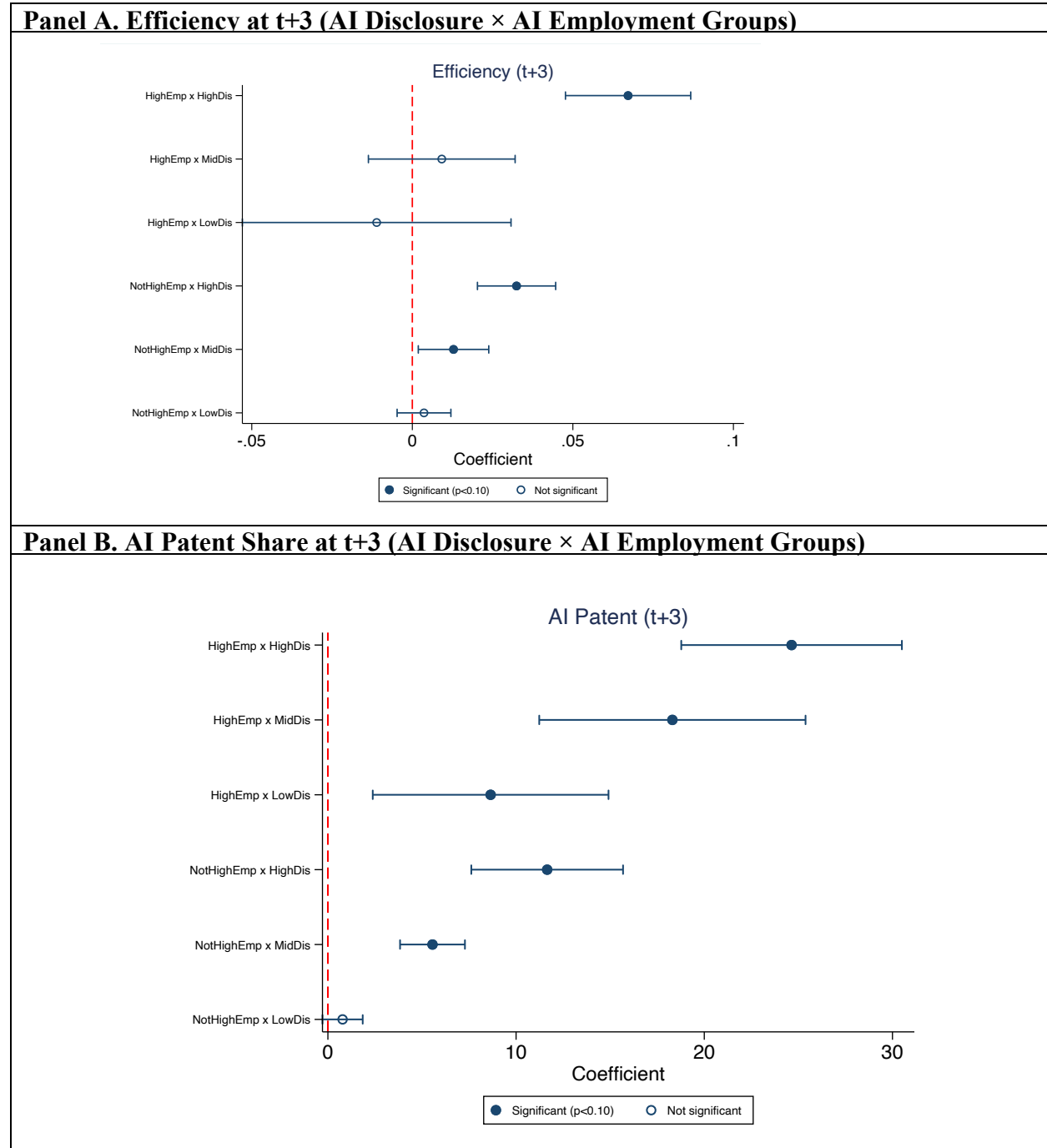
### Figure OA 3. Efficiency and AI Patent at t+2: AI Disclosure $\times$ AI Employment Groups

This figure plots 90% confidence intervals for the coefficients of interest from re-estimating Table 8 of the main text at the t+2 horizon. Panel A reports estimates for efficiency and Panel B reports estimates for the AI patent share. Each row represents a combination of AI employment level (HighEmp versus NotHighEmp) and AI disclosure intensity (High, Mid, or Low). The omitted reference group is firms with not-high AI employment and no AI disclosure. Filled markers indicate significance at the 10% level.



### Figure OA 4. Efficiency and AI Patent at t+3: AI Disclosure $\times$ AI Employment Groups

This figure plots 90% confidence intervals for the coefficients of interest from re-estimating Table 8 of the main text at the t+3 horizon. Panel A reports estimates for efficiency and Panel B reports estimates for the AI patent share. Each row represents a combination of AI employment level (HighEmp versus NotHighEmp) and AI disclosure intensity (High, Mid, or Low). The omitted reference group is firms with not-high AI employment and no AI disclosure. Filled markers indicate significance at the 10% level.



**Table OA 1. Expanded Determinants of AI Disclosure**

This table reports the results of regressions of AI disclosure levels (AI\_10K, AI\_EA, and AI\_CC) on firm characteristics, informational/external monitoring, managerial characteristics, litigation risk, industry total AI disclosure, and industry fixed effects and year fixed effects. Variables employed in this table are standardized. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, and \* denoted significance at the 1%, 5%, and 10% levels (two-tailed).

**Panel A. AI 10K Disclosure**

| VARIABLES          | (1)<br>AI_10K       | (2)<br>AI_10K       | (3)<br>AI_10K       | (4)<br>AI_10K       | (5)<br>AI_10K       |
|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| AI Employee%       | 0.39***<br>(8.14)   | 0.39***<br>(7.20)   | 0.46***<br>(4.67)   | 0.39***<br>(7.52)   | 0.35***<br>(3.14)   |
| R&D                | 0.09**<br>(2.36)    | 0.06<br>(1.26)      | 0.38**<br>(2.58)    | 0.10**<br>(2.35)    | 0.50***<br>(2.79)   |
| Size               | 0.07*<br>(1.81)     | -0.02<br>(-0.30)    | 0.14**<br>(2.21)    | 0.04<br>(0.79)      | -0.04<br>(-0.34)    |
| HHI_Seg            | -0.01<br>(-0.45)    | -0.01<br>(-0.15)    | 0.07<br>(1.28)      | -0.02<br>(-0.61)    | 0.00<br>(0.07)      |
| Tangibility        | -0.26***<br>(-8.71) | -0.30***<br>(-9.10) | -0.26***<br>(-7.02) | -0.26***<br>(-8.30) | -0.24***<br>(-5.93) |
| Leverage           | 0.00<br>(0.07)      | 0.01<br>(0.18)      | -0.03<br>(-0.76)    | 0.01<br>(0.35)      | -0.05<br>(-1.21)    |
| Age                | 0.03<br>(0.91)      | 0.06*<br>(1.78)     | 0.14**<br>(2.20)    | 0.02<br>(0.62)      | 0.19***<br>(2.69)   |
| Investment         | 0.02<br>(0.79)      | 0.03<br>(1.04)      | 0.04<br>(0.77)      | 0.04*<br>(1.86)     | 0.04<br>(0.64)      |
| CFO                | 0.29***<br>(5.06)   | 0.34***<br>(5.19)   | 0.19<br>(1.57)      | 0.30***<br>(4.59)   | 0.11<br>(0.77)      |
| Profitability      | -0.26***<br>(-4.39) | -0.30***<br>(-4.09) | -0.19*<br>(-1.73)   | -0.24***<br>(-3.76) | -0.09<br>(-0.68)    |
| MTB                | 0.06*<br>(1.78)     | 0.01<br>(0.15)      | 0.04<br>(0.76)      | 0.06*<br>(1.75)     | -0.05<br>(-0.78)    |
| Restatement        | 0.03*<br>(1.73)     | 0.01<br>(0.94)      | 0.02<br>(1.29)      | 0.03*<br>(1.65)     | 0.02<br>(1.14)      |
| Volatility         | -0.03<br>(-1.10)    | -0.06**<br>(-2.11)  | -0.07*<br>(-1.74)   | -0.03<br>(-1.19)    | -0.09*<br>(-1.88)   |
| N_Analyst          |                     | 0.11**<br>(2.35)    |                     |                     | 0.15***<br>(2.58)   |
| Share_ByFund       |                     | -0.01<br>(-0.36)    |                     |                     | 0.02<br>(0.38)      |
| Insti_Own          |                     | -0.02<br>(-0.40)    |                     |                     | -0.06<br>(-0.93)    |
| Board_Indep        |                     | 0.02<br>(0.98)      |                     |                     | 0.05**<br>(2.00)    |
| Managerial_Ability |                     |                     | -0.05<br>(-1.44)    |                     | -0.10**<br>(-2.47)  |
| CEO_Stock          |                     |                     | 0.07**<br>(2.14)    |                     | 0.07*<br>(1.81)     |

|                 |                   |                   |                   |                   |                    |
|-----------------|-------------------|-------------------|-------------------|-------------------|--------------------|
| Litigation_Risk |                   |                   |                   | 0.03<br>(0.55)    | 0.04<br>(0.63)     |
| AI_Dis_Ind      |                   |                   |                   |                   | 0.54***<br>(4.93)  |
| Constant        | 0.43***<br>(9.59) | 0.40***<br>(7.12) | 0.54***<br>(6.58) | 0.44***<br>(9.21) | 1.17***<br>(17.14) |
| Observations    | 11,870            | 9,377             | 6,122             | 10,483            | 5,188              |
| Adj. R-squared  | 0.39              | 0.38              | 0.43              | 0.39              | 0.47               |
| Year FE         | Yes               | Yes               | Yes               | Yes               | Yes                |
| Industry FE     | Yes               | Yes               | Yes               | Yes               | Yes                |
| Cluster         | Firm              | Firm              | Firm              | Firm              | Firm               |

**Panel B. AI Earnings Announcements**

| VARIABLES     | (1)<br>AI_EA        | (2)<br>AI_EA        | (3)<br>AI_EA        | (4)<br>AI_EA        | (5)<br>AI_EA        |
|---------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| AI Employee%  | 0.43***<br>(8.07)   | 0.42***<br>(7.47)   | 0.45***<br>(5.13)   | 0.44***<br>(7.40)   | 0.39***<br>(4.12)   |
| R&D           | 0.17***<br>(3.48)   | 0.17***<br>(2.78)   | 0.58***<br>(3.50)   | 0.16***<br>(3.16)   | 0.61***<br>(3.14)   |
| Size          | -0.06<br>(-1.42)    | -0.15***<br>(-2.59) | 0.00<br>(0.03)      | -0.23***<br>(-3.37) | -0.32***<br>(-3.01) |
| HHI_Seg       | -0.02<br>(-0.73)    | -0.01<br>(-0.44)    | 0.00<br>(0.01)      | -0.03<br>(-1.04)    | -0.04<br>(-0.74)    |
| Tangibility   | -0.20***<br>(-6.90) | -0.22***<br>(-6.97) | -0.18***<br>(-5.65) | -0.19***<br>(-6.27) | -0.17***<br>(-4.96) |
| Leverage      | -0.03<br>(-1.14)    | -0.03<br>(-0.87)    | -0.08**<br>(-1.99)  | -0.01<br>(-0.49)    | -0.09**<br>(-2.13)  |
| Age           | -0.06*<br>(-1.88)   | -0.05<br>(-1.43)    | -0.08<br>(-0.98)    | -0.06<br>(-1.62)    | -0.05<br>(-0.60)    |
| Investment    | 0.06**<br>(2.50)    | 0.07**<br>(2.32)    | 0.15**<br>(2.04)    | 0.08***<br>(3.04)   | 0.13*<br>(1.77)     |
| CFO           | 0.28***<br>(3.87)   | 0.36***<br>(4.08)   | 0.12<br>(0.86)      | 0.31***<br>(3.83)   | 0.16<br>(0.99)      |
| Profitability | -0.23***<br>(-3.39) | -0.32***<br>(-3.73) | -0.21<br>(-1.65)    | -0.25***<br>(-3.32) | -0.23<br>(-1.45)    |
| MTB           | 0.09**<br>(2.38)    | 0.05<br>(1.19)      | 0.01<br>(0.17)      | 0.09**<br>(2.23)    | -0.05<br>(-1.01)    |
| Restatement   | 0.02<br>(1.48)      | 0.01<br>(0.78)      | 0.02<br>(0.90)      | 0.02<br>(1.10)      | 0.01<br>(0.32)      |
| Volatility    | 0.08**<br>(2.50)    | 0.04<br>(1.16)      | -0.00<br>(-0.05)    | 0.03<br>(0.76)      | -0.07<br>(-1.13)    |
| N_Analyst     |                     | 0.17***<br>(4.00)   |                     |                     | 0.17***<br>(3.28)   |

|                    |                   |                   |                   |                   |                    |
|--------------------|-------------------|-------------------|-------------------|-------------------|--------------------|
| Share_ByFund       |                   | -0.04<br>(-1.17)  |                   |                   | 0.01<br>(0.32)     |
| Insti_Own          |                   | -0.03<br>(-0.72)  |                   |                   | -0.05<br>(-0.77)   |
| Board_Indep        |                   | 0.01<br>(0.71)    |                   |                   | 0.03<br>(1.25)     |
| Managerial_Ability |                   |                   | 0.01<br>(0.23)    |                   | -0.04<br>(-1.09)   |
| CEO_Stock          |                   |                   | 0.05*<br>(1.80)   |                   | 0.07**<br>(1.96)   |
| Litigation_Risk    |                   |                   |                   | 0.15***<br>(2.70) | 0.16**<br>(2.09)   |
| AI_Dis_Ind         |                   |                   |                   |                   | 0.27**<br>(2.55)   |
| Constant           | 0.54***<br>(8.61) | 0.58***<br>(7.26) | 0.80***<br>(7.77) | 0.59***<br>(8.96) | 1.21***<br>(15.13) |
| Observations       | 11,870            | 9,377             | 6,122             | 10,483            | 5,188              |
| Adj. R-squared     | 0.36              | 0.36              | 0.34              | 0.35              | 0.34               |
| Year FE            | Yes               | Yes               | Yes               | Yes               | Yes                |
| Industry FE        | Yes               | Yes               | Yes               | Yes               | Yes                |
| Cluster            | Firm              | Firm              | Firm              | Firm              | Firm               |

#### Panel C AI Conference Call

| VARIABLES     | (1)<br>AI_CC        | (2)<br>AI_CC        | (3)<br>AI_CC        | (4)<br>AI_CC        | (5)<br>AI_CC        |
|---------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| AI Employee%  | 0.57***<br>(8.04)   | 0.55***<br>(7.04)   | 0.53***<br>(4.31)   | 0.60***<br>(7.54)   | 0.43***<br>(3.10)   |
| R&D           | 0.24***<br>(4.61)   | 0.25***<br>(3.60)   | 0.87***<br>(4.84)   | 0.28***<br>(4.91)   | 0.97***<br>(4.65)   |
| Size          | 0.18***<br>(3.38)   | -0.07<br>(-1.00)    | 0.28***<br>(3.52)   | 0.06<br>(0.78)      | -0.19<br>(-1.37)    |
| HHI_Seg       | -0.06*<br>(-1.74)   | -0.08**<br>(-2.21)  | -0.05<br>(-0.83)    | -0.07**<br>(-1.98)  | -0.15**<br>(-2.38)  |
| Tangibility   | -0.24***<br>(-6.71) | -0.28***<br>(-7.16) | -0.24***<br>(-5.55) | -0.24***<br>(-6.29) | -0.21***<br>(-4.52) |
| Leverage      | -0.04<br>(-1.17)    | -0.03<br>(-0.86)    | -0.09*<br>(-1.81)   | -0.02<br>(-0.76)    | -0.14***<br>(-2.58) |
| Age           | -0.10**<br>(-2.25)  | -0.10**<br>(-2.08)  | -0.10<br>(-1.03)    | -0.09*<br>(-1.76)   | -0.14<br>(-1.12)    |
| Investment    | 0.05*<br>(1.71)     | 0.05<br>(1.52)      | 0.17*<br>(1.90)     | 0.05<br>(1.63)      | 0.10<br>(1.11)      |
| CFO           | 0.50***<br>(5.21)   | 0.59***<br>(5.33)   | 0.59***<br>(3.08)   | 0.52***<br>(4.72)   | 0.53**<br>(2.48)    |
| Profitability | -0.35***            | -0.39***            | -0.47***            | -0.32***            | -0.37*              |

|                    |         |         |         |         |         |
|--------------------|---------|---------|---------|---------|---------|
|                    | (-3.93) | (-3.62) | (-2.73) | (-3.29) | (-1.73) |
| MTB                | 0.12*** | 0.05    | 0.06    | 0.14*** | -0.02   |
|                    | (2.70)  | (1.01)  | (0.85)  | (2.73)  | (-0.23) |
| Restatement        | 0.05**  | 0.03    | 0.03    | 0.04    | 0.02    |
|                    | (2.12)  | (1.29)  | (0.99)  | (1.57)  | (0.78)  |
| Volatility         | 0.03    | 0.00    | 0.01    | 0.00    | 0.01    |
|                    | (0.89)  | (0.05)  | (0.23)  | (0.11)  | (0.08)  |
| N_Analyst          |         | 0.28*** |         |         | 0.36*** |
|                    |         | (4.73)  |         |         | (4.45)  |
| Share_ByFund       |         | -0.04   |         |         | 0.02    |
|                    |         | (-0.90) |         |         | (0.29)  |
| Insti_Own          |         | 0.08    |         |         | 0.07    |
|                    |         | (1.52)  |         |         | (1.00)  |
| Board_Indep        |         | 0.02    |         |         | 0.04    |
|                    |         | (0.83)  |         |         | (1.26)  |
| Managerial_Ability |         |         | -0.04   |         | -0.09   |
|                    |         |         | (-0.78) |         | (-1.52) |
| CEO_Stock          |         |         | 0.11*** |         | 0.12*** |
|                    |         |         | (2.83)  |         | (2.78)  |
| Litigation_Risk    |         |         |         | 0.10*   | 0.11    |
|                    |         |         |         | (1.72)  | (1.32)  |
| AI_Dis_Ind         |         |         |         |         | 0.53*** |
|                    |         |         |         |         | (3.57)  |
| Constant           | 0.95*** | 0.93*** | 1.23*** | 1.00*** | 1.67*** |
|                    | (10.33) | (8.03)  | (7.93)  | (10.74) | (16.13) |
| Observations       | 11,870  | 9,377   | 6,122   | 10,483  | 5,188   |
| Adj. R-squared     | 0.38    | 0.39    | 0.38    | 0.38    | 0.41    |
| Year FE            | Yes     | Yes     | Yes     | Yes     | Yes     |
| Industry FE        | Yes     | Yes     | Yes     | Yes     | Yes     |
| Cluster            | Firm    | Firm    | Firm    | Firm    | Firm    |

**Table OA 2 Outsourcing, AI employee, and Disclosure Informativeness in Conference Calls and Earnings Announcements**

This table replicates the results in Table 7 for the conference call (Panel A) and earnings announcement (Panel B) channels. For each channel we estimate:  $Y_{i,t+h} = \alpha_l + \beta_1 AI\_Disclosure + \beta_2 AI\_Outsource + \beta_3 (AI\_Disclosure \times AI\_Outsource) + \gamma Controls + Industry\ FE + Year\ FE + \varepsilon_{i,t}$ . In Panel A, AI\_Disclosure is AI\_CC and AI\_Outsource\_CC is the FinBERT-based outsourcing indicator constructed from conference call transcripts. In Panel B, AI\_Disclosure is AI\_EA and AI\_Outsource\_EA is constructed from earnings announcement press releases. Y is Efficiency (cols. 1-3), AI\_Patent (cols. 4-6), or AI\_Patent\_Cite (cols. 7-9), measured at t+1, t+2, and t+3. All variables are defined in Appendix 1. Continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

**Panel A Conference Calls Channel**

| VARIABLES               | (1)                 | (2)                 | (3)                 | (4)                  | (5)                  | (6)                  | (7)                 | (8)                 | (9)                 |
|-------------------------|---------------------|---------------------|---------------------|----------------------|----------------------|----------------------|---------------------|---------------------|---------------------|
|                         | Efficiency          |                     |                     | AI_Patent (%)        |                      |                      | AI_Patent Cite(Log) |                     |                     |
|                         | t+1                 | t+2                 | t+3                 | t+1                  | t+2                  | t+3                  | t+1                 | t+2                 | t+3                 |
| AI_CC                   | 0.01***<br>(5.98)   | 0.01***<br>(6.05)   | 0.01***<br>(6.60)   | 2.61***<br>(7.11)    | 2.90***<br>(6.55)    | 3.07***<br>(5.81)    | 0.04***<br>(5.97)   | 0.03***<br>(4.62)   | 0.04***<br>(4.82)   |
| AI_Outsource_CC         | 0.02***<br>(2.69)   | 0.01<br>(0.86)      | 0.00<br>(0.51)      | 3.88*<br>(1.92)      | 4.44*<br>(1.89)      | 6.57**<br>(2.38)     | 0.08*<br>(1.76)     | 0.04<br>(0.91)      | 0.04<br>(0.88)      |
| AI_CC x AI_Outsource_CC | -0.00***<br>(-3.29) | -0.00***<br>(-2.68) | -0.00***<br>(-2.78) | -0.47<br>(-1.14)     | -0.68<br>(-1.41)     | -1.03*<br>(-1.71)    | -0.01<br>(-0.97)    | 0.00<br>(0.12)      | -0.01<br>(-0.57)    |
| AI_Employee(%)          | 0.02**<br>(2.29)    | 0.02***<br>(2.68)   | 0.02***<br>(2.72)   | 13.43***<br>(6.75)   | 16.05***<br>(6.95)   | 17.16***<br>(6.09)   | 0.15***<br>(3.99)   | 0.17***<br>(4.12)   | 0.15***<br>(3.50)   |
| Constant                | 0.78***<br>(43.11)  | 0.75***<br>(35.48)  | 0.74***<br>(29.49)  | -15.34***<br>(-4.50) | -16.10***<br>(-4.31) | -16.93***<br>(-4.19) | -0.41***<br>(-5.30) | -0.39***<br>(-4.68) | -0.35***<br>(-4.23) |
| Observations            | 8,478               | 6,609               | 5,123               | 8,598                | 6,704                | 5,206                | 8,598               | 6,704               | 5,206               |
| Adjusted R-squared      | 0.26                | 0.26                | 0.25                | 0.31                 | 0.33                 | 0.34                 | 0.22                | 0.21                | 0.20                |
| Controls                | Yes                 | Yes                 | Yes                 | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 |
| Year FE                 | Yes                 | Yes                 | Yes                 | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 |
| Industry FE             | Yes                 | Yes                 | Yes                 | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 |
| Cluster                 | Firm                | Firm                | Firm                | Firm                 | Firm                 | Firm                 | Firm                | Firm                | Firm                |
| $\beta_1 + \beta_3$     | 14.09***            | 15.75***            | 17.43***            | 41.87***             | 35.70***             | 22.71***             | 17.09***            | 15.54***            | 9.78***             |

### Panel B Earnings Announcements Channel

| VARIABLES                      | (1)                | (2)                | (3)                | (4)                  | (5)                  | (6)                  | (7)                 | (8)                 | (9)                 |
|--------------------------------|--------------------|--------------------|--------------------|----------------------|----------------------|----------------------|---------------------|---------------------|---------------------|
|                                | Efficiency         |                    |                    | AI Patent (%)        |                      |                      | AI Patent Cite(Log) |                     |                     |
|                                | t+1                | t+2                | t+3                | t+1                  | t+2                  | t+3                  | t+1                 | t+2                 | t+3                 |
| AI_EA                          | 0.01***<br>(4.48)  | 0.01***<br>(4.53)  | 0.01***<br>(3.55)  | 2.20***<br>(4.96)    | 1.97***<br>(3.99)    | 2.43***<br>(4.05)    | 0.05***<br>(5.22)   | 0.04***<br>(4.47)   | 0.04***<br>(3.73)   |
| AI_Outsource_EA                | -0.00<br>(-0.27)   | -0.00<br>(-0.02)   | -0.01<br>(-0.93)   | 4.29**<br>(2.05)     | 2.13<br>(0.86)       | 3.25<br>(1.17)       | 0.09**<br>(2.31)    | 0.06<br>(1.30)      | 0.04<br>(0.81)      |
| AI_EA x AI_Outsource_EA        | -0.00<br>(-0.60)   | -0.00<br>(-0.95)   | 0.00<br>(0.54)     | -0.67<br>(-1.22)     | -0.19<br>(-0.30)     | -0.42<br>(-0.61)     | -0.02*<br>(-1.68)   | -0.01<br>(-0.97)    | -0.01<br>(-0.67)    |
| AI_Employee(%)                 | 0.02**<br>(2.54)   | 0.03***<br>(3.00)  | 0.03***<br>(3.04)  | 14.67***<br>(7.18)   | 17.87***<br>(7.67)   | 19.01***<br>(6.84)   | 0.16***<br>(4.20)   | 0.18***<br>(4.35)   | 0.16***<br>(3.96)   |
| Constant                       | 0.78***<br>(43.08) | 0.75***<br>(35.57) | 0.74***<br>(29.76) | -15.66***<br>(-4.44) | -16.36***<br>(-4.22) | -17.13***<br>(-4.10) | -0.43***<br>(-5.39) | -0.41***<br>(-4.86) | -0.36***<br>(-4.29) |
| Observations                   | 8,473              | 6,606              | 5,121              | 8,593                | 6,701                | 5,204                | 8,593               | 6,701               | 5,204               |
| Adjusted R-squared             | 0.26               | 0.26               | 0.25               | 0.30                 | 0.31                 | 0.32                 | 0.21                | 0.21                | 0.19                |
| Controls                       | Yes                | Yes                | Yes                | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 |
| Year FE                        | Yes                | Yes                | Yes                | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 |
| Industry FE                    | Yes                | Yes                | Yes                | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 |
| Cluster                        | Firm               | Firm               | Firm               | Firm                 | Firm                 | Firm                 | Firm                | Firm                | Firm                |
| F-Test ( $\beta_1 + \beta_3$ ) | 26.49***           | 13.17***           | 22.01***           | 11.25***             | 11.52***             | 13.12***             | 8.27***             | 8.82***             | 7.63***             |

### Table OA 3. Sample Composition of Different Groups

This table reports the sample composition for the four AI-disclosure groups (Low, Mid, and High terciles among firms with non-zero composite scores) crossed with the two AI-employment groups (High versus Not-High), as defined in Section 4.4 of the main text. Panel A corresponds to the sample used in Panel A of Table 8 in the main manuscript; Panel B reports sample composition for product disclosure groups.

#### Panel A

| Variable           | N    | Mean | SD   | P25 | Median | P75 |
|--------------------|------|------|------|-----|--------|-----|
| HighDis_HighEmp    | 3425 | 0.36 | 0.48 | 0   | 0      | 1   |
| MidDis_HighEmp     | 2574 | 0.14 | 0.35 | 0   | 0      | 0   |
| LowDis_HighEmp     | 2466 | 0.11 | 0.31 | 0   | 0      | 0   |
| HighDis_NotHighEmp | 3934 | 0.44 | 0.5  | 0   | 0      | 1   |
| MidDis_NotHighEmp  | 4692 | 0.53 | 0.5  | 0   | 1      | 1   |
| LowDis_NotHighEmp  | 5783 | 0.62 | 0.49 | 0   | 1      | 1   |

#### Panel B

| Variable               | N    | Mean | SD   | P25 | Median | P75 |
|------------------------|------|------|------|-----|--------|-----|
| HighProdDis_HighEmp    | 7504 | 0.11 | 0.31 | 0   | 0      | 0   |
| MidProdDis_HighEmp     | 7138 | 0.06 | 0.24 | 0   | 0      | 0   |
| LowProdDis_HighEmp     | 6980 | 0.04 | 0.19 | 0   | 0      | 0   |
| HighProdDis_NotHighEmp | 7508 | 0.11 | 0.31 | 0   | 0      | 0   |
| MidProdDis_NotHighEmp  | 7875 | 0.15 | 0.35 | 0   | 0      | 0   |
| LowProdDis_NotHighEmp  | 8038 | 0.16 | 0.37 | 0   | 0      | 0   |

#### Table OA 4. Robustness Checks: Alternative AI Worker Measures

This table reports robustness checks using alternative AI workforce measures: the share of mid-to-senior AI employees (Revelio seniority levels 3 to 5), the share of junior AI employees (Revelio seniority levels 1 to 2), and two compensation-weighted AI employment share. One is constructed based on the estimated salary provided by Revelio (Weighted\_AI\_Share(Revelio)) and another one is constructed by mapping Revelio job titles to BLS Standard Occupational Classification codes and merging in median annual wages from the BLS Occupational Employment and Wage Statistics survey (Weighted\_AI\_Share(BLS)). Panel A reports descriptive statistics. Panel B reports pairwise correlations. Panel C reports determinants regressions of AI disclosure on each alternative workforce measure together with the controls in Equation (1). Panels D and E report regressions of one-year-ahead efficiency and AI patent ratio, respectively, on each alternative measure together with AI disclosure and controls. Variables employed in Panel C are standardized. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

##### Panel A. Descriptive Statistics

| Variable                   | N     | Mean | SD   | P25  | Median | P75  |
|----------------------------|-------|------|------|------|--------|------|
| AI_Employee(%)             | 11927 | 0.15 | 0.35 | 0.00 | 0.00   | 0.10 |
| AI_Employee_1to2(%)        | 11927 | 0.06 | 0.17 | 0.00 | 0.00   | 0.03 |
| AI_Employee_3to5(%)        | 11927 | 0.08 | 0.20 | 0.00 | 0.00   | 0.04 |
| Weighted_AI_Share(Revelio) | 11268 | 0.19 | 0.42 | 0.00 | 0.00   | 0.17 |
| Weighted_AI_Share(BLS)     | 11871 | 0.19 | 0.37 | 0.00 | 0.00   | 0.19 |

##### Panel B. Pairwise Correlation

| Variables                  | (1)  | (2)  | (3)  | (4)  | (5)  |
|----------------------------|------|------|------|------|------|
| AI_Employee(%)             | 1.00 |      |      |      |      |
| AI_Employee_1to2(%)        | 0.87 | 1.00 |      |      |      |
| AI_Employee_3to5(%)        | 0.91 | 0.62 | 1.00 |      |      |
| Weighted_AI_Share(Revelio) | 0.90 | 0.77 | 0.83 | 1.00 |      |
| Weighted_AI_Share(BLS)     | 0.84 | 0.74 | 0.77 | 0.96 | 1.00 |

# Panel C. Determinants Test

| VARIABLES                  | (1)<br>AI_10K       | (2)<br>AI_CC        | (3)<br>AI_EA        | (4)<br>AI_10K       | (5)<br>AI_CC        | (6)<br>AI_EA        | (7)<br>AI_10K       | (8)<br>AI_CC        | (9)<br>AI_EA        | (10)<br>AI_10K      | (11)<br>AI_CC       | (12)<br>AI_EA       |
|----------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| AI_Employee_1to2           | 0.28***<br>(6.44)   | 0.41***<br>(6.53)   | 0.33***<br>(6.76)   |                     |                     |                     |                     |                     |                     |                     |                     |                     |
| AI_Employee_3to5           |                     |                     |                     | 0.32***<br>(6.32)   | 0.50***<br>(6.76)   | 0.38***<br>(6.73)   |                     |                     |                     |                     |                     |                     |
| Weighted_AI_Share(Revelio) |                     |                     |                     |                     |                     |                     | 0.31***<br>(7.36)   | 0.47***<br>(7.16)   | 0.37***<br>(7.39)   |                     |                     |                     |
| Weighted_AI_Share(BLS)     |                     |                     |                     |                     |                     |                     |                     |                     |                     | 0.37***<br>(7.54)   | 0.60***<br>(8.45)   | 0.39***<br>(7.50)   |
| R&D                        | 0.09**<br>(2.56)    | 0.22***<br>(4.46)   | 0.16***<br>(3.34)   | 0.10***<br>(2.80)   | 0.23***<br>(4.67)   | 0.17***<br>(3.58)   | 0.10***<br>(2.80)   | 0.19***<br>(4.07)   | 0.14***<br>(3.20)   | 0.09**<br>(2.53)    | 0.22***<br>(4.40)   | 0.16***<br>(3.47)   |
| Investment                 | 0.03<br>(1.33)      | 0.07**<br>(2.38)    | 0.07***<br>(3.00)   | 0.02<br>(0.91)      | 0.05*<br>(1.91)     | 0.06**<br>(2.50)    | 0.01<br>(0.40)      | 0.05*<br>(1.86)     | 0.05**<br>(2.12)    | 0.02<br>(0.90)      | 0.05*<br>(1.87)     | 0.06**<br>(2.52)    |
| Age                        | 0.07**<br>(2.12)    | -0.07<br>(-1.61)    | -0.05<br>(-1.58)    | 0.06*<br>(1.93)     | -0.07*<br>(-1.77)   | -0.06*<br>(-1.77)   | 0.04<br>(1.38)      | -0.11***<br>(-2.72) | -0.06*<br>(-1.71)   | 0.07**<br>(2.10)    | -0.07<br>(-1.55)    | -0.06*<br>(-1.68)   |
| Size                       | -0.03<br>(-0.57)    | -0.07<br>(-1.18)    | -0.19***<br>(-3.99) | -0.02<br>(-0.52)    | -0.06<br>(-1.12)    | -0.19***<br>(-3.98) | -0.11***<br>(-2.66) | -0.17***<br>(-3.05) | -0.23***<br>(-4.91) | -0.08*<br>(-1.67)   | -0.15**<br>(-2.54)  | -0.24***<br>(-5.06) |
| Profitability              | 0.01<br>(0.17)      | 0.07<br>(1.49)      | -0.00<br>(-0.05)    | 0.01<br>(0.38)      | 0.08*<br>(1.77)     | 0.01<br>(0.15)      | 0.03<br>(1.12)      | 0.08*<br>(1.78)     | 0.01<br>(0.17)      | 0.02<br>(0.63)      | 0.09**<br>(2.06)    | 0.01<br>(0.31)      |
| Tangibility                | -0.23***<br>(-8.13) | -0.20***<br>(-5.54) | -0.18***<br>(-6.42) | -0.22***<br>(-7.79) | -0.18***<br>(-5.15) | -0.17***<br>(-6.10) | -0.20***<br>(-7.47) | -0.16***<br>(-4.77) | -0.17***<br>(-6.08) | -0.21***<br>(-7.81) | -0.17***<br>(-5.06) | -0.17***<br>(-6.15) |
| HHI_Seg                    | -0.06**<br>(-2.22)  | -0.12***<br>(-3.62) | -0.05<br>(-1.64)    | -0.06**<br>(-2.16)  | -0.12***<br>(-3.57) | -0.05<br>(-1.58)    | -0.06**<br>(-2.29)  | -0.11***<br>(-3.45) | -0.05<br>(-1.63)    | -0.06**<br>(-2.05)  | -0.11***<br>(-3.43) | -0.04<br>(-1.43)    |
| MTB                        | 0.02<br>(0.65)      | 0.05<br>(1.13)      | 0.05<br>(1.33)      | 0.02<br>(0.71)      | 0.05<br>(1.16)      | 0.05<br>(1.38)      | 0.02<br>(0.62)      | 0.01<br>(0.22)      | 0.04<br>(1.25)      | 0.01<br>(0.34)      | 0.03<br>(0.71)      | 0.04<br>(1.12)      |

|                |                    |                     |                    |                    |                     |                    |                    |                    |                    |                    |                    |                   |
|----------------|--------------------|---------------------|--------------------|--------------------|---------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-------------------|
| Leverage       | -0.03<br>(-1.51)   | -0.08***<br>(-2.69) | -0.05*<br>(-1.80)  | -0.04*<br>(-1.69)  | -0.08***<br>(-2.82) | -0.05*<br>(-1.92)  | -0.02<br>(-1.24)   | -0.06**<br>(-2.28) | -0.04<br>(-1.47)   | -0.03<br>(-1.28)   | -0.07**<br>(-2.44) | -0.04*<br>(-1.66) |
| N_Analyst      | 0.16***<br>(3.67)  | 0.33***<br>(5.71)   | 0.20***<br>(4.58)  | 0.14***<br>(3.39)  | 0.31***<br>(5.41)   | 0.19***<br>(4.30)  | 0.16***<br>(3.83)  | 0.27***<br>(4.87)  | 0.18***<br>(3.88)  | 0.12***<br>(2.85)  | 0.27***<br>(4.84)  | 0.17***<br>(3.77) |
| Insti_Own      | -0.02<br>(-0.67)   | 0.01<br>(0.36)      | -0.09**<br>(-2.45) | -0.02<br>(-0.71)   | 0.01<br>(0.28)      | -0.09**<br>(-2.51) | -0.01<br>(-0.29)   | 0.05<br>(1.53)     | -0.07**<br>(-2.31) | 0.00<br>(0.01)     | 0.04<br>(1.12)     | -0.06*<br>(-1.78) |
| AI_Dis_Ind     | 0.55***<br>(7.81)  | 0.67***<br>(7.21)   | 0.35***<br>(5.02)  | 0.53***<br>(7.50)  | 0.63***<br>(6.64)   | 0.32***<br>(4.58)  | 0.52***<br>(7.49)  | 0.62***<br>(6.92)  | 0.38***<br>(5.11)  | 0.54***<br>(7.68)  | 0.64***<br>(6.97)  | 0.35***<br>(4.90) |
| Constant       | 0.60***<br>(11.07) | 1.13***<br>(10.98)  | 0.59***<br>(8.25)  | 0.61***<br>(11.28) | 1.14***<br>(11.34)  | 0.60***<br>(8.39)  | 0.59***<br>(10.94) | 1.10***<br>(10.95) | 0.59***<br>(8.02)  | 0.62***<br>(11.61) | 1.16***<br>(11.71) | 0.61***<br>(8.52) |
| Observations   | 11,927             | 11,927              | 11,927             | 11,927             | 11,927              | 11,927             | 11,268             | 11,268             | 11,268             | 11,871             | 11,871             | 11,871            |
| Adj. R-squared | 0.41               | 0.40                | 0.36               | 0.42               | 0.41                | 0.37               | 0.41               | 0.40               | 0.37               | 0.42               | 0.42               | 0.37              |
| Year FE        | Yes                | Yes                 | Yes                | Yes                | Yes                 | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes               |
| Industry FE    | Yes                | Yes                 | Yes                | Yes                | Yes                 | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes               |
| Cluster        | Firm               | Firm                | Firm               | Firm               | Firm                | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               | Firm              |

**Panel D. Efficiency Test (Efficiency (t+1))**

| VARIABLES                   | (1)                | (2)                | (3)                | (4)                | (5)                | (6)                | (7)                | (8)                | (9)                | (10)               | (11)               | (12)               |
|-----------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                             | Efficiency(t+1)    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
| AI_Worker_1to2(%)           | 0.04***<br>(2.90)  | 0.04***<br>(2.58)  | 0.04***<br>(2.75)  |                    |                    |                    |                    |                    |                    |                    |                    |                    |
| AI_Worker_3to5(%)           |                    |                    |                    | 0.04***<br>(2.82)  | 0.03**<br>(2.40)   | 0.03***<br>(2.63)  |                    |                    |                    |                    |                    |                    |
| Weighted_AI_Share(Rev elio) |                    |                    |                    |                    |                    |                    | 0.02***<br>(3.22)  | 0.02***<br>(2.75)  | 0.02***<br>(3.02)  |                    |                    |                    |
| Weighted_AI_Share(BL S)     |                    |                    |                    |                    |                    |                    |                    |                    |                    | 0.02***<br>(2.92)  | 0.01**<br>(2.32)   | 0.02***<br>(2.81)  |
| AI_10K                      | 0.01***<br>(4.17)  |                    |                    | 0.01***<br>(4.13)  |                    |                    | 0.01***<br>(3.70)  |                    |                    | 0.01***<br>(4.09)  |                    |                    |
| AI_CC                       |                    | 0.01***<br>(7.37)  |                    |                    | 0.01***<br>(7.44)  |                    |                    | 0.01***<br>(7.17)  |                    |                    | 0.01***<br>(7.42)  |                    |
| AI_EA                       |                    |                    | 0.01***<br>(6.51)  |                    |                    | 0.01***<br>(6.58)  |                    |                    | 0.01***<br>(6.22)  |                    |                    | 0.01***<br>(6.61)  |
| Constant                    | 0.79***<br>(46.62) | 0.78***<br>(46.87) | 0.78***<br>(46.79) | 0.79***<br>(46.83) | 0.78***<br>(47.05) | 0.78***<br>(46.97) | 0.79***<br>(45.33) | 0.78***<br>(45.45) | 0.78***<br>(45.41) | 0.79***<br>(46.55) | 0.78***<br>(46.74) | 0.78***<br>(46.67) |
| Observations                | 9,516              | 9,516              | 9,516              | 9,516              | 9,516              | 9,516              | 9,046              | 9,046              | 9,046              | 9,470              | 9,470              | 9,470              |
| Adj. R-squared              | 0.26               | 0.26               | 0.26               | 0.26               | 0.26               | 0.26               | 0.26               | 0.26               | 0.26               | 0.26               | 0.26               | 0.26               |
| Controls                    | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Year FE                     | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Industry FE                 | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Cluster                     | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               |

**Panel E. AI Patent (t+1)**

| VARIABLES                  | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  | (8)                 | (9)                  | (10)                 | (11)                 | (12)                 |
|----------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|---------------------|----------------------|----------------------|----------------------|----------------------|
|                            | AI_Patent(t+1)       |                      |                      |                      |                      |                      |                      |                     |                      |                      |                      |                      |
| AI_Employee_1to2(%)        | 28.68***<br>(7.54)   | 27.51***<br>(7.30)   | 29.23***<br>(7.47)   |                      |                      |                      |                      |                     |                      |                      |                      |                      |
| AI_Employee_3to5(%)        |                      |                      |                      | 23.91***<br>(6.60)   | 22.49***<br>(6.25)   | 24.27***<br>(6.55)   |                      |                     |                      |                      |                      |                      |
| Weighted_AI_Share(Revelio) |                      |                      |                      |                      |                      |                      | 10.00***<br>(6.65)   | 9.50***<br>(6.42)   | 10.23***<br>(6.64)   |                      |                      |                      |
| Weighted_AI_Share(BLS)     |                      |                      |                      |                      |                      |                      |                      |                     |                      | 14.80***<br>(7.60)   | 13.89***<br>(7.17)   | 15.30***<br>(7.86)   |
| AI_10K                     | 2.84***<br>(6.03)    |                      |                      | 2.85***<br>(5.82)    |                      |                      | 3.05***<br>(5.41)    |                     |                      | 2.68***<br>(5.65)    |                      |                      |
| AI_CC                      |                      | 2.35***<br>(9.04)    |                      |                      | 2.33***<br>(8.60)    |                      |                      | 2.34***<br>(8.37)   |                      |                      | 2.17***<br>(7.97)    |                      |
| AI_EA                      |                      |                      | 2.17***<br>(6.77)    |                      |                      | 2.16***<br>(6.51)    |                      |                     | 2.22***<br>(6.64)    |                      |                      | 2.05***<br>(6.24)    |
| Constant                   | -14.86***<br>(-4.53) | -16.43***<br>(-5.15) | -16.65***<br>(-5.07) | -14.65***<br>(-4.52) | -16.19***<br>(-5.12) | -16.42***<br>(-5.05) | -11.36***<br>(-3.95) | 12.84***<br>(-4.56) | -12.55***<br>(-4.35) | -13.34***<br>(-4.14) | -14.83***<br>(-4.72) | -15.01***<br>(-4.66) |
| Observations               | 9,646                | 9,646                | 9,646                | 9,646                | 9,646                | 9,646                | 9,171                | 9,171               | 9,171                | 9,597                | 9,597                | 9,597                |
| Adj. R-squared             | 0.29                 | 0.29                 | 0.28                 | 0.29                 | 0.29                 | 0.28                 | 0.27                 | 0.26                | 0.25                 | 0.29                 | 0.30                 | 0.29                 |
| Controls                   | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  |
| Year FE                    | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  |
| Industry FE                | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  |
| Cluster                    | Firm                 | Firm                 | Firm                 | Firm                 | Firm                 | Firm                 | Firm                 | Firm                | Firm                 | Firm                 | Firm                 | Firm                 |

**Table OA 5. Efficiency and AI Patent in Alternative Functional Forms**

This table reports estimates of the main efficiency and AI patent regressions using alternative functional forms. Panels A and B replace the scaled AI disclosure and AI employment measures with their log transformations, computed as  $\ln(1 + \text{variable})$ . Panel A reports one-year-ahead efficiency regressions and Panel B reports one-year-ahead log AI patent regressions. Panel C estimates a negative binomial model for raw one-year-ahead AI patent counts, with log AI employment included as a control. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

**Panel A. Efficiency**

| VARIABLES        | (1)                | (2)                | (3)                | (4)                | (5)                | (6)                |
|------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                  | Efficiency(t+1)    |                    |                    |                    |                    |                    |
| AI_10K(log)      | 0.01***<br>(7.75)  |                    |                    | 0.01***<br>(6.81)  |                    |                    |
| AI_CC(log)       |                    | 0.02***<br>(8.86)  |                    |                    | 0.02***<br>(8.06)  |                    |
| AI_EA(log)       |                    |                    | 0.02***<br>(7.16)  |                    |                    | 0.02***<br>(6.33)  |
| AI_Employee(log) |                    |                    |                    | 0.00<br>(1.50)     | 0.00*<br>(1.71)    | 0.00***<br>(2.71)  |
| Constant         | 0.77***<br>(47.21) | 0.78***<br>(47.09) | 0.78***<br>(47.03) | 0.78***<br>(44.51) | 0.78***<br>(44.69) | 0.79***<br>(44.83) |
| Observations     | 9,516              | 9,516              | 9,516              | 9,516              | 9,516              | 9,516              |
| Adj. R-squared   | 0.26               | 0.26               | 0.26               | 0.26               | 0.27               | 0.26               |
| Controls         | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Year FE          | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Industry FE      | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Cluster          | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               |

**Panel B. AI Patent (in Log)**

| VARIABLES        | (1)                | (2)               | (3)               | (4)               | (5)               | (6)               |
|------------------|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                  | AI Patent log(t+1) |                   |                   |                   |                   |                   |
| AI_10K(log)      | 0.35***<br>(10.35) |                   |                   | 0.28***<br>(8.20) |                   |                   |
| AI_CC(log)       |                    | 0.48***<br>(8.47) |                   |                   | 0.36***<br>(6.24) |                   |
| AI_EA(log)       |                    |                   | 0.46***<br>(6.53) |                   |                   | 0.33***<br>(4.60) |
| AI_Employee(log) |                    |                   |                   | 0.25***           | 0.27***           | 0.30***           |

|                |          |          |          |          |          |          |
|----------------|----------|----------|----------|----------|----------|----------|
|                |          |          |          | (6.41)   | (6.67)   | (7.29)   |
| Constant       | -2.49*** | -2.22*** | -2.11*** | -1.86*** | -1.58*** | -1.44*** |
|                | (-8.13)  | (-7.07)  | (-6.60)  | (-6.04)  | (-5.04)  | (-4.55)  |
| Observations   | 3,845    | 3,845    | 3,845    | 3,845    | 3,845    | 3,845    |
| Adj. R-squared | 0.54     | 0.53     | 0.51     | 0.56     | 0.55     | 0.55     |
| Controls       | Yes      | Yes      | Yes      | Yes      | Yes      | Yes      |
| Year FE        | Yes      | Yes      | Yes      | Yes      | Yes      | Yes      |
| Industry FE    | Yes      | Yes      | Yes      | Yes      | Yes      | Yes      |
| Cluster        | Firm     | Firm     | Firm     | Firm     | Firm     | Firm     |

**Panel C. AI Patent in Raw Forms using Negative Binomial Distribution Model**

| VARIABLES        | (1)                | (2)      | (3)      | (4)      | (5)      | (6)      |
|------------------|--------------------|----------|----------|----------|----------|----------|
|                  | AI Patent Raw(t+1) |          |          |          |          |          |
| AI_10K(log)      | 0.62***            |          |          | 0.53***  |          |          |
|                  | (11.79)            |          |          | (10.44)  |          |          |
| AI_CC(log)       |                    | 0.63***  |          |          | 0.48***  |          |
|                  |                    | (8.55)   |          |          | (6.20)   |          |
| AI_EA(log)       |                    |          | 0.68***  |          |          | 0.55***  |
|                  |                    |          | (8.65)   |          |          | (6.51)   |
| AI_Employee(log) |                    |          |          | 0.35***  | 0.40***  | 0.43***  |
|                  |                    |          |          | (6.21)   | (6.25)   | (7.06)   |
| Constant         | -4.68***           | -3.73*** | -3.66*** | -3.73*** | -2.78*** | -2.66*** |
|                  | (-7.73)            | (-5.66)  | (-5.41)  | (-5.88)  | (-4.10)  | (-3.92)  |
| Observations     | 3,845              | 3,845    | 3,845    | 3,845    | 3,845    | 3,845    |
| Controls         | Yes                | Yes      | Yes      | Yes      | Yes      | Yes      |
| Year FE          | Yes                | Yes      | Yes      | Yes      | Yes      | Yes      |
| Industry FE      | Yes                | Yes      | Yes      | Yes      | Yes      | Yes      |
| Cluster          | Firm               | Firm     | Firm     | Firm     | Firm     | Firm     |

**Table OA 6. Control for Lagged Dependent Variables**

This table augments the main efficiency and AI patent specifications with the lagged dependent variable. Panel A reports one-year-ahead efficiency regressions including lagged efficiency, and Panel B reports one-year-ahead AI patent regressions including the lagged AI patent share. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

**Panel A. Efficiency**

| VARIABLES      | (1)                | (2)                | (3)                | (4)                | (5)                | (6)                |
|----------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                |                    |                    | t+1                |                    |                    |                    |
| Efficiency     | 0.80***<br>(41.58) | 0.79***<br>(41.28) | 0.79***<br>(41.55) | 0.79***<br>(41.35) | 0.79***<br>(41.10) | 0.79***<br>(41.34) |
| AI_10K         | 0.00***<br>(3.71)  |                    |                    | 0.00***<br>(3.58)  |                    |                    |
| AI_CC          |                    | 0.00***<br>(5.56)  |                    |                    | 0.00***<br>(5.54)  |                    |
| AI_EA          |                    |                    | 0.00***<br>(4.20)  |                    |                    | 0.00***<br>(3.93)  |
| AI_Employee(%) |                    |                    |                    | 0.00<br>(0.67)     | 0.00<br>(0.38)     | 0.00<br>(0.58)     |
| Constant       | 0.13***<br>(8.33)  | 0.13***<br>(8.32)  | 0.13***<br>(8.31)  | 0.13***<br>(8.32)  | 0.13***<br>(8.30)  | 0.13***<br>(8.29)  |
| Observations   | 9,516              | 9,516              | 9,516              | 9,516              | 9,516              | 9,516              |
| Adj. R-squared | 0.68               | 0.68               | 0.68               | 0.68               | 0.68               | 0.68               |
| Year FE        | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Industry FE    | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Cluster        | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               |

**Panel B AI Patent**

| VARIABLES      | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
|----------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                |                     |                     | t+1                 |                     |                     |                     |
| AI_Patent%     | 0.64***<br>(29.79)  | 0.63***<br>(29.26)  | 0.64***<br>(30.00)  | 0.62***<br>(28.37)  | 0.62***<br>(28.05)  | 0.63***<br>(28.64)  |
| AI_10K         | 1.11***<br>(5.09)   |                     |                     | 0.91***<br>(4.67)   |                     |                     |
| AI_CC          |                     | 1.00***<br>(7.15)   |                     |                     | 0.83***<br>(5.81)   |                     |
| AI_EA          |                     |                     | 0.94***<br>(5.86)   |                     |                     | 0.71***<br>(4.30)   |
| AI_Employee(%) |                     |                     |                     | 6.44***<br>(6.56)   | 6.14***<br>(6.23)   | 6.40***<br>(6.43)   |
| Constant       | -6.41***<br>(-4.01) | -7.19***<br>(-4.59) | -7.19***<br>(-4.53) | -7.33***<br>(-4.62) | -7.93***<br>(-5.05) | -7.86***<br>(-4.96) |
| Observations   | 9,646               | 9,646               | 9,646               | 9,646               | 9,646               | 9,646               |
| Adj. R-squared | 0.55                | 0.55                | 0.55                | 0.56                | 0.56                | 0.56                |
| Controls       | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Year FE        | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Industry FE    | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Cluster        | Firm                | Firm                | Firm                | Firm                | Firm                | Firm                |

**Table OA 7. Efficiency and AI Disclosure over Longer Horizons**

This table reports the relationship between AI disclosure and operational efficiency at longer horizons. Panel A reports two-year-ahead efficiency regressions and Panel B reports three-year-ahead efficiency regressions. Each panel reports specifications across the three disclosure channels (AI\_10K, AI\_CC, AI\_EA), with and without AI employment included. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

**Panel A Efficiency (t+2)**

| VARIABLES      | (1)                | (2)                | (3)                | (4)                | (5)                | (6)                |
|----------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                | Efficiency(t+2)    |                    |                    |                    |                    |                    |
| AI_10K         | 0.01***<br>(4.27)  |                    |                    | 0.01***<br>(3.92)  |                    |                    |
| AI_CC          |                    | 0.01***<br>(7.52)  |                    |                    | 0.01***<br>(6.34)  |                    |
| AI_EA          |                    |                    | 0.01***<br>(6.55)  |                    |                    | 0.01***<br>(5.67)  |
| AI_Employee(%) |                    |                    |                    | 0.03***<br>(3.38)  | 0.02***<br>(2.92)  | 0.03***<br>(3.19)  |
| Constant       | 0.76***<br>(38.60) | 0.76***<br>(38.80) | 0.75***<br>(38.67) | 0.76***<br>(38.49) | 0.75***<br>(38.66) | 0.75***<br>(38.57) |
| Observations   | 7,377              | 7,377              | 7,377              | 7,377              | 7,377              | 7,377              |
| Adj. R-squared | 0.26               | 0.26               | 0.26               | 0.26               | 0.26               | 0.26               |
| Controls       | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Year FE        | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Industry FE    | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Cluster        | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               |

**Panel B Efficiency (t+3)**

| VARIABLES      | (1)                | (2)                | (3)                | (4)                | (5)                | (6)                |
|----------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                |                    |                    |                    | t+3                |                    |                    |
| AI_10K         | 0.01***<br>(3.36)  |                    |                    | 0.01***<br>(3.15)  |                    |                    |
| AI_CC          |                    | 0.01***<br>(7.96)  |                    |                    | 0.01***<br>(6.73)  |                    |
| AI_EA          |                    |                    | 0.01***<br>(6.55)  |                    |                    | 0.01***<br>(5.75)  |
| AI_Employee(%) |                    |                    |                    | 0.03***<br>(3.56)  | 0.03***<br>(2.97)  | 0.03***<br>(3.28)  |
| Constant       | 0.75***<br>(32.45) | 0.75***<br>(32.50) | 0.74***<br>(32.63) | 0.75***<br>(32.25) | 0.74***<br>(32.32) | 0.74***<br>(32.47) |
| Observations   | 5,690              | 5,690              | 5,690              | 5,690              | 5,690              | 5,690              |
| Adj. R-squared | 0.24               | 0.25               | 0.24               | 0.25               | 0.25               | 0.25               |
| Year FE        | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Industry FE    | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Cluster        | Firm               | Firm               | Firm               | Firm               | Firm               | Firm               |

**Table OA 8. AI Patent and AI Disclosure over Longer Horizons**

This table reports the relationship between AI disclosure and AI patent ratio at longer horizons. Panel A reports two-year-ahead AI patent regressions and Panel B reports three-year-ahead AI patent regressions. Each panel reports specifications across the three disclosure channels (AI\_10K, AI\_CC, AI\_EA), with and without AI employment included. All continuous variables are winsorized at the 1% and 99% levels. T-statistics (based on firm-clustered standard errors) are reported in parentheses. \*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels (two-tailed).

**Panel A. AI Patent (t+2)**

| VARIABLES      | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|----------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                | t+2                  |                      |                      |                      |                      |                      |
| AI_10K         | 3.53***<br>(6.12)    |                      |                      | 2.86***<br>(5.74)    |                      |                      |
| AI_CC          |                      | 3.13***<br>(10.72)   |                      |                      | 2.52***<br>(8.25)    |                      |
| AI_EA          |                      |                      | 2.83***<br>(7.99)    |                      |                      | 2.08***<br>(5.58)    |
| AI_Employee(%) |                      |                      |                      | 17.66***<br>(8.01)   | 16.52***<br>(7.39)   | 17.98***<br>(7.97)   |
| Constant       | -14.02***<br>(-3.73) | -15.93***<br>(-4.41) | -16.32***<br>(-4.35) | -16.20***<br>(-4.50) | -17.57***<br>(-5.01) | -17.81***<br>(-4.94) |
| Observations   | 7,479                | 7,479                | 7,479                | 7,479                | 7,479                | 7,479                |
| Adj. R-squared | 0.28                 | 0.29                 | 0.27                 | 0.31                 | 0.32                 | 0.30                 |
| Controls       | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Year FE        | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Industry FE    | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Cluster        | Firm                 | Firm                 | Firm                 | Firm                 | Firm                 | Firm                 |

**Panel B. AI Patent (t+3)**

| VARIABLES      | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|----------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                |                      |                      | t+3                  |                      |                      |                      |
| AI_10K         | 3.34***<br>(4.94)    |                      |                      | 2.69***<br>(4.83)    |                      |                      |
| AI_CC          |                      | 3.35***<br>(10.41)   |                      |                      | 2.68***<br>(7.80)    |                      |
| AI_EA          |                      |                      | 3.25***<br>(7.92)    |                      |                      | 2.51***<br>(6.05)    |
| AI_Employee(%) |                      |                      |                      | 19.02***<br>(7.17)   | 17.07***<br>(6.20)   | 18.77***<br>(6.96)   |
| Constant       | -14.55***<br>(-3.53) | -16.93***<br>(-4.32) | -16.95***<br>(-4.18) | -16.61***<br>(-4.21) | -18.27***<br>(-4.78) | -18.38***<br>(-4.71) |
| Observations   | 5,780                | 5,780                | 5,780                | 5,780                | 5,780                | 5,780                |
| Adj. R-squared | 0.28                 | 0.30                 | 0.28                 | 0.32                 | 0.33                 | 0.31                 |
| Controls       | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Year FE        | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Industry FE    | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Cluster        | Firm                 | Firm                 | Firm                 | Firm                 | Firm                 | Firm                 |

## References

- Ahn, J., R. Hoitash, U. Hoitash, and E. Krause. 2025. Diversity and career trajectories: Evidence from LinkedIn data on race, ethnicity, and gender in auditing. *The Accounting Review* 100 (4): 1–31.
- Baker, A. C., D. F. Larcker, C. McClure, D. Saraph, and E. M. Watts. 2024. Diversity washing. *Journal of Accounting Research* 62 (5): 1661–1709.
- Barrios, J. M. 2022. Occupational licensing and accountant quality: Evidence from the 150-hour rule. *Journal of Accounting Research* 60 (1): 3–43.
- Lin, Y., M. Shen, and J. Zeng. 2026. The falling Roe and relocation of skilled women. *Contemporary Accounting Research*, forthcoming
- Yang, Y., Uy, M.C.S. and Huang, A., 2020. Finbert: A pretrained language model for financial communications. arXiv preprint arXiv:2006.08097.